Mathematics

Class 2 (English Medium)

Solved Exercises — Sindh Textbook Board

15 lessons — 203 questions

Unit 1: Numbers

Activity: Count forward and write the missing numbers. (i) 1, 2, 3, 4, _____, _____, 7

Answer: 5,, 6, 7.

Activity: Count forward and write the missing numbers. (ii) 32, 33, 34, _____, _____, 37, _____

Answer: 35,, 36,, 38.

Activity: Count forward and write the missing numbers. (iii) 16, 26, 36, _____, 56, _____, 76

Answer: 46,, 66, 76.

Activity: Count backward and write the missing numbers. (i) 6, 5, _____, 3, _____, 1

Answer: 4,, 2, 1.

Activity: Count backward and write the missing numbers. (ii) 88, _____, 86, 85, _____, 83

Answer: 87,, 84, 83.

Activity: Count backward and write the missing numbers. (iii) _____, 24, _____, 22, 21, 20

Answer: Step 1: The row goes backward:, then 23,

Question: Example: Count hundreds, tens and ones. Also write the number. The picture shows 2 hundred-flats (100 each), 3 tens-rods and 6 ones-cubes.

Answer: Step 1: Hundreds: 2 × 100 = 200. Step 2: Tens: 3 × 10 = 30. Step 3: Ones: 6 × 1 = 6. Step 4: Add all: 200 + 30 + 6 = 236. Final answer: The number is 236 — two hundred thirty six.

Question: Exercise 1, Question 1(a): Count hundreds, tens and ones. Also

write the numbers. The picture shows 1 hundred-flat, 4 tens-rods and 5 ones-cubes.

Answer: Step 1: Hundreds: 1 × 100 = 100. Step 2: Tens: 4 × 10 = 40. Step 3: Ones: 5 × 1 = 5. Step 4: Add: 100 + 40 + 5 = 145. Final answer: The number is 145.

Question: Exercise 1, Question 1(b): Count hundreds, tens and ones. Also write the numbers. The picture shows 2 hundred-flats, 2 tens-rods and 9 ones-cubes.

Answer: Step 1: Hundreds: 2 × 100 = 200. Step 2: Tens: 2 × 10 = 20. Step 3: Ones: 9 × 1 = 9. Step 4: Add: 200 + 20 + 9 = 229. Final answer: The number is 229.

Exercise 1, Question 2(a): Count forward and write the missing numbers.

0, 1, 2, _____, _____, 5, _____

Answer: numbers are, 3, 4, and 6.

Exercise 1, Question 2(b): Count forward and write the missing numbers.

20, 21, 22, _____, 24, _____, 26

Answer: numbers are, 23 and 25.

Exercise 1, Question 2(c): Count forward and write the missing numbers.

11, 22, 33, _____, _____, 66, _____

Answer: 77.

Exercise 1, Question 3(a): Count backward and write the missing numbers.

3, 4, 5, _____, 7, _____, 9

Answer: numbers are, 6 and 8.

Exercise 1, Question 3(b): Count backward and write the missing numbers.

56, 55, _____, _____, 52, _____, _____

Answer: numbers, are, 54,, 53, 51 and 50.

Exercise 1, Question 3(c): Count backward and write the missing numbers.

85, 75, _____, _____, 45, _____, _____

Answer: numbers, are, 65,, 55, 35 and 25.

Question: Reading chart: Read the numbers from 100 to 999. For example, how do we read 100, 250 and 999?

Answer: Step 1: First read the hundreds digit, then the tens and ones part. Step 2: 100 = one hundred; 250 = two hundred fifty; 999 = nine hundred ninety-nine. Final answer: 100 is read "one hundred", 250 is "two hundred fifty" and 999 is "nine hundred ninety-nine". (The book's charts show the rows 100-499 and 800-999.)

Activity: Write the missing numbers. (i) 341, 342, _____, _____, _____, _____, _____, _____, _____, 350 (ii) 261, 262, _____, _____, _____, _____, _____, _____, _____, 270 (iii) 401, 402, _____, _____, _____, _____, _____, _____, _____, 410 (iv) 981, _____, _____, _____, _____, _____, _____, _____, 990 (v) 721, _____, _____, _____, _____, _____, _____, _____, _____, 730

Answer: After, 342:, 343,, 344,, 345,, 346,, 347, 348, 349. Step 3:, After, 262:, 263,, 264,, 265,, 266,, 267, 268, 269. Step 4:, After, 402:, 403,, 404,, 405,, 406,, 407, 408, 409. Step 5:, After, 981:, 982,, 983,, 984,, 985,, 986,, 987, 988, 989. Step 6:, After 721:, 722, 723,, 724, 725,, 726, 727,, 728, 729., Final answer:, (i) 343-349, (ii) 263-269 (iii) 403-409 (iv) 982-989 (v) 722-729.

Question: Activity: Read and trace the numbers from 1 to 50 in words.

Answer: Step 1: 1-10: One, Two, Three, Four, Five, Six, Seven, Eight, Nine, Ten. Step 2: 11-20: Eleven, Twelve, Thirteen, Fourteen, Fifteen, Sixteen, Seventeen, Eighteen, Nineteen, Twenty. Step 3: 21-30: Twenty one, Twenty two, Twenty three, Twenty four, Twenty five, Twenty six, Twenty seven, Twenty eight, Twenty nine, Thirty. Step 4: 31-40: Thirty one, Thirty two,

Thirty three, Thirty four, Thirty five, Thirty six, Thirty seven, Thirty eight, Thirty nine, Forty. Step 5: 41-50: Forty one, Forty two, Forty three, Forty four, Forty five, Forty six, Forty seven, Forty eight, Forty nine, Fifty. Final answer: The number names from 1 to 50 are written above.

Exercise 2, Question 1: Write the missing numbers. (a) 200, 201, _____, _____, 204, _____, _____, 207, _____, _____, 210 (b) 311, 312, _____, _____, _____, 316, _____, _____, 319, _____, 321 (c) 415, _____, _____, 418, _____, _____, _____, 422, _____, _____, 425 (d) 509, _____, _____, _____, 513, _____, _____, _____, 517, _____, _____ (e) 625, _____, _____, _____, _____, 630, _____, _____, _____, 634, _____ (f) 895, _____, _____, _____, 899, _____, _____, 902, _____, _____, _____ (g) 899, _____, _____, 902, _____, _____, _____, 906, _____, _____, 909 (h) 989, _____, 991, _____, _____, _____, 995, _____, _____, _____, _____

Answer: 202,, 203,, 205,, 206,, 208,, 209, 313,, 314,, 315,, 317,, 318,, 320, 416,, 417,, 419,, 420,, 421,, 423,, 424, 510,, 511,, 512,, 514,, 515,, 516,, 518,, 519, 520, 626,, 627,, 628,, 629,, 631,, 632,, 633,, 635, 896,, 897,, 898,, 900,, 901,, 903,, 904,, 905, 900,, 901,, 903,, 904,, 905,, 907,, 908, 990,, 992,, 993,, 994,, 996,, 997,, 998,, 999.

Question: Exercise 2, Question 2: Match the numbers in words with numbers in figures. Words: Nineteen, Twelve, Twenty three, Twenty, Thirty seven, Forty nine, Twenty seven, Fifty. Figures: 37, 49, 12, 27, 50, 19, 20, 23.

Answer: Step 1: Read each number name and find its figure. Step 2: Nineteen = 19, Twelve = 12, Twenty three = 23, Twenty = 20, Thirty seven

= 37, Forty nine = 49, Twenty seven = 27, Fifty = 50. Final answer: The matching is given above.

Question: Exercise 2, Question 3: Write the numbers in figures. Eighteen, Twelve, Twenty two, Thirty eight, Forty three, Forty, Thirty nine, Seven.

Answer: Step 1: Split each name into its tens word and ones word. Step 2: Eighteen = 18, Twelve = 12, Twenty two = 22, Thirty eight = 38, Forty three = 43, Forty = 40, Thirty nine = 39, Seven = 7. Final answer: 18, 12, 22, 38, 43, 40, 39, 7.

Question: Exercise 2, Question 4: Write the numbers in words. 47, 36, 11, 17, 26, 44, 8, 30.

Answer: Step 1: Look at the tens and ones of each number. Step 2: 47 = Forty seven, 36 = Thirty six, 11 = Eleven, 17 = Seventeen, 26 = Twenty six, 44 = Forty four, 8 = Eight, 30 = Thirty. Final answer: The words are written above.

Question: Place value: Write the place value of each digit in 537, and the place value of the encircled digits in the examples: 5(6)7, (9)78, 42(3).

Answer: Step 1: From the right, the first digit is ones, the second is tens and the third is hundreds. Step 2: In 537: 7 is 7 ones = 7, 3 is 3 tens = 30, 5 is 5 hundreds = 500, so 537 = 500 + 30 + 7. Step 3: In 567 the encircled 6 is in tens = 60; in 978 the 9 is in hundreds = 900; in 423 the 3 is in ones = 3. Final answer: 537 = 500 + 30 + 7; encircled digits: 60, 900, 3.

Question: Activity: Count hundreds, tens, ones on the abacus and write the numbers. (i) Hundreds 1, Tens 1, Ones 2 (ii) Hundreds 2, Tens 0, Ones 5 (iii) Hundreds 3, Tens 4, Ones 0

Answer: Step 1: Count the beads on each abacus rod: hundreds, tens, ones. Step 2: (i) 1, 1, 2 → 112. Step 3: (ii) 2, 0, 5 → 205. Step 4: (iii) 3, 4, 0 → 340. Final answer: (i) 112 (ii) 205 (iii) 340. (The first example, 423, is

solved in the book.)

Question: Exercise 3, Question 1: Write the place value of the encircled digit. (a) 3(5)9 (b) 71(3) (c) (5)98 (d) (6)98 (e) 2(4)6 (f) 80(1)

Answer: Step 1: See the place of the encircled digit. Step 2: (a) 5 is in tens = 5 tens = 50 (solved in the book). (b) 3 is in ones = 3. (c) 5 is in hundreds = 500. (d) 6 is in hundreds = 600. (e) 4 is in tens = 40. (f) 1 is in ones = 1. Final answer: (a) 50 (b) 3 (c) 500 (d) 600 (e) 40 (f) 1.

Question: Exercise 3, Question 2: Count hundreds, tens, ones in the abacus pictures and write the numbers. (i) Hundreds 2, Tens 5, Ones 3 (ii) Hundreds 3, Tens 1, Ones 5 (iii) Hundreds 3, Tens 2, Ones 0 (iv) Hundreds 4, Tens 0, Ones 3 (v) Hundreds 1, Tens 2, Ones 3 (vi) Hundreds 5, Tens 0, Ones 0

Answer: Step 1: Count the beads on the three rods of each picture. Step 2: Write the digits in hundreds-tens-ones order: (i) 253 (ii) 315 (iii) 320 (iv) 403 (v) 123 (vi) 500. Final answer: (i) 253 (ii) 315 (iii) 320 (iv) 403 (v) 123 (vi) 500.

Question: Example: For the given number 346 — tick the number which is more, cross the number which is less and encircle the number which is same: 596, 346, 299. And for the given number 998, colour: 998, 899, 999 (red = greater, green = smaller, blue = equal).

Answer: Step 1: Compare each number with 346: 596 > 346 so tick ✓; 346 = 346 so encircle it; 299 < 346 so cross ✗. Step 2: Compare with 998: 998 is equal = blue; 899 is smaller = green; 999 is greater = red. Final answer: Tick 596, encircle 346, cross 299; 998 blue, 899 green, 999 red.

Question: Example: Write the numbers in ascending and descending order. (i) 526, 176, 985, 645, 396 in ascending order (ii) 127, 476, 896, 645, 255 in descending order

Answer: Step 1: Ascending order goes from smallest to largest: 176, 396,

526, 645, 985. Step 2: Descending order goes from largest to smallest: 896, 645, 476, 255, 127. Final answer: (i) 176, 396, 526, 645, 985 (ii) 896, 645, 476, 255, 127.

Question: Exercise 4, Question 1: Tick the number which is more, cross the less number and encircle the number which is same as the given number. (a) Given 236: 512, 236, 117 (b) Given 525: 525, 496, 552 (c) Given 815: 684, 920, 815 (d) Given 305: 503, 305, 130 (e) Given 114: 108, 141, 114

Answer: Step 1: Compare each group with the given number; the bigger gets a tick, the smaller a cross, and the equal one is encircled. Final answer: (a) Tick 512, encircle 236, cross 117 (b) Encircle 525, cross 496, tick 552 (c) Cross 684, tick 920, encircle 815 (d) Tick 503, encircle 305, cross 130 (e) Cross 108, tick 141, encircle 114.

Question: Exercise 4, Question 2: Colour red for greater, colour green for smaller and colour blue for equal to the given number. (a) Given 419: 914, 419, 276 (b) Given 675: 675, 705, 576 (c) Given 829: 829, 452, 928 (d) Given 800: 825, 800, 450 (e) Given 101: 100, 102, 101

Answer: Step 1: Compare with the given number and choose the colour. Final answer: (a) 914 red, 419 blue, 276 green (b) 675 blue, 705 red, 576 green (c) 829 blue, 452 green, 928 red (d) 825 red, 800 blue, 450 green (e) 100 green, 102 red, 101 blue.

Question: Exercise 4, Question 3: Write the numbers in ascending order. (a) 565, 200, 432, 152 (b) 910, 676, 408, 326 (c) 136, 108, 296, 300 (d) 686, 700, 940, 850

Answer: Step 1: Find the smallest number first, then the next bigger one, and so on till the end. Final answer: (a) 152, 200, 432, 565 (b) 326, 408, 676, 910 (c) 108, 136, 296, 300 (d) 686, 700, 850, 940.

Question: Exercise 4, Question 4: Write the numbers in descending order. (a) 316, 205, 596, 601 (b) 444, 555, 333, 666 (c) 199, 999, 812, 777 (d)

512, 627, 196, 952

Answer: Step 1: Find the largest number first, then the next smaller one, and so on till the end. Final answer: (a) 601, 596, 316, 205 (b) 666, 555, 444, 333 (c) 999, 812, 777, 199 (d) 952, 627, 512, 196.

Question: Example: Round the given numbers to the nearest tens. (a) 243 (b) 426 (c) 126 (d) 124

Answer: Step 1: Rule: if the ones digit is 5 or more, increase the tens by 1; if it is 4 or less, keep the tens the same and write 0 in the ones place. Step 2: (a) 243 has ones digit 3 (less than 4... it is 4 or less) → 240. Step 3: (b) 426 has ones digit 6 (5 or more) → 430. Step 4: (c) 126 has ones digit 6 → 130. Step 5: (d) 124 has ones digit 4 → 120. Final answer: (a) 240 (b) 430 (c) 130 (d) 120.

Question: Exercise 5, Question 1: Round the given numbers (shown on the abacus) to the nearest tens. (a) 327 (b) 248 (c) 232

Answer: Step 1: Look at the ones digit of each number. Step 2: (a) 327: ones digit 7 (5 or more) → tens 2 becomes 3 → 330 (solved in the book). Step 3: (b) 248: ones digit 8 → tens 4 becomes 5 → 250. Step 4: (c) 232: ones digit 2 (4 or less) → tens stay the same → 230. Final answer: (a) 330 (b) 250 (c) 230.

Question: Exercise 5, Question 2: Round the given numbers to the nearest tens. (a) 569 (b) 392 (c) 105 (d) 237 (e) 621

Answer: Step 1: Apply the rule: ones digit 5 or more rounds up, 4 or less rounds down. Step 2: (a) 569: ones 9 → 570 (solved in the book). (b) 392: ones 2 → 390. (c) 105: ones 5 → 110. (d) 237: ones 7 → 240. (e) 621: ones 1 → 620. Final answer: (a) 570 (b) 390 (c) 110 (d) 240 (e) 620.

Question: Activity: Read the positions of the bikes in the race (1st to 20th) and write the ordinal numbers in words.

Answer: Step 1: The bike nearest the finish is 1st; count on in order. Step

2: Positions: 1st green, 2nd orange, 3rd light purple, 4th light green, 5th purple, 6th orange-brown, 7th white, 8th cream, 9th pink, 10th red, 11th teal-blue, 12th teal, 13th light blue, 14th light pink, 15th light green, 16th hot pink, 17th red, 18th light blue, 19th dark blue, 20th yellow. Step 3: In words: First, Second, Third, Fourth, Fifth, Sixth, Seventh, Eighth, Ninth, Tenth, Eleventh, Twelfth, Thirteenth, Fourteenth, Fifteenth, Sixteenth, Seventeenth, Eighteenth, Nineteenth, Twentieth. Final answer: The positions and words are written above.

Question: Activity: Match the ordinal numbers with their words. 1st, 2nd, 3rd, 4th, 5th, 6th, 7th, 8th, 9th, 10th, 11th, 12th, 13th, 14th, 15th, 16th, 17th, 18th, 19th, 20th.

Answer: Step 1: Find the correct word for each ordinal number. Final answer: 1st = First, 2nd = Second, 3rd = Third, 4th = Fourth, 5th = Fifth, 6th = Sixth, 7th = Seventh, 8th = Eighth, 9th = Ninth, 10th = Tenth, 11th = Eleventh, 12th = Twelfth, 13th = Thirteenth, 14th = Fourteenth, 15th = Fifteenth, 16th = Sixteenth, 17th = Seventeenth, 18th = Eighteenth, 19th = Nineteenth, 20th = Twentieth.

Question: Exercise 6, Question 1: Look at the positions of the vehicles and write the missing positions. Given positions: 1st, 2nd, 4th, 6th, 9th, 10th, 13th, 14th, 16th, 18th, 19th, 20th.

Answer: Step 1: Write the positions in order from 1 to 20. Step 2: The positions not given are missing: 3rd, 5th, 7th, 8th, 11th, 12th, 15th, 17th. Step 3: By colour: 3rd pink car, 5th green car, 7th purple car, 8th blue car, 11th green car, 12th orange car, 15th light pink car, 17th purple car. Final answer: The missing positions are 3rd, 5th, 7th, 8th, 11th, 12th, 15th, 17th.

Question: Exercise 6, Question 2: Write down the ordinal numbers for English letters (left to right): A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R,

S, T.

Answer: Step 1: Count the letters in order. Step 2: A = 1st, B = 2nd, C = 3rd, D = 4th, E = 5th, F = 6th, G = 7th, H = 8th, I = 9th, J = 10th. Step 3: K = 11th, L = 12th, M = 13th, N = 14th, O = 15th, P = 16th, Q = 17th, R = 18th, S = 19th, T = 20th. Final answer: Written above; A, B, E, F, I, K and Q are already given in the book, the rest are to be filled.

Question: Roman numbers: Write the Roman numbers from 1 to 12 and tell why there is no Roman number for 0.

Answer: Step 1: The basic symbols are: I = 1, V = 5, X = 10. Step 2: 1 = I, 2 = II, 3 = III, 4 = IV, 5 = V, 6 = VI, 7 = VII, 8 = VIII, 9 = IX, 10 = X, 11 = XI, 12 = XII. Step 3: In small letters: i, ii, iii, iv, v, vi, vii, viii, ix, x, xi, xii. Final answer: There is no symbol for 0 (zero) in Roman numbers, so 0 has no Roman number.

Question: Exercise 7, Question 1: Write in Roman numbers. 3, 8, 12, 1, 5, 4.

Answer: Step 1: Make each number from the symbols: 3 = I three times = III (solved in the book). Step 2: 8 = V + III = VIII; 12 = X + II = XII; 1 = I; 5 = V; 4 = one less than V = IV. Final answer: 3 = III, 8 = VIII, 12 = XII, 1 = I, 5 = V, 4 = IV.

Question: Exercise 7, Question 2: Write in Arabic numbers. II, VI, VII, IX, XI.

Answer: Step 1: Add or subtract the values of the symbols: II = 2 (solved in the book). Step 2: VI = 5 + 1 = 6; VII = 5 + 2 = 7; IX = 10 − 1 = 9; XI = 10 + 1 = 11. Final answer: II = 2, VI = 6, VII = 7, IX = 9, XI = 11.

Exercise 7, Question 3: Write the missing Roman numbers. I, II, _____, _____, _____, VI, VII, _____, _____, X, _____, _____

Answer: III, IV,, V, VI., Step 2: Continue:, VIII,, IX,, XI, XII. Final answer: The missing ones, are III, IV, V, VIII, IX, XI, XII.

Question: Exercise 7, Question 4: Match the numbers with their Roman numbers (small letters). Numbers: 5, 9, 12, 3, 11, 6, 2, 8. Roman: iii, ii, viii, v, ix, xii, vi, xi.

Answer: Step 1: Find the Roman number for each number. Final answer: 5 = v, 9 = ix, 12 = xii, 3 = iii, 11 = xi, 6 = vi, 2 = ii, 8 = viii.

Unit 2: Number Operations

Question: Example: Find the following. (i) 60 + 20 (ii) 72 + 9 (iii) 39 + 11 (iv) 77 − 37 (v) 96 − 24 (vi) 58 − 20

Answer: Step 1: (i) Ones: 0 + 0 = 0; Tens: 6 + 2 = 8 → 80. Step 2: (ii) Ones: 2 + 9 = 11, write 1 and carry 1; Tens: 7 + 1 = 8 → 81. Step 3: (iii) Ones: 9 + 1 = 10, write 0 and carry 1; Tens: 3 + 1 + 1 = 5 → 50. Step 4: (iv) Ones: 7 − 7 = 0; Tens: 7 − 3 = 4 → 40. Step 5: (v) Ones: 6 − 4 = 2; Tens: 9 − 2 = 7 → 72. Step 6: (vi) Ones: 8 − 0 = 8; Tens: 5 − 2 = 3 → 38. Final answer: (i) 80 (ii) 81 (iii) 50 (iv) 40 (v) 72 (vi) 38.

Activity: Complete the following addition and subtraction facts to 100. 32 + 50 = _____; 69 + 11 = _____; 80 + _____ = 100; _____ + 15 = 65; 92 + 8 = _____; 82 − 32 = _____; 80 − 11 = _____; 100 − _____ = 80; 65 − _____ = 15; 100 − 8 = _____

Answer: 82, given). Step 2:, 10, tens 6 + 1 + 1 = 8 → 80. Step 3: 100 − 80 = 20, so, 20, 100. Step 4: 65 − 15 = 50, so 50, 100. Step 6: Subtraction:, 50;, 69;, 20, 50, 92. Final answer: 82; 80; 20; 50; 100; 50; 69; 20; 50; 92.

Question: Exercise 1 (Unit 2), Question 1: Match the following (equal sums). Left: 32 + 64, 19 + 21, 61 + 10, 91 + 9, 58 + 12, 50 + 50. Right: 9 + 91, 80 + 20, 60 + 10, 60 + 11, 30 + 10, 64 + 32.

Answer: Step 1: Find each sum: 32 + 64 = 96; 19 + 21 = 40; 61 + 10 = 71; 91 + 9 = 100; 58 + 12 = 70; 50 + 50 = 100. Step 2: Right side sums: 9 + 91 = 100; 80 + 20 = 100; 60 + 10 = 70; 60 + 11 = 71; 30 + 10 = 40; 64 + 32 = 96. Final answer: 32 + 64 ↔ 64 + 32 (96); 19 + 21 ↔ 30 + 10 (40); 61 + 10 ↔ 60 + 11 (71); 91 + 9 ↔ 9 + 91 (100); 58 + 12 ↔ 60 + 10 (70); 50 + 50 ↔ 80 + 20 (100).

Exercise 1 (Unit 2), Question 2: Complete the following addition and subtraction facts. 63 + 7 = _____; 90 − _____ = 80; 50 − 22 = _____; 100 + 0 = _____; 21 + 41 = _____; 70 − 63 = _____; 80 + _____ = 90; 22 + 28 = _____; 100 − _____ = 100; 62 − 41 = _____

Answer: 10 → 70. Step 2:, 80, 8; tens 4 − 2 = 2 → 28. Step 4:, 100. Step 5:, 62. Step 6:, 7. Step 7: 90 − 80 = 10, so, 10, 50. Step 9:, 0, 21. Final answer: 70; 10; 28; 100; 62; 7; 10; 50; 0; 21.

Question: Examples (mental addition): (i) 245 + 3 (ii) 326 + 7 (iii) 534 + 20 (iv) 937 + 40 (v) 400 + 40 (vi) 564 + 17 (vii) 520 + 260 (viii) 208 + 300

Answer: Step 1: (i) Only the ones change: 245 + 3 = 248. Step 2: (ii) Split 7: 326 + 4 = 330, 330 + 3 = 333. Step 3: (iii) Only tens: 534 + 20 = 554. Step 4: (iv) 937 + 40: tens 3 + 4 = 7 → 977. Step 5: (v) 400 + 40 = 440. Step 6: (vi) Split 17: 564 + 6 = 570, 570 + 11 = 581. Step 7: (vii) 520 + 260: hundreds 5 + 2 = 7, tens 2 + 6 = 8 → 780. Step 8: (viii) 208 + 300 = 8 + 500 = 508. Final answer: (i) 248 (ii) 333 (iii) 554 (iv) 977 (v) 440 (vi) 581 (vii) 780 (viii) 508.

Activity: Add the following mentally. 720 + 350 = _____; 110 + 760 = _____; 226 + 7 = _____; 196 + 14 = _____; 180 + 30 = _____; 180 + 9 = _____; 560 + 7 = _____; 972 + 18 = _____; 921 + 11 = _____; 442 + 5 = _____; 192 + 28 = _____; 500 + 18 = _____

Answer: 10, tens 2 + 5 = 7 → 1070. Step 2:, 870. Step 3:, 230, + 3 = 233. Step 4:, 200, + 10 = 210. Step 5:, 210. Step 6:, 189. Step 7:, 567. Step 8:, 980, + 10 = 990. Step 9:, 932. Step 10:, 447. Step 11:, 200, + 20 = 220. Step 12:, 518. Final answer: 1070; 870; 233; 210; 210; 189; 567; 990; 932; 447; 220; 518. (206 + 5 = 211 and 500 + 400 = 900 are solved in the

book.)

Question: Examples (addition in written form): (i) 523 + 5 (ii) 627 + 5 (iii) 423 + 60 (the book's heading says 523 but the working uses 423) (iv) 634 + 80 (v) 721 + 36 (vi) 567 + 58 (vii) 543 + 316 (viii) 475 + 348

Answer: Step 1: (i) Ones: 3 + 5 = 8; other digits stay the same → 528. Step 2: (ii) Ones: 7 + 5 = 12, write 2, carry 1; Tens: 2 + 1 = 3 → 632. Step 3: (iii) Tens: 2 + 6 = 8 → 483. Step 4: (iv) Tens: 3 + 8 = 11, write 1, carry 1; Hundreds: 6 + 1 = 7 → 714. Step 5: (v) Ones: 1 + 6 = 7; Tens: 2 + 3 = 5 → 757. Step 6: (vi) Ones: 7 + 8 = 15, write 5, carry 1; Tens: 6 + 5 + 1 = 12, write 2, carry 1; Hundreds: 5 + 1 = 6 → 625. Step 7: (vii) Ones: 3 + 6 = 9; Tens: 4 + 1 = 5; Hundreds: 5 + 3 = 8 → 859. Step 8: (viii) Ones: 5 + 8 = 13, write 3, carry 1; Tens: 7 + 4 + 1 = 12, write 2, carry 1; Hundreds: 4 + 3 + 1 = 8 → 823. Final answer: (i) 528 (ii) 632 (iii) 483 (iv) 714 (v) 757 (vi) 625 (vii) 859 (viii) 823.

Question: Activity: Add the following (in columns). 236 + 8; 529 + 80; 783 + 16; 537 + 152; 562 + 396; 918 + 32; 639 + 80; 696 + 23

Answer: Step 1: 236 + 8: ones 6 + 8 = 14, write 4, carry 1; tens 3 + 1 = 4 → 244. Step 2: 529 + 80: tens 2 + 8 = 10, write 0, carry 1; hundreds 5 + 1 = 6 → 609. Step 3: 783 + 16: ones 3 + 6 = 9; tens 8 + 1 = 9 → 799. Step 4: 537 + 152: ones 7 + 2 = 9; tens 3 + 5 = 8; hundreds 5 + 1 = 6 → 689. Step 5: 562 + 396: ones 2 + 6 = 8; tens 6 + 9 = 15, write 5, carry 1; hundreds 5 + 3 + 1 = 9 → 958. Step 6: 918 + 32: ones 8 + 2 = 10, write 0, carry 1; tens 1 + 3 + 1 = 5 → 950. Step 7: 639 + 80: tens 3 + 8 = 11, write 1, carry 1; hundreds 6 + 1 = 7 → 719. Step 8: 696 + 23: ones 6 + 3 = 9; tens 9 + 2 = 11, write 1, carry 1; hundreds 6 + 1 = 7 → 719. Final answer: 244; 609; 799; 689; 958; 950; 719; 719. (125 + 3 = 128, 346 + 30 = 376, 296 + 25 = 321 and 289 + 346 = 635 are solved in the book.)

Question: Exercise 2 (Unit 2), Question 1: Add mentally. (i) 123 + 5 (ii) 634

+ 8 (iii) 579 + 20 (iv) 440 + 80 (v) 810 + 23 (vi) 913 + 29 (vii) 638 + 151 (viii) 785 + 136 (ix) 513 + 64 (x) 345 + 9

Answer: Step 1: (i) 123 + 5 = 128. (ii) 634 + 8: 634 + 6 = 640, + 2 = 642. (iii) 579 + 20 = 599. (iv) 440 + 80: tens 4 + 8 = 12 → 520. (v) 810 + 23 = 833. Step 2: (vi) 913 + 29: 913 + 9 = 922, + 20 = 942. (vii) 638 + 151 = 789. (viii) 785 + 136: 785 + 100 = 885, + 36 = 921. (ix) 513 + 64 = 577. (x) 345 + 9: 345 + 5 = 350, + 4 = 354. Final answer: (i) 128 (ii) 642 (iii) 599 (iv) 520 (v) 833 (vi) 942 (vii) 789 (viii) 921 (ix) 577 (x) 354.

Question: Exercise 2 (Unit 2), Question 2: Add the following (in columns). (i) 236 + 53 (ii) 986 + 3 (iii) 508 + 6 (iv) 518 + 43 (v) 675 + 20 (vi) 853 + 70 (vii) 834 + 76 (viii) 456 + 168 (ix) 345 + 632 (x) 278 + 369 (xi) 509 + 198 (xii) 467 + 298

Answer: Step 1: (i) Ones 6 + 3 = 9; Tens 3 + 5 = 8 → 289. (ii) Ones 6 + 3 = 9 → 989. (iii) Ones 8 + 6 = 14, write 4, carry 1; Tens 0 + 1 = 1 → 514. (iv) Ones 8 + 3 = 11, write 1, carry 1; Tens 1 + 4 + 1 = 6 → 561. (v) Tens 7 + 2 = 9 → 695. (vi) Tens 5 + 7 = 12, write 2, carry 1; Hundreds 8 + 1 = 9 → 923. Step 2: (vii) Ones 4 + 6 = 10, write 0, carry 1; Tens 3 + 7 + 1 = 11, write 1, carry 1; Hundreds 8 + 1 = 9 → 910. (viii) Ones 6 + 8 = 14, write 4, carry 1; Tens 5 + 6 + 1 = 12, write 2, carry 1; Hundreds 4 + 1 + 1 = 6 → 624. (ix) Ones 5 + 2 = 7; Tens 4 + 3 = 7; Hundreds 3 + 6 = 9 → 977. (x) Ones 8 + 9 = 17, write 7, carry 1; Tens 7 + 6 + 1 = 14, write 4, carry 1; Hundreds 2 + 3 + 1 = 6 → 647. (xi) Ones 9 + 8 = 17, write 7, carry 1; Tens 0 + 9 + 1 = 10, write 0, carry 1; Hundreds 5 + 1 + 1 = 7 → 707. (xii) Ones 7 + 8 = 15, write 5, carry 1; Tens 6 + 9 + 1 = 16, write 6, carry 1; Hundreds 4 + 2 + 1 = 7 → 765. Final answer: (i) 289 (ii) 989 (iii) 514 (iv) 561 (v) 695 (vi) 923 (vii) 910 (viii) 624 (ix) 977 (x) 647 (xi) 707 (xii) 765.

Question: Examples (mental and written subtraction): (i) 325 − 4 (ii) 534 − 6 (iii) 678 − 50 (iv) 624 − 70 (v) 978 − 35 (vi) 542 − 65 (vii) 879 − 567

(viii) 620 − 255 (ix) 398 − 5 (x) 275 − 8 (xi) 694 − 20 (xii) 526 − 70 (xiii) 289 − 57 (xiv) 532 − 64 (xv) 457 − 321 (xvi) 613 − 425

Answer: Step 1: (i) Only ones: 5 − 4 = 1 → 321. (ii) 6 cannot be taken from 4: 534 − 6 = 524 + (10 − 6) = 528. (iii) Only tens: 7 − 5 = 2 → 628. (iv) 7 tens cannot be taken from 2 tens: 624 − 70 = 524 + (100 − 70) = 524 + 30 = 554. (v) Ones 8 − 5 = 3; Tens 7 − 3 = 4 → 943. (vi) 542 − 65 = 442 + (100 − 65) = 442 + 35 = 477. (vii) Each place separately: 8−5, 7−6, 9−7 → 312. (viii) 620 − 255 = (520 − 200) + (100 − 55) = 320 + 45 = 365. Step 2: (ix) Ones 8 − 5 = 3 → 393. (x) 8 cannot be taken from 5, borrow 1 ten: 15 − 8 = 7; tens become 6 → 267. (xi) Tens 9 − 2 = 7 → 674. (xii) 7 tens cannot be taken from 2 tens, borrow 1 hundred: 12 − 7 = 5; hundreds become 4 → 456. (xiii) Ones 9 − 7 = 2; Tens 8 − 5 = 3 → 232. (xiv) Ones: 4 cannot be taken from 2, borrow: 12 − 4 = 8; Tens: 6 cannot be taken from 2, borrow: 12 − 6 = 6; Hundreds 4 → 468. (xv) Each place: 7−1, 5−2, 4−3 → 136. (xvi) Ones: 5 cannot be taken from 3, borrow: 13 − 5 = 8; Tens: 2 cannot be taken from 0, borrow: 10 − 2 = 8; Hundreds: 5 − 4 = 1 → 188. Final answer: (i) 321 (ii) 528 (iii) 628 (iv) 554 (v) 943 (vi) 477 (vii) 312 (viii) 365 (ix) 393 (x) 267 (xi) 674 (xii) 456 (xiii) 232 (xiv) 468 (xv) 136 (xvi) 188.

Activity: Subtract mentally. 735 − 14 = _____; 340 − 165 = _____; 916 − 9 = _____; 790 − 280 = _____; 506 − 30 = _____; 515 − 40 = _____; 796 − 87 = _____; 800 − 210 = _____

Answer: 1; tens 3 − 1 = 2 → 721. Step 2:, 240; 240 − 65 = 175. Step 3:, 910; 910 − 3 = 907. Step 4:, 510. Step 5:, 7; hundreds become 4 → 476. Step 6:, 475. Step 7:, 716; 716 − 7 = 709. Step 8:, 590. Final answer: 721; 175; 907; 510; 476; 475; 709; 590. (527 − 3 = 524, 478 − 50 = 428, 690 − 25 = 665 and 500 − 130 = 370 are solved in the book.)

Question: Activity: Subtract the following (in columns). (e) 723 − 6 (f) 510

− 23 (g) 780 − 123 (h) 645 − 60 (i) 412 − 123 (j) 678 − 289 (k) 600 − 78 (l) 962 − 518 (m) 100 − 8 (n) 500 − 62 (o) 666 − 277 (p) 852 − 31 (q) 600 − 60 (r) 223 − 154 (s) 784 − 10 (t) 602 − 214

Answer: Step 1: (e) Ones: 6 cannot be taken from 3, borrow: 13 − 6 = 7; tens become 1 → 717. (f) Ones: 10 − 3 = 7 (borrow); Tens: 2 cannot be taken from 0, borrow: 10 − 2 = 8; Hundreds 4 → 487. (g) Ones 0 − 3: borrow, 10 − 3 = 7; Tens 7 − 2 = 5; Hundreds 7 − 1 = 6 → 657. (h) Tens 4 − 6: borrow, 14 − 6 = 8; Hundreds 5 → 585. (i) Ones: 12 − 3 = 9 (borrow); Tens 0 − 2: borrow, 10 − 2 = 8; Hundreds 3 → 289. (j) Ones: 18 − 9 = 9 (borrow); Tens 6 − 8: borrow, 16 − 8 = 8; Hundreds 5 − 2 = 3 → 389. Step 2: (k) Ones: 10 − 8 = 2 (borrow through); Tens 9 − 7 = 2; Hundreds 5 → 522. (l) Ones 2 − 8: borrow, 12 − 8 = 4; Tens 5 − 1 = 4; Hundreds 9 − 5 = 4 → 444. (m) 100 − 8: 10 − 8 = 2, tens 9 → 92. (n) 500 − 62: ones 10 − 2 = 8; tens 9 − 6 = 3; hundreds 4 → 438. (o) Ones 6 − 7: borrow, 16 − 7 = 9; Tens 5 − 7: borrow, 15 − 7 = 8; Hundreds 5 − 2 = 3 → 389. (p) Ones 2 − 1 = 1; Tens 5 − 3 = 2 → 821. (q) Tens 0 − 6: borrow, 10 − 6 = 4; Hundreds 5 → 540. (r) Ones 3 − 4: borrow, 13 − 4 = 9; Tens 1 − 5: borrow, 11 − 5 = 6; Hundreds 1 − 1 = 0 → 69. (s) Tens 8 − 1 = 7 → 774. (t) Ones 2 − 4: tens are 0, so borrow from hundreds: ones 12 − 4 = 8; tens 9 − 1 = 8; hundreds 5 − 2 = 3 → 388. Final answer: (e) 717 (f) 487 (g) 657 (h) 585 (i) 289 (j) 389 (k) 522 (l) 444 (m) 92 (n) 438 (o) 389 (p) 821 (q) 540 (r) 69 (s) 774 (t) 388. ((a) 845 − 9 = 836, (b) 608 − 15 = 593, (c) 745 − 30 = 715 and (d) 677 − 288 = 389 are solved in the book.)

Question: Exercise 3 (Unit 2), Question 1: Subtract mentally. (i) 529 − 3 (ii) 643 − 30 (iii) 715 − 9 (iv) 800 − 35 (v) 692 − 215 (vi) 750 − 60 (vii) 752 − 121 (viii) 845 − 369 (ix) 509 − 36 (x) 555 − 10

Answer: Step 1: (i) Only ones: 9 − 3 = 6 → 526. (ii) Only tens: 4 − 3 = 1 → 613. (iii) 715 − 9: 715 − 5 = 710, − 4 = 706. (iv) 800 − 35: 800 − 30 =

770, − 5 = 765. (v) 692 − 215: 692 − 200 = 492, − 15 = 477. (vi) Tens 5 − 6 is not possible, borrow: 15 − 6 = 9 → 690. Step 2: (vii) 752 − 121: each place: 2−1, 5−2, 7−1 → 631. (viii) 845 − 369: 845 − 300 = 545, − 69 = 476. (ix) 509 − 36: 509 − 30 = 479, − 6 = 473. (x) 555 − 10 = 545. Final answer: (i) 526 (ii) 613 (iii) 706 (iv) 765 (v) 477 (vi) 690 (vii) 631 (viii) 476 (ix) 473 (x) 545.

Question: Exercise 3 (Unit 2), Question 2: Subtract the following (in columns). (i) 235 − 2 (ii) 506 − 39 (iii) 785 − 196 (iv) 417 − 250 (v) 345 − 60 (vi) 715 − 8 (vii) 405 − 28 (viii) 715 − 426 (ix) 600 − 101 (x) 800 − 175 (xi) 200 − 99 (xii) 754 − 189

Answer: Step 1: (i) Ones 5 − 2 = 3 → 233. (ii) Ones: 16 − 9 = 7 (borrow); Tens: 9 − 3 = 6; Hundreds 4 → 467. (iii) Ones: 15 − 6 = 9 (borrow); Tens: 9 cannot be taken from 7, borrow: 17 − 9 = 8; Hundreds: 6 − 1 = 5 → 589. (iv) Ones 7 − 0 = 7; Tens 5 cannot be taken from 1, borrow: 11 − 5 = 6; Hundreds 3 − 2 = 1 → 167. (v) Tens 6 cannot be taken from 4, borrow: 14 − 6 = 8; Hundreds 2 → 285. (vi) Ones: 15 − 8 = 7 (borrow); tens become 0 → 707. Step 2: (vii) Ones: 15 − 8 = 7 (borrow); Tens: 9 − 2 = 7; Hundreds 3 → 377. (viii) Ones: 15 − 6 = 9 (borrow); Tens: 2 cannot be taken from 0, borrow: 10 − 2 = 8; Hundreds: 6 − 4 = 2 → 289. (ix) Ones 0 − 1: borrow, 10 − 1 = 9; Tens 9 − 0 = 9; Hundreds 5 − 1 = 4 → 499. (x) Ones: 10 − 5 = 5; Tens: 9 − 7 = 2; Hundreds: 7 − 1 = 6 → 625. (xi) Ones: 10 − 9 = 1; Tens: 9 − 9 = 0; Hundreds 1 → 101. (xii) Ones: 14 − 9 = 5 (borrow); Tens: 8 cannot be taken from 4, borrow: 14 − 8 = 6; Hundreds: 6 − 1 = 5 → 565. Final answer: (i) 233 (ii) 467 (iii) 589 (iv) 167 (v) 285 (vi) 707 (vii) 377 (viii) 289 (ix) 499 (x) 625 (xi) 101 (xii) 565.

Question: Examples (real life word problems): (1) Ayesha purchased a book for Rs. 560 and a note-book for Rs. 280. How much did she spend in total? (2) The weights of two packets of spices are 235g and 427g.

Calculate the total weight. (3) There are 567 students in a school. If 229 are girls, how many are boys? (4) A factory produces 562 bikes in a month. If 491 are sold, how many bikes are left?

Answer: Step 1: (1) Add: 560 + 280: tens 6 + 8 = 14, write 4, carry 1; hundreds 5 + 2 + 1 = 8 → 840. Step 2: (2) 235 + 427: ones 5 + 7 = 12, write 2, carry 1; tens 3 + 2 + 1 = 6; hundreds 2 + 4 = 6 → 662 g. Step 3: (3) Subtract: 567 − 229: ones 9 cannot be taken from 7, borrow: 17 − 9 = 8; tens 5 − 2 = 3; hundreds 5 − 2 = 3 → 338 boys. Step 4: (4) 562 − 491: ones 2 − 1 = 1; tens 9 cannot be taken from 6, borrow: 16 − 9 = 7; hundreds 4 − 4 = 0 → 71. Final answer: (1) Rs. 840 (2) 662 g (3) 338 boys (4) 71 bikes left.

Question: Exercise 4 (Unit 2), Question 1: Bisma has two number cards having numbers 278 and 356. What is their sum?

Answer: Step 1: Write both numbers in columns: 278 + 356. Step 2: Ones: 8 + 6 = 14, write 4 and carry 1. Step 3: Tens: 7 + 5 + 1 = 13, write 3 and carry 1. Step 4: Hundreds: 2 + 3 + 1 = 6. Final answer: The sum is 634.

Question: Exercise 4 (Unit 2), Question 2: Abdul Rafay purchased a pen for Rs. 135 and a bag for Rs. 675. How much did he spend?

Answer: Step 1: Add the costs: 135 + 675. Step 2: Ones: 5 + 5 = 10, write 0, carry 1. Step 3: Tens: 3 + 7 + 1 = 11, write 1, carry 1. Step 4: Hundreds: 1 + 6 + 1 = 8. Final answer: He spent Rs. 810 in total.

Question: Exercise 4 (Unit 2), Question 3: 236 people visited the zoo on Saturday and 417 people visited on Sunday. How many people visited the zoo altogether?

Answer: Step 1: Add both days: 236 + 417. Step 2: Ones: 6 + 7 = 13, write 3, carry 1. Step 3: Tens: 3 + 1 + 1 = 5. Step 4: Hundreds: 2 + 4 = 6. Final answer: 653 people visited altogether.

Question: Exercise 4 (Unit 2), Question 4: There were 157 balloons in the

birthday party of Taha. If 129 balloons were burst by the children, how many are left?

Answer: Step 1: Subtract: 157 − 129. Step 2: Ones: 9 cannot be taken from 7, borrow: 17 − 9 = 8. Step 3: Tens: 4 − 2 = 2. Step 4: Hundreds: 1 − 1 = 0. Final answer: 28 balloons are left.

Question: Exercise 4 (Unit 2), Question 5: The sum of two numbers is 786. If one is 345, what is the other number?

Answer: Step 1: To find the other number, subtract the given number from the sum: 786 − 345. Step 2: Ones: 6 − 5 = 1. Step 3: Tens: 8 − 4 = 4. Step 4: Hundreds: 7 − 3 = 4. Final answer: The other number is 441. (Check: 345 + 441 = 786.)

Question: Exercise 4 (Unit 2), Question 6: Bushra and Aafia arranged a party for children. Total expenses were 950 rupees. How much did Bushra spend if Aafia spent 460 rupees?

Answer: Step 1: Subtract: 950 − 460. Step 2: Ones: 0 − 0 = 0. Step 3: Tens: 6 cannot be taken from 5, borrow: 15 − 6 = 9. Step 4: Hundreds: 8 − 4 = 4. Final answer: Bushra spent 490 rupees.

Question: Exercise 4 (Unit 2), Question 7: A shopkeeper sold 135 liters of milk on the first day and 204 liters on the second day. How much milk did he sell in two days?

Answer: Step 1: Add: 135 + 204. Step 2: Ones: 5 + 4 = 9. Step 3: Tens: 3 + 0 = 3. Step 4: Hundreds: 1 + 2 = 3. Final answer: He sold 339 liters of milk in two days.

Question: Exercise 4 (Unit 2), Question 8: Tanveer had 875 rupees. He spent 350 rupees on ice-cream. How much amount is left with him?

Answer: Step 1: Subtract: 875 − 350. Step 2: Ones: 5 − 0 = 5. Step 3: Tens: 7 − 5 = 2. Step 4: Hundreds: 8 − 3 = 5. Final answer: 525 rupees are left with him.

Question: Examples (estimation): (1) Estimate the sum 241 + 36 by rounding to the nearest tens. (2) Estimate the difference 675 − 34 by rounding to the nearest tens. (3) Estimate the sum and difference of 583 and 216 by front-end estimation.

Answer: Step 1: (1) 241 → 240 (round down); 36 → 40 (round up); 240 + 40 = 280. Step 2: (2) 675 → 680 (round up); 34 → 30 (round down); 680 − 30 = 650. Step 3: (3) 583 → 600 (round up); 216 → 200 (round down); estimated sum: 600 + 200 = 800; estimated difference: 600 − 200 = 400. Final answer: (1) 280 (2) 650 (3) sum 800, difference 400. (Estimation means to get close enough to the exact answer.)

Question: Exercise 5 (Unit 2), Question 1: Estimate the following by rounding nearest to tens. (i) 46 + 58 (ii) 462 + 34 (iii) 421 + 567 (iv) 588 + 461 (v) 46 − 34 (vi) 372 − 18 (vii) 884 − 625 (viii) 965 − 486

Answer: Step 1: (i) 46 → 50; 58 → 60; 50 + 60 = 110. (ii) 462 → 460; 34 → 30; 460 + 30 = 490. (iii) 421 → 420; 567 → 570; 420 + 570 = 990. (iv) 588 → 590; 461 → 460; 590 + 460 = 1050. Step 2: (v) 46 → 50; 34 → 30; 50 − 30 = 20. (vi) 372 → 370; 18 → 20; 370 − 20 = 350. (vii) 884 → 880; 625 → 630 (5 rounds up); 880 − 630 = 250. (viii) 965 → 970; 486 → 490; 970 − 490 = 480. Final answer: (i) 110 (ii) 490 (iii) 990 (iv) 1050 (v) 20 (vi) 350 (vii) 250 (viii) 480.

Question: Exercise 5 (Unit 2), Question 2: Estimate the following sum and difference by front-end estimation. (i) 49 + 38 (ii) 32 + 75 (iii) 615 + 456 (iv) 776 + 220 (v) 56 − 31 (vi) 55 − 11 (vii) 708 − 178 (viii) 333 − 266

Answer: Step 1: Round a 2-digit number to the nearest ten and a 3-digit number to the nearest hundred (the highest place value). (i) 49 → 50; 38 → 40; 50 + 40 = 90. (ii) 32 → 30; 75 → 80; 30 + 80 = 110. (iii) 615 → 600; 456 → 500; 600 + 500 = 1100. (iv) 776 → 800; 220 → 200; 800 + 200 = 1000. Step 2: (v) 56 → 60; 31 → 30; 60 − 30 = 30. (vi) 55 → 60; 11 → 10;

60 − 10 = 50. (vii) 708 → 700; 178 → 200; 700 − 200 = 500. (viii) 333 → 300; 266 → 300; 300 − 300 = 0. Final answer: (i) 90 (ii) 110 (iii) 1100 (iv) 1000 (v) 30 (vi) 50 (vii) 500 (viii) 0.

Question: Even and odd numbers: (1) Example: Write odd numbers between 20 and 30. (2) Example: Write even numbers from 50 to 60. (3) Activity: Write even or odd for the following: 20, 99, 64, 71, 82, 19, 24. (56 = Even and 39 = Odd are solved in the book.)

Answer: Step 1: Rule: A number with 0, 2, 4, 6 or 8 at the unit place is even; a number with 1, 3, 5, 7 or 9 at the unit place is odd. Even numbers make pairs with no remainder; odd numbers leave one over. Step 2: (1) Between 20 and 30: 21, 23, 25, 27, 29 are odd. (2) From 50 to 60: 50, 52, 54, 56, 58, 60 are even. Step 3: (3) 20: unit 0 → Even; 99: 9 → Odd; 64: 4 → Even; 71: 1 → Odd; 82: 2 → Even; 19: 9 → Odd; 24: 4 → Even. Final answer: (1) 21, 23, 25, 27, 29 (2) 50, 52, 54, 56, 58, 60 (3) 20 Even, 99 Odd, 64 Even, 71 Odd, 82 Even, 19 Odd, 24 Even.

Question: Exercise 6 (Unit 2), Question 1: Separate even and odd numbers written on the train coaches (5, 17, 16, 29, 32, 23, 82, 8, 91, 18, 70, 61, 57, 41, 11, 87).

Answer: Step 1: Look at the unit digit of each number. Even (unit 0, 2, 4, 6, 8): 16, 32, 82, 8, 18, 70. Step 2: Odd (unit 1, 3, 5, 7, 9): 5, 17, 29, 23, 91, 61, 57, 41, 11, 87. Final answer: Even numbers: 8, 16, 18, 32, 70, 82; Odd numbers: 5, 11, 17, 23, 29, 41, 57, 61, 87, 91.

Question: Exercise 6 (Unit 2), Question 2: Encircle even numbers from the following: 31, 42, 17, 27, 64, 89, 96, 18, 13, 100

Answer: Step 1: Find numbers with 0, 2, 4, 6 or 8 at the unit place. Step 2: 42 (2), 64 (4), 96 (6), 18 (8), 100 (0) are even; the rest (31, 17, 27, 89, 13) are odd. Final answer: Encircle: 42, 64, 96, 18, 100.

Question: Exercise 6 (Unit 2), Question 3: Encircle odd numbers from the

following: 16, 91, 83, 78, 22, 41, 58, 69, 76

Answer: Step 1: Find numbers with 1, 3, 5, 7 or 9 at the unit place. Step 2: 91 (1), 83 (3), 41 (1), 69 (9) are odd; the rest (16, 78, 22, 58, 76) are even. Final answer: Encircle: 91, 83, 41, 69.

Question: Exercise 6 (Unit 2), Question 4: Write odd numbers between 30 and 40. Question 5: Write even numbers between 51 and 61.

Answer: Step 1: After 30, every second number is odd: 31, 33, 35, 37, 39. Step 2: Even numbers between 51 and 61: 52, 54, 56, 58, 60. Final answer: Q4: 31, 33, 35, 37, 39; Q5: 52, 54, 56, 58, 60.

Activities (counting in 3s, 4s, 5s, 10s and 100s): (1) Count in 3s and write the missing numbers: 15, 18, _____, 24, _____, _____, 33; and 9, 12, _____, _____, 21, 24, _____ (2) Count in 4s and write the missing numbers: 16, 20, _____, 28, _____, _____; and 28, _____, 36, _____, 44, _____ (3) Count in 5s and write the missing numbers: 40, 45, _____, _____, 60, _____, 70; and 105, _____, 115, 120, _____, 130, _____ (4) Count in 10s and write the missing numbers: 50, 60, _____, _____, 90, _____, _____ / 110, _____, _____, 140, _____, _____, 170

Answer: 21,, 27,, 30,, 15,, 18,, 27. Missing: 21, 27, 30; 15, 18, 27. Step 2:, 24,, 32,, 36;, 32,, 40,, 48. Missing: 24, 32, 36; 32, 40, 48. Step 3:, 50,, 55,, 65,, 110,, 125,, 135. Missing: 50, 55, 65; 110, 125, 135. Step 4:, in, 100s:, 100,, 200,, ...,, 600, and, 105, 205, ..., 605.)

Activity and Exercise 7 (Unit 2): Counting in 100s and missing numbers. (a) Activity: 500, _____, _____, _____, 900; and 200, _____, _____, 400, _____ (b) Exercise 7 Q1: in 3s: 15, _____, 21, _____, _____, 30; and 30, _____, 36, _____, 42, _____ Q2: in 4s: 8, _____, 16, _____, 24, _____; and 40, _____, 48, _____, _____, 60 Q3: in 5s: 15, _____, 25,

_____, 35, _____; and 110, _____, 120, _____, 130, _____ Q4: in 10s: 40, _____, 60, _____, 80, _____; and 220, _____, 240, _____, _____, 270 Q5: in 100s: 300, _____, 500, _____, _____, 800; and 220, _____, 420, _____, 620, _____

Answer: 600,, 700,, 800,, 300, 400 sequence — the book prints 400 in, the fourth box; the completed row is 200, 300,, 500, 600. Missing: 600, 700, 800; 300, 500, 600. Step 2:, 18,, 24,, 27,, 33,, 39,, 45. Missing: 18, 24, 27; 33, 39, 45., 12,, 20,, 28;, 44,, 52, 56, 60. Missing: 12, 20,, 28; 44, 52, 56. Step 3:, 20,, 30,, 40;, 115,, 125,, 135. Missing: 20, 30, 40; 115, 125, 135., 50,, 70,, 90;, 230,, 250, 260, 270. Missing: 50, 70,, 90; 230, 250, 260. Step 4:, 400,, 600,, 700,, 320,, 520,, 720. Missing: 400, 600, 700; 320, 520, 720. Final answer: The completed sequences are given above.

Question: Multiplication tables (given in the book): Write the tables of 2, 3, 4, 5 and 10. (Activities: groups of 3 pencils and groups of 4 apples are counted: 1 three is 3 … 10 threes are 30; 1 four is 4 … 10 fours are 40.)

Answer: Step 1: Table of 2: 1 × 2 = 2, 2 × 2 = 4, 3 × 2 = 6, 4 × 2 = 8, 5 × 2 = 10, 6 × 2 = 12, 7 × 2 = 14, 8 × 2 = 16, 9 × 2 = 18, 10 × 2 = 20. Step 2: Table of 3: 1 × 3 = 3, 2 × 3 = 6, 3 × 3 = 9, 4 × 3 = 12, 5 × 3 = 15, 6 × 3 = 18, 7 × 3 = 21, 8 × 3 = 24, 9 × 3 = 27, 10 × 3 = 30. Step 3: Table of 4: 1 × 4 = 4, 2 × 4 = 8, 3 × 4 = 12, 4 × 4 = 16, 5 × 4 = 20, 6 × 4 = 24, 7 × 4 = 28, 8 × 4 = 32, 9 × 4 = 36, 10 × 4 = 40. Step 4: Table of 5: 1 × 5 = 5, 2 × 5 = 10, 3 × 5 = 15, 4 × 5 = 20, 5 × 5 = 25, 6 × 5 = 30, 7 × 5 = 35, 8 × 5 = 40, 9 × 5 = 45, 10 × 5 = 50. Step 5: Table of 10: 1 × 10 = 10, 2 × 10 = 20, 3 × 10 = 30, 4 × 10 = 40, 5 × 10 = 50, 6 × 10 = 60, 7 × 10 = 70, 8 × 10 = 80, 9 × 10 = 90, 10 × 10 = 100. Final answer: All five tables are complete above. Each table is repeated addition: for example 4 × 3 = 3 + 3 + 3 + 3 = 12.

Exercise 8 (Unit 2): Q1: Match the threes: 3 threes, 5 threes, 6 threes, 9 threes, 10 threes with the answers 15, 30, 9, 27, 18. Q2: Match the fours: 3 fours, 5 fours, 7 fours, 8 fours, 9 fours with the answers 32, 36, 12, 20, 28. Q3: Complete: 6 fours = _____; 5 threes = _____; 9 threes = _____; 5 fours = _____; 8 fours = _____; 2 threes = _____ (3 fours = 12 is solved in the book.)

Answer: 24;, 15;, 27;, 20;, 32;, 6. Final answer: Q1: 3 threes ↔ 9, 5 threes ↔ 15, 6 threes ↔ 18, 9 threes ↔ 27, 10 threes ↔ 30; Q2: 3

Question: Multiplication as repeated addition: (1) Example: There are 3 cars and each car has 4 wheels — how many wheels are there in total? (2) Activity: Count the objects and write: (a) 4 groups of 2 (oranges) (b) 5 groups of 3 (pencils) (c) 5 groups of 4 (apples). (3) Activity: Match the repeated additions with the multiplications: 5 + 5 + 5 + 5; 2 + 2 + 2 + 2 + 2; 3 + 3 + 3 + 3 + 3 + 3; 4 + 4 + 4 + 4 + 4 + 4 + 4 + 4; 6 + 6 + 6 + 6 + 6 with 5 × 6, 8 × 4, 4 × 5, 5 × 2, 6 × 3.

Answer: Step 1: (1) Add 4 three times: 4 + 4 + 4 = 12; as multiplication: 3 × 4 = 12. "×" is the sign of multiplication. Step 2: (2) (a) 2 + 2 + 2 + 2 = 8, or 4 times 2 = 8, or 4 × 2 = 8. (b) 3 + 3 + 3 + 3 + 3 = 15, or 5 × 3 = 15. (c) 4 + 4 + 4 + 4 + 4 = 20, or 5 × 4 = 20. (These are solved in the book.) Step 3: (3) Number of groups × the number in each group: 5 + 5 + 5 + 5 = four groups → 4 × 5 (matched in the book); 2 five times → 5 × 2; 3 six times → 6 × 3; 4 eight times → 8 × 4; 6 five times → 5 × 6. Final answer: (1) 12 wheels (2) 8; 15; 20 (3) 4 × 5; 5 × 2; 6 × 3; 8 × 4; 5 × 6.

Exercise 9 (Unit 2), Question 1: Complete the following (repeated addition, "times" and multiplication forms). Row 1: 6 + 6 + 6 + 6 | 4 times 6 | 4 × 6. Row 2: 2 + 2 + 2 + 2 + 2 | 5 times 2 | _____. Row 3: 3 + 3 + 3 + 3 + 3 + 3 | _____ | 6 × 3. Row 4: _____ | 3 times 7 | 3 × 7. Row 5: 5 + 5 + 5 + 5 | _____ | _____.

Answer: = 10;, 6 times 3,, =, 20.

Exercise 9 (Unit 2), Question 2: Fill in the boxes. 3 × 2 = _____; 6 × 4 = _____; 7 × _____ = 70; 8 × _____ = 32; _____ × 3 = 27; _____ × 5 = 30; 6 × 3 = _____; 7 × 3 = _____; 9 × 4 = _____; 3 × 5 = _____; 6 × 10 = _____; 7 × _____ = 14; _____ × 4 = 40 (2 × 5 = 10 is solved in the book.) Also the Activity: Write the multiplication statements: (a) 6 eggs in 2 rows of 3 (2 × 3) (b) 12 faces in 3 columns × 4 rows (3 × 4).

Answer: 6;, 24;, 18;, 21;, 36;, 15;, 60. Step 2: Missing numbers:, 10, 70 → 10; 8 × 4 = 32 → 4; 9 × 3 = 27 → 9; 6 × 5 = 30 → 6; 7 × 2 = 14 → 2; 10

Question: Activity and Exercise 10 (Unit 2): Match the figures with the multiplication statements and count and write statements. Figure 1: stars, 4 rows × 5 (total 20). Figure 2: smiley faces, 5 rows × 3 (total 15). Figure 3: watermelon slices, 3 rows × 4 (total 12). Statements: 3 × 5, 4 × 3, 5 × 4. Exercise 10: (1) Strawberries: 3 rows × 4 (given in the book) (2) Tents: 2 rows × 6 (3) Cars: 4 rows × 5 (4) Spinning tops: 5 rows × 4.

Answer: Step 1: Count the rows and the objects in each row. Figure 1: 4 groups of 5 = 20 → 5 × 4. Figure 2: 5 groups of 3 = 15 → 3 × 5 (matched by an arrow in the book). Figure 3: 3 groups of 4 = 12 → 4 × 3. Step 2: Exercise 10 (1): Four times three is 12: 4 × 3 = 12 (given in the book). (2) Tents: two rows of 6: six times two is 12: 6 × 2 = 12. (3) Cars: four rows of 5: five times four is 20: 5 × 4 = 20. (4) Spinning tops: five rows of 4: four times five is 20: 4 × 5 = 20. Final answer: Matching: 3 × 5 ↔ Figure 2, 4 × 3 ↔ Figure 3, 5 × 4 ↔ Figure 1; Exercise 10: (1) 4 × 3 = 12 (2) 6 × 2 = 12 (3) 5 × 4 = 20 (4) 4 × 5 = 20.

Order in multiplication and Exercise 11 (Unit 2): (a) Activity: Count 15 apples in two ways (3 × 5 and 5 × 3) and compare 2 × 3 with 3 × 2. (b) Exercise 11 Q1: Complete: (i) 5 × 3 = _____ × 5 (ii) 10 × 4 = 4 ×

_____ (iii) 3 × 6 = _____ × 3 (iv) _____ × 4 = 4 × 5 (v) 7 × _____ = 2 × 7. Q2: Match: 3 × 4, 2 × 10, 4 × 6, 5 × 7, 10 × 4 with 6 × 4, 4 × 10, 7 × 5, 4 × 3, 10 × 2.

Answer: 15. So 3, 10. Final answer: (a), 5, 2; matching as above.

Question: Multiplication using tables, Activity and Exercise 12 (Unit 2): Examples: Multiply 5 by 2; multiply 6 by 3. Activity: Solve 8 × 5; 9 × 2; 7 × 3 (5 × 4 = 20 is solved in the book). Exercise 12 Q1: Multiply: (i) 6 by 3 (ii) 7 by 2 (iii) 4 by 5 (iv) 5 by 5 (v) 9 by 2 (vi) 8 by 4. Q2: Fill in the blanks: (i) 5 × 3 (ii) 6 × 4 (iii) 9 × 5 (iv) 7 × 2 (v) 3 × 4 (vi) 9 × 3. Q3: Solve in columns: 4 × 2; 5 × 3; 9 × 4; 8 × 5; 7 × 4; 8 × 2; 4 × 3; 9 × 2; 5 × 5; 7 × 5; 3 × 3; 4 × 4.

Answer: Step 1: Examples: Read the table of 2 up to 5: 5 × 2 = 10. Read the table of 3 up to 6: 6 × 3 = 18. Activity: 8 × 5: table of 5 up to 8 = 40; 9 × 2 = 18; 7 × 3 = 21. Step 2: Exercise 12 Q1: (i) 6 × 3 = 18 (ii) 7 × 2 = 14 (iii) 4 × 5 = 20 (iv) 5 × 5 = 25 (v) 9 × 2 = 18 (vi) 8 × 4 = 32. Q2: (i) 15 (ii) 24 (iii) 45 (iv) 14 (v) 12 (vi) 27. Step 3: Q3: 4 × 2 = 8; 5 × 3 = 15; 9 × 4 = 36; 8 × 5 = 40; 7 × 4 = 28; 8 × 2 = 16; 4 × 3 = 12; 9 × 2 = 18; 5 × 5 = 25; 7 × 5 = 35; 3 × 3 = 9; 4 × 4 = 16. Final answer: Examples: 10; 18; Activity: 40; 18; 21; Q1: 18; 14; 20; 25; 18; 32; Q2: 15; 24; 45; 14; 12; 27; Q3: 8; 15; 36; 40; 28; 16; 12; 18; 25; 35; 9; 16.

Question: Multiplication grid and multiplying by 0 and 1 — Examples and Exercise 13 (Unit 2): Examples: (a) Using the grid: multiply 25 with 3 and 34 by 2. (b) Multiply 6 by 0 and 5 by 1 (banana example: 3 children, each has 0 bananas → 3 × 0; then each gets 1 banana → 3 × 1). Exercise 13 Q1: Multiply using the multiplication grid: (i) 17 by 2 (ii) 32 by 3 (iii) 26 by 4 (iv) 12 by 5 (v) 43 by 2 (vi) 24 by 3 (vii) 27 by 2 (viii) 35 by 2. Q2: Multiply: (i) 6 by 0 (ii) 7 by 1 (iii) 8 by 0 (iv) 4 by 1 (v) 5 by 0 (vi) 8 by 1. Q3: Fill in the blanks: (i) 5 × 0 (ii) 7 × 0 (iii) 0 × 4 (iv) 0 × 2 (v) 2 × 1 (vi) 9

× 1 (vii) 1 × 5 (viii) 1 × 3.

Answer: Step 1: Grid method: Partition the 2-digit number into tens and ones, multiply each part, then add. Examples: 25 = 20 + 5; 3 × 20 = 60; 3 × 5 = 15; 60 + 15 = 75. 34 = 30 + 4; 2 × 30 = 60; 2 × 4 = 8; 60 + 8 = 68. Step 2: Q1: (i) 17 × 2 = 20 + 14 = 34 (ii) 32 × 3 = 90 + 6 = 96 (iii) 26 × 4 = 80 + 24 = 104 (iv) 12 × 5 = 50 + 10 = 60 (v) 43 × 2 = 80 + 6 = 86 (vi) 24 × 3 = 60 + 12 = 72 (vii) 27 × 2 = 40 + 14 = 54 (viii) 35 × 2 = 60 + 10 = 70. Step 3: Rules for 0 and 1: Any number multiplied by 0 is zero; any number multiplied by 1 is that number itself. Examples: 3 × 0 = 0 (0 + 0 + 0); 3 × 1 = 3 (1 + 1 + 1); 6 × 0 = 0; 5 × 1 = 5. Q2: (i) 0 (ii) 7 (iii) 0 (iv) 4 (v) 0 (vi) 8. Q3: (i) 0 (ii) 0 (iii) 0 (iv) 0 (v) 2 (vi) 9 (vii) 5 (viii) 3. Final answer: Grid examples: 75; 68; Q1: 34; 96; 104; 60; 86; 72; 54; 70; Q2: 0; 7; 0; 4; 0; 8; Q3: 0; 0; 0; 0; 2; 9; 5; 3.

Question: Multiplication word problems: Examples: (1) The price of one pencil is Rs. 10 — find the price of 6 such pencils. (2) There are 4 wheels in a van — how many wheels are there in 7 vans? Exercise 14 (Unit 2): 1. The price of a chocolate is Rs. 5 — what is the price of 9 such chocolates? 2. There are 4 fans in a room — how many fans are there in 8 such rooms? 3. The price of an egg is Rs. 22 — what is the price of 6 eggs? 4. Abdul Rafay spends Rs. 35 in a day — how much does he spend in 7 days? 5. There are 4 persons in a car — how many persons are in 5 cars? 6. A shirt has 8 buttons — how many buttons are there in 5 shirts? 7. A dog has 4 legs — how many legs do 7 dogs have? 8. The price of a sharpener is Rs. 8 — what is the price of 9 such sharpeners?

Answer: Step 1: Example (1): 6 × 10 = 60 rupees. Example (2): 7 × 4 = 28 wheels. Step 2: Exercise 14: 1. 9 × 5 = 45 rupees. 2. 8 × 4 = 32 fans. 3. 6 × 22: grid: 6 × 20 = 120; 6 × 2 = 12; 120 + 12 = 132 rupees. 4. 7 × 35: 7 × 30 = 210; 7 × 5 = 35; 210 + 35 = 245 rupees. Step 3: 5. 5 × 4 = 20

persons. 6. 5 × 8 = 40 buttons. 7. 7 × 4 = 28 legs. 8. 9 × 8 = 72 rupees. Final answer: Examples: Rs. 60; 28 wheels; Exercise 14: 1. Rs. 45 2. 32 fans 3. Rs. 132 4. Rs. 245 5. 20 persons 6. 40 buttons 7. 28 legs 8. Rs. 72. Division: Activity: Distribute 12 mangoes equally among 3 children (Bushra/Sonia, Tanveer and Ayesha). Example: Divide 10 eggs equally in 2 baskets by repeated subtraction. Exercise 15 Q1: Divide 12 balls equally in 3 ball stands by repeated subtraction: 12 − 3 = _____; 9 − 3 = _____; 6 − _____ = 3; 3 − _____ = 0; so 12 ÷ 3 = _____. Activity: Write division statements: (a) 8 objects in 2 groups of 4 (b) 6 cups in 2 groups of 3.

Answer: 9 (given);, 6 (given);, 3, 3, 4; each stand has 4 balls. Step 5: Activity

Division statements and order — Exercise 16 and 17 and Activity (Unit 2): Exercise 16: Write division statements for the groupings: 1. 18 bottles in 3 groups of 6 (given in the book). 2. 16 flowers in 4 groups of 4: 16 divided by 4 is _____; _____ ÷ _____ = _____. 3. 9 ice creams in 3 groups of 3: 9 divided by 3 is _____; _____ ÷ _____ = _____. Activity: Divide 6 balls in two baskets; then see if 2 balls can be divided in six baskets. Exercise 17: Answer Yes or No: 1. Can we divide 8 books equally among 12 students? 2. Can we divide 5 balls equally among 5 students? 3. Can we divide 8 pens equally among 16 students? 4. Can we divide 15 pencils equally among 5 students? 5. Can we divide 12 chocolates equally among 20 students?

Answer: not, possible;

Question: Division without remainder — Examples and Exercise 18 (Unit 2): Examples: Divide: 8 ÷ 2; 9 ÷ 3; 15 ÷ 5; 20 ÷ 4; and in long form: 6 ÷ 3; 4 ÷ 2; 21 ÷ 3; 24 ÷ 4. Exercise 18 Q1: Divide and give the reason: (i) 6 ÷ 3 (in the book: 2, because 2 × 3 = 6) (ii) 8 ÷ 4 (iii) 24 ÷ 6 (iv) 28 ÷ 4 (v)

36 ÷ 6. Q2: Divide in long form: 1. 8 ÷ 2 2. 8 ÷ 2 (printed the same twice in the book) 3. 24 ÷ 4 4. 20 ÷ 10.

Answer: Step 1: Method: Read the table of the divisor until the dividend comes; that count is the quotient; the remainder is 0. Division is the reverse process of multiplication. Example: 6 ÷ 2 = 3 because 2 × 3 = 6 (2 = divisor, 6 = dividend, 3 = quotient, 0 = remainder). Step 2: Examples: 8 ÷ 2 = 4 because 2 × 4 = 8; 9 ÷ 3 = 3 because 3 × 3 = 9; 15 ÷ 5 = 3 because 5 × 3 = 15; 20 ÷ 4 = 5 because 4 × 5 = 20. Long form: 6 ÷ 3 = 2 (3 × 2 = 6, 6 − 6 = 0); 4 ÷ 2 = 2; 21 ÷ 3 = 7; 24 ÷ 4 = 6. Step 3: Exercise 18 Q1: (ii) 8 ÷ 4 = 2 because 4 × 2 = 8 (iii) 24 ÷ 6 = 4 because 6 × 4 = 24 (iv) 28 ÷ 4 = 7 because 4 × 7 = 28 (v) 36 ÷ 6 = 6 because 6 × 6 = 36. Q2: 1. 8 ÷ 2 = 4, remainder 0 2. 8 ÷ 2 = 4, remainder 0 3. 24 ÷ 4 = 6, remainder 0 4. 20 ÷ 10 = 2, remainder 0. Final answer: Examples: 4; 3; 3; 5; 2; 2; 7; 6; Q1: (ii) 2 (iii) 4 (iv) 7 (v) 6; Q2: 4; 4; 6; 2 (remainder 0 in each).

Question: Mixed word problems: Example 1: A student gets 15 marks in each of 5 subjects — what are his total marks? Example 2: 50 chocolates are distributed equally among 10 students — how many chocolates does each child get? Exercise 19 (Unit 2): 1. 18 apples are equally distributed to 3 students — how many apples does each student get? 2. There are 12 trees; on each tree 4 birds are sitting — how many birds are sitting on all the trees? 3. 16 books are distributed equally among 4 children — how many books does each child get? 4. The price of a pen is 25 rupees — what will be the price of 12 such pens? 5. Shazia gets 20 rupees as her pocket money daily — how much will she get after 15 days?

Answer: Step 1: Example 1: Multiply: 15 × 5; grid: 5 × 10 = 50; 5 × 5 = 25; 50 + 25 = 75 marks. Example 2: Divide: 50 ÷ 10 = 5 chocolates for each child (because 10 × 5 = 50). Step 2: Exercise 19: 1. 18 ÷ 3 = 6

apples (3 × 6 = 18). 2. 12 × 4 = 48 birds. 3. 16 ÷ 4 = 4 books (4 × 4 = 16). Step 3: 4. 25 × 12; grid: 25 = 20 + 5; 12 × 20 = 240; 12 × 5 = 60; 240 + 60 = 300 rupees. 5. 20 × 15; 20 × 10 = 200; 20 × 5 = 100; 200 + 100 = 300 rupees. Final answer: Example 1: 75 marks; Example 2: 5 chocolates; Exercise 19: 1. 6 apples 2. 48 birds 3. 4 books 4. Rs. 300 5. Rs. 300.

Unit 3: Money

Currency and exchanging money: (1) What is the currency of Pakistan? Ten notes of ten rupees and five twenty-rupee notes are equal to how many rupees? (2) Complete the exchange table: 20 Rupees = _____; 100 Rupees = _____; 8 Rupees = _____. (3) The dollar ($) is the currency of which country? Yuan is China's currency and Riyal is Saudi Arabia's — Example: Mahi wants change for $100 — what notes can she get? (4) Table: 20 Dollars = _____; 20 Yuan = _____; 20 Riyal = _____.

Answer: two 10-rupee notes;, ten 10-rupee notes;, a 5-rupee coin + a 2-rupee coin + a 1-rupee coin (5 + 2 + 1 = 8). Step 3:, four five-dollar notes (4 × 5 = 20);, four five-Yuan notes;, four five-Riyal notes. Final answer: (1) Rupee; 100 rupees; 100 rupees (2) two 10s; ten 10s; 5 + 2 + 1 coins (3) United States; change: 5 × 20 or 2 × 50 or 10 × 10 (4) four five-notes in each.

Exercise 1 (Unit 3): 1. A chair costs 50 dollars — how many 5-dollar notes will you pay for the chair? 2. The cost of a pen is 30 rupees — how many 10-rupee notes are needed to pay the cost? 3. A toy costs 10 dollars — tick on the dollar notes shown (a 1-dollar note and a 20-dollar note) that you will pay. 4. Fill in the blanks: (i) 10 rupees = _____ five-rupee coins (ii) 100 dollars = _____ 20 dollars (iii) 10 Yuan = _____ five-yuan notes (iv) 20 Riyal = two _____ Riyals.

Answer: two, five, two, 10.

Question: Addition and subtraction of money: Example (book): a 50-rupee note + a 20-rupee note = ? Example 1: Rohan received two 20-dollar notes from his father and one 10-dollar note from his mother — find the

total amount he received. Example 2: Amina has 10 Riyal, a 20-Riyal and a 5 Riyal — what is the total amount? Subtraction example: Amna had 50 Riyal; she bought a pencil that cost 2 Riyal — how much did the shopkeeper return? Activity: Amina has two notes of Rs. 50, and Shamsa has a note of Rs. 100 — who has got less money?

Answer: Step 1: Book example: To find the total, add the amounts written on the notes: 50 + 20 = Rs. 70. Step 2: Example 1: Rohan: 20 + 20 + 10; first 20 + 20 = 40; then 40 + 10 = 50 dollars. (Note: the book's printed line shows only 20 $ + 10 $ = 30 $, but the question says two 20-dollar notes, so the correct total is 50 dollars.) Step 3: Example 2: 10 + 20 + 5; 10 + 20 = 30; 30 + 5 = 35 Riyal. Step 4: Subtraction: subtract the amount spent from the total: 50 − 2 = 48 Riyal returned by the shopkeeper. Step 5: Activity: Amina: 50 + 50 = 100 rupees; Shamsa: 100 rupees. Both have equal money — neither has less. Final answer: Rs. 70; Example 1: 50 dollars; Example 2: 35 Riyal; 48 Riyal returned; Activity: both equal (Rs. 100 each).

Exercise 2 (Unit 3): Q1: Count the notes shown and write the total amount: notes of Rs. 20, Rs. 10, Rs. 100, Rs. 50, Rs. 500 and Rs. 1000 (one of each). Q2: Fill in the blanks: (i) Rs. 50 + Rs. 20 = _____ (ii) 100 yuan − 80 yuan = _____ (iii) 50 Riyal = _____ 10 Riyals. Q3: Musa has $40; he bought a toy for $10 — what is the amount of money he has left? Q4: Moona bought four pens, each costing 5 riyals — how much money did she spend? Q5: A father gives 400 yuan to his children every month; he has four sons; if the money is divided equally, how much does each son get? Q6: Neha wants to buy a toy costing 200 riyal; she had saved 150 riyal — how much more money does she need to save to buy the toy? Q7: Raj buys a new purse for $25; he gives the cashier a $20 note and a $10 note — how much change does he get back?

Answer: yuan (iii) five; 3. $30 4. 20 riyals 5., 6.

Unit 4: Fractions

Question: What is a fraction? Example: A circle is divided into 6 equal parts and 5 parts are shaded — write the fraction and tell its numerator and denominator. Unit fractions: What fractions are made when an apple is divided into 2 equal parts and into 3 equal parts? Pizzas show 1/4, 1/5 and 1/6. Activity: A circle is divided into 10 equal parts, each labelled 1/10 — how many 1/10 are in the figure? Non-unit fractions: 2/3 (2 pink parts), and pizzas showing 2/3, 3/4, 2/5.

Answer: Step 1: A fraction is a part of a whole, like 1/4. The number at the top is the Numerator (the parts we have) and the number at the bottom is the Denominator (the total parts in the whole). Step 2: Example: Shaded parts 5, total parts 6, so the fraction is 5/6; numerator = 5, denominator = 6. Step 3: A unit fraction has numerator 1: an apple in 2 equal parts → each part is 1/2; in 3 equal parts → each part is 1/3; likewise 1/4, 1/5, 1/6 are unit fractions. Step 4: Activity: Count the parts: 10 parts, each 1/10 — so there are ten 1/10 in the figure, and together they make 1 whole. Step 5: A non-unit fraction has a numerator greater than 1: 2/3, 3/4, 2/5 are non-unit; in 2/3, two of the 3 parts are pink. Final answer: 5/6 (numerator 5, denominator 6); unit: 1/2, 1/3, 1/4, 1/5, 1/6; Activity: ten 1/10; non-unit: 2/3, 3/4, 2/5.

Question: Exercise 1 (Unit 4): (i) A circle is in 4 equal parts: 3 parts pink, 1 part blue — which colour represents a unit fraction? (ii) A square is in 4 equal parts: 1 part blue, 3 parts yellow — which colour represents a non-unit fraction? (iii) A circle is in 8 equal parts: 7 parts blue, 1 part pink — what fraction of the circle is blue? (iv) Which fraction represents a unit fraction: 1/3, 2/5 or 5/8? (v) Circle the non-unit fractions: 1/4, 2/3, 3/5, 1/8,

1/6, 1/7.

Answer: Step 1: A unit fraction has numerator 1. (i) Blue is only 1 part: 1/4 — so the blue colour represents the unit fraction (pink 3/4 is non-unit). Step 2: (ii) Yellow is 3 parts: 3/4, whose numerator 3 is greater than 1 — so the yellow colour represents the non-unit fraction. Step 3: (iii) Count the coloured (blue) parts: 7; total parts: 8; so the blue fraction is 7/8. Step 4: (iv) The one with numerator 1 is the unit fraction: 1/3. Step 5: (v) The ones with numerators greater than 1: 2/3 and 3/5 — circle these; the rest (1/4, 1/8, 1/6, 1/7) are unit fractions. Final answer: (i) Blue (1/4) (ii) Yellow (3/4) (iii) 7/8 (iv) 1/3 (v) 2/3 and 3/5.

Question: Like fractions and Unlike fractions: (1) What are like fractions? Shown on circles: 1/8, 2/8, 3/8, 4/8, 7/8, 8/8 and on strips: 2/5, 3/5, 1/5. More examples: (a) 2/9, 3/9, 5/9, 9/9 (b) 1/7, 2/7, 4/7, 5/7, 7/7. Activity: What is the denominator in 1/5, 2/5, 3/5, 4/5? (2) What are unlike fractions? Example: 3/8, 2/3, 3/5, 2/7; more: (a) 1/2, 1/4, 2/3, 5/6 (b) 1/9, 2/7, 3/4, 2/5. Activity: Figure A (1/4, 2/4, 3/4 — same denominators) and Figure B (1/2, 2/3, 2/4 — different denominators) — which figure is an example of unlike fractions?

Answer: Step 1: (1) When two or more fractions have the same denominator, they are like fractions. All the circle fractions have denominator 8; the strips have 5; (a) has 9 and (b) has 7 — so all are like. Activity: The denominator in all four is 5. Step 2: (2) Fractions with different denominators are unlike fractions: in the example 8, 3, 5, 7 are all different; in (a) and (b) also the denominators are all different. Step 3: Activity: Figure A has the same denominators (4) — it shows like fractions; Figure B has different denominators (2, 3, 4) — so Figure B is the example of unlike fractions (as also shown in the book). Final answer: (1) Like = same denominator; Activity: 5 (2) Unlike = different denominators (3)

Figure B.

Question: Exercise 2 (Unit 4): Q1: Write whether the following sets of fractions are like or unlike: (i) 4/7, 3/7 (ii) 5/6, 13/9 (iii) 8/5, 1/2 (iv) 4/3, 5/3 (v) 1/4, 5/4 (vi) 1/12, 6/11 (vii) 9/5, 7/6 (viii) 13/13, 13/13. Q2: Circle the pairs of like fractions: (i) 8/11, 5/11 (ii) 7/4, 2/3 (iii) 1/6, 7/6 (iv) 9/7, 9/8. Q3: Circle the pairs of unlike fractions: (i) 6/5, 8/7 (ii) 1/9, 3/2 (iii) 10/9, 2/9 (iv) 3/14, 7/15.

Answer: Step 1: Rule: Look at the denominators — same means like, different means unlike. Q1: (i) 7 = 7 → like (ii) 6 ≠ 9 → unlike (iii) 5 ≠ 2 → unlike (iv) 3 = 3 → like (v) 4 = 4 → like (vi) 12 ≠ 11 → unlike (vii) 5 ≠ 6 → unlike (viii) 13 = 13 → like. Step 2: Q2: Circle only the pairs with the same denominators: (i) 11 = 11 → circle (iii) 6 = 6 → circle; (ii) and (iv) are unlike. Step 3: Q3: Circle the pairs with different denominators: (i) 5 ≠ 7 → circle (ii) 9 ≠ 2 → circle (iv) 14 ≠ 15 → circle; (iii) 9 = 9 is like, do not circle. Final answer: Q1: (i) like (ii) unlike (iii) unlike (iv) like (v) like (vi) unlike (vii) unlike (viii) like; Q2: (i) and (iii); Q3: (i), (ii) and (iv).

Question: Comparing fractions and Exercise 3 (Unit 4): State the rules. Example: Ali ate 6 parts of a pizza divided into 8 parts, and his friend ate 7 parts — who ate more? Example: Compare 5/9 and 4/9. Exercise 3: 1. Choose any two of the unit fractions shown (1/2, 1/3, 1/4, 1/5, 1/6, 1/7, 1/8, 1/9, 1/10) and compare which is smaller and which is greater. 2. Which one is greater: 1/10 or 3/10? 3. Tick the smaller fraction: 1/4 (strip of 4 parts) or 1/5 (strip of 5 parts). 4. Reema ate 3/5 of the pizza, and Anum ate 4/5 — who ate more pizza? 5. Raj lives 1/4 km from the school, and Pirya lives 2/4 km from the school — who lives closest to the school? 6. Danial's garden has 2/10 of flowers and 5/10 of vegetables — are there more flowers or vegetables in his garden?

Answer: Step 1: Rule 1: In like fractions (same denominator), the fraction

with the greater numerator is greater. Rule 2: In unit fractions (same numerator 1), the fraction with the greater denominator is smaller: 1/2 > 1/3 > 1/4 > 1/5. (Note: the book's "Remember" box prints the ascending/descending definitions reversed; correctly: ascending = smaller to greater, descending = greater to smaller.) Step 2: Examples: Both denominators are 8; 7 > 6, so the friend ate more: 7/8 > 6/8. And 5/9 > 4/9 (5 > 4). Step 3: Exercise 3: 1. For example 1/2 and 1/10: the numerators are the same (1); the denominator 10 is greater, so 1/10 is smaller and 1/2 is greater; the full order is: 1/2 > 1/3 > 1/4 > 1/5 > 1/6 > 1/7 > 1/8 > 1/9 > 1/10. 2. Same denominator 10: 3 > 1, so 3/10 is greater (3 parts are shaded in the picture). 3. 1/5 is smaller — tick it (denominator 5 is greater). Step 4: 4. 4 > 3, so Anum ate more pizza. 5. 1/4 < 2/4, so Raj lives closest to the school. 6. 5 > 2, so there are more vegetables (5/10 > 2/10). Final answer: Examples: the friend (7/8); 5/9 > 4/9; 1. 1/2 greatest, 1/10 smallest; 2. 3/10; 3. 1/5; 4. Anum; 5. Raj; 6. Vegetables.

Question: Addition and subtraction of like fractions and Exercise 4 and 5 (Unit 4): Rules and examples: Anisa ate 1/4 of a loaf of bread and her friend ate 2/4 — how much did they eat together? Example: 3/8 + 2/8. Activity: 4/9 + 3/9. Example 5: Alisha got 3/4 of a pizza and ate 2/4 of it — how much pizza is left? Exercise 4 Q1: Add: 3/5 + 1/5 and 1/5 + 3/5; 5/7 + 1/7 and 1/7 + 5/7; 2/8 + 3/8 and 3/8 + 2/8; 3/10 + 4/10 and 4/10 + 3/10. Subtraction Activity: (1) 3/6 − 2/6 (2) 5/10 − 2/10 (3) 4/5 − 1/5 (4) 6/8 − 4/8 (5) 8/12 − 5/12 (6) 2/4 − 1/4. Exercise 5: 1. Ali had 5/8 of pizza; he ate 2/8 (the book's text prints 2/5 but its working uses 2/8) — what fraction is left (solved in the book)? 2. Amina had 4/6 of the pizza; she ate 1/6 — what fraction is left? 3. Sapna had 4/5 of the pizza; she ate 3/5 of it — what fraction is left? 4. Solve: (i) 5/8 − 3/8 (ii) 2/3 − 1/3 (iii) 6/6 − 3/6 (iv) 1/2 − 1/2 (v) 2/10 − 1/10 (vi) 5/7 − 4/7.

Answer: Step 1: Addition rule: For like fractions, add only the numerators and keep the denominator the same: Anisa: 1/4 + 2/4 = (1 + 2)/4 = 3/4. Example: 3/8 + 2/8 = 5/8. Activity: 4/9 + 3/9 = 7/9 (solved in the book). Step 2: Exercise 4: 3/5 + 1/5 = 4/5 and 1/5 + 3/5 = 4/5; 5/7 + 1/7 = 6/7 and 1/7 + 5/7 = 6/7; 2/8 + 3/8 = 5/8 and 3/8 + 2/8 = 5/8; 3/10 + 4/10 = 7/10 and 4/10 + 3/10 = 7/10. (Changing the order gives the same answer.) Step 3: Subtraction rule: Subtract the numerators and keep the denominator the same: Alisha: 3/4 − 2/4 = 1/4 left. Activity: (1) 3/6 − 2/6 = 1/6 (2) 5/10 − 2/10 = 3/10 (3) 4/5 − 1/5 = 3/5 (4) 6/8 − 4/8 = 2/8 (5) 8/12 − 5/12 = 3/12 (6) 2/4 − 1/4 = 1/4. Step 4: Exercise 5: 1. 5/8 − 2/8 = 3/8 (solved in the book). 2. 4/6 − 1/6 = 3/6. 3. 4/5 − 3/5 = 1/5. 4. (i) 5/8 − 3/8 = 2/8 (ii) 2/3 − 1/3 = 1/3 (iii) 6/6 − 3/6 = 3/6 (iv) 1/2 − 1/2 = 0 (v) 2/10 − 1/10 = 1/10 (vi) 5/7 − 4/7 = 1/7. Final answer: Addition: 3/4; 5/8; 7/9; Exercise 4: 4/5; 4/5; 6/7; 6/7; 5/8; 5/8; 7/10; 7/10; Subtraction: Alisha 1/4; Activity: 1/6; 3/10; 3/5; 2/8; 3/12; 1/4; Exercise 5: 3/8; 3/6; 1/5; (i) 2/8 (ii) 1/3 (iii) 3/6 (iv) 0 (v) 1/10 (vi) 1/7.

Unit 5: Algebra (Patterns)

What is a pattern? Example: blue cube, red cube, blue cube, red cube. Extended pattern: circle, triangle, pentagon, circle, triangle, pentagon — what are the next two shapes after the pentagon? Example 1 (3D shapes): cone, cube, cuboid, cone, cube, cuboid — describe the pattern. Exercise 1 Q1: rectangle, pentagon, pentagon, rectangle, pentagon, pentagon, _____ — what shape will appear next (solution given in the book)? Answer: followed by two pentagons (the book's solution says the same). Final answer: Cores: (blue, red); (circle, triangle, pentagon); (cone, cube, cuboid);

Exercise 1 (Unit 5) continued: Q2: Find the missing emoji in the pattern: angel face, sunglasses face, winking face, angel face, sunglasses face, winking face, angel face, _____, winking face. Q3: pizza, pizza, cake, pizza, pizza, cake, pizza, pizza, _____ — what follows next, the pizza or the cake? Q4: Extend the patterns (10 boxes in each row; the first boxes are given): (a) yellow triangle, green square, ... (b) green circle, orange rectangle, ... (c) big green square, small red square, yellow triangle, ... (d) green circle, red square, yellow triangle, green square, ...

Answer: cake. It completes twice and two pizzas have come again, so next comes

Question: Exercise 1 (Unit 5) continued: Q5: Match the pattern in Column A with the same pattern in Column B. Column A: (a) star, square face, circle face (b) square face, circle face, star (c) circle face, star, square face. Column B: (i) circle face, star, square face (ii) star, square face, circle face (iii) square face, circle face, star. Q6: You have spheres, cones, cubes and cuboids, four sets of each 3D shape — what patterns could you make?

Answer: Step 1: Q5: Read each pattern's order carefully and find the same order in B: (a) star, square, circle = B (ii). (b) square, circle, star = B (iii). (c) circle, star, square = B (i). Step 2: Q6: This is an open question — any repeating core is correct. Example patterns: (a) sphere, cone, cube, cuboid, sphere, cone, cube, cuboid ... (core of all four shapes). (b) sphere, sphere, cube, cube, sphere, sphere, cube, cube ... (c) cube, cone, cube, cone ... Final answer: Q5: a ↔ ii, b ↔ iii, c ↔ i; Q6: example patterns are given above (any repeating order is fine).

Question: Patterns of a Hundred Chart, Activity and Exercise 2 (Unit 5): (1) State the pattern of the rows and columns in the hundred chart. (2) Activity/Examples: Make a pattern by adding plus two to the initial number: (i) 2, 4, 6, 8, ... (ii) 23, 25, 27, 29, ... (3) Exercise 2 Q1: Using the hundred-point chart, make a pattern by adding plus three to the initial number. Q2: Make a pattern by adding plus four. Q3: What would be the next number: (a) 41, 45, 49, ... (b) 34, 44, 54, ...

Answer: Step 1: (1) In each row, left to right, the numbers increase by 1 (1, 2, 3, ...). In each column, top to bottom, the numbers increase by 10 (1, 11, 21, ...; 2, 12, 22, ...). Step 2: (2) Add 2 each time: (i) 2, 4, 6, 8, 10, 12, 14, ... (even numbers) (ii) 23, 25, 27, 29, 31, 33, 35, ... Step 3: (3) Q1: From any start, add 3: e.g. from 1: 1, 4, 7, 10, 13, 16, 19, 22, ... Q2: Add 4: e.g. from 1: 1, 5, 9, 13, 17, 21, 25, ... (Any starting point is fine as long as the same amount is added each step.) Step 4: Q3: (a) Find the difference: 45 − 41 = 4 and 49 − 45 = 4, so it is a +4 pattern: next = 49 + 4 = 53. (b) 44 − 34 = 10 and 54 − 44 = 10, so +10: next = 54 + 10 = 64. Final answer: (1) Rows +1, columns +10 (2) (i) 10, 12, 14 ... (ii) 31, 33, 35 ... (3) Q1: e.g. 1, 4, 7, 10, 13 ...; Q2: e.g. 1, 5, 9, 13, 17 ...; Q3: (a) 53 (b) 64.

Unit 6: Measurement

Question: Length and standard units: What does length tell us? What is a meter (m) and where is it used? What is a centimeter (cm)? The meter rod is marked from 0 to 100 — how many centimeters are in 1 meter? What are the lengths of the rulers shown? Examples: a table is 1 meter; a bus is about 16 meters; a pencil is about 15 centimeters; a book is about 20 centimeters. Activity: In which unit is each measured: (i) length of a mobile phone (ii) length of a long pipe/roll (iii) your height (iv) height of a cupboard?

Answer: Step 1: Length tells us whether an object is long or short; the long pencil has a larger length than the short pencil. Step 2: A meter (m) is a standard unit of length; it is used for larger things like trains, streets and jogging tracks; a meter tape is used to measure length. Step 3: A centimeter (cm) is a smaller unit; it is used for smaller things like pencils, books and our own height. The meter rod is divided from 0 to 100, so 1 meter = 100 centimeters. The rulers are 10 cm, 15 cm and 20 cm long. Step 4: Examples: table = 1 meter; bus = 16 meters; pencil = 15 cm; book = 20 cm. Step 5: Activity: (i) A mobile phone is small → centimeters (ii) A long pipe is big → meters (iii) Height → centimeters (the book uses cm for our height) (iv) A cupboard is big → meters. Final answer: 1 m = 100 cm; rulers 10, 15, 20 cm; Activity: (i) cm (ii) m (iii) cm (iv) m.

Question: Exercise 1 (Unit 6): Q1: Using the 15 cm ruler scale printed on the page, measure the length of the objects and write in centimeters: (i) paint brush (ii) paper clip (iii) spoon (iv) comb (v) pencil (vi) toothbrush (vii) toothpaste tube (viii) key. Q2: Measure the length of objects in your classroom and write it in meters (rounded in units): 1. Table in the

classroom 2. Door in the classroom 3. Black board in the classroom 4. Desk in the classroom.

Answer: Step 1: Method: Match the left end of the object with the 0 mark and read the mark at its right end; count up to the end of each object. Step 2: (i) The brush ends at 4 → 4 cm (ii) The clip ends at 3 → 3 cm (iii) The spoon ends at 8 → 8 cm (iv) The comb ends at 11 → 11 cm. Step 3: (v) The pencil tip is at 8 → 8 cm (vi) The toothbrush ends at 10 → 10 cm (vii) The toothpaste ends at 11 → 11 cm (viii) The key ends at 4 → 4 cm. (These are read from the book's printed scale to the nearest centimeter.) Step 4: Q2 is practical work — measurements differ from classroom to classroom; example values (rounded meters): table ≈ 1 m, door ≈ 2 m, black board ≈ 3 m, desk ≈ 1 m. Measure your own classroom with a meter tape and write. Final answer: Q1: (i) 4 cm (ii) 3 cm (iii) 8 cm (iv) 11 cm (v) 8 cm (vi) 10 cm (vii) 11 cm (viii) 4 cm; Q2: example: 1 m, 2 m, 3 m, 1 m (measure your own).

Comparison of length, Examples and Exercise 2 (Unit 6): Example 1: car (4 meter) and bike (2 meter). Example 2: pencil (15 cm) and mobile (12 cm). Example 3: wall (5 meter) and bus (16 meter). Example 4: book (20 cm) and an arm of a child (45 cm). Example 5: pencil (15 cm) and pen (15 cm). Example 6: red shirt (50 cm) and blue shirt (50 cm). Exercise 2: 1. Which is longer, a pencil (15 cm) or a ruler (30 cm)? _____ is longer than _____. 2. Is a bookshelf (2 m) longer or shorter than a bed (3 m)? _____ is shorter than _____. 3. Compare a football field (100 m) and a basketball court (28 m): _____ is shorter than _____. 4. Which is longer, a shoe (25 cm) or a book (15 cm)? 5. Football field (100 m) and cricket ground (100 m): length of both is _____. 6. Mobile 15 cm and pen 15 cm: length of both is _____.

Answer: the pencil., the bed., the football field., equal., equal. Final

answer: Examples: car longer; pencil longer; wall less; book less; equal; equal. Exercise 2: 1. Ruler 2. Bookshelf shorter 3. Basketball court 4. Shoe 5. Equal 6. Equal.

Question: Length word problems, Examples and Exercise 3 (Unit 6): Example 1: A path in a garden is 8 meters long; if we extend it 2 more meters, how long will it be? Example 2: The length of a toy train is 80 cm and one more coach of 8 cm is added — what is the total length? Example 3: Hussain draws a line 25 cm long, then removes a 10 cm part — what is the length of the remaining line? Example 4: A toy train track is 18 meters long; if 5 meters are removed, what is the length now? Exercise 3: (1) A pencil is 15 cm long and a ruler is 20 cm long — what is the total length? (2) A highway is 500 meters long and a connector road is 250 meters long — what is the total length? (3) Ali runs 100 meters and his friend runs 75 meters — how much farther did Ali run? (4) A dress is 90 cm long and a tailor shortens it by 18 cm — how long is the dress now? (5) Ayesha has a 60 cm ribbon; she used 40 cm — what is the length of the remaining ribbon? (6) Shahid's father is 185 cm tall and Awais is 85 cm shorter than his father — what is Awais's height?

Answer: Step 1: Example 1: Add: ones column 8 + 2 = 10 → total 10 meters. Example 2: 80 + 8: ones 0 + 8 = 8; tens 8 → 88 cm (the book misprints 88m; the correct unit is cm). Example 3: Subtract: 25 − 10 = 15 → remaining line 15 cm. Example 4: 18 − 5 = 13 → track 13 meters. Step 2: Exercise 3 (1): 15 + 20: ones 5 + 0 = 5; tens 1 + 2 = 3 → 35 cm. (2) 500 + 250: ones 0; tens 0 + 5 = 5; hundreds 5 + 2 = 7 → 750 meters. Step 3: (3) 100 − 75: ones: 0 cannot take 5, borrow → 10 − 5 = 5; tens: 9 − 7 = 2 → 25 meters farther. (4) 90 − 18: ones: 10 − 8 = 2 (borrow); tens: 8 − 1 = 7 → 72 cm. Step 4: (5) 60 − 40 = 20 cm remaining. (6) 185 − 85: ones 5 − 5 = 0; tens 8 − 8 = 0; hundreds 1 → 100 cm. Final answer:

Examples: 10 m; 88 cm; 15 cm; 13 m. Exercise 3: (1) 35 cm (2) 750 m (3) 25 m (4) 72 cm (5) 20 cm (6) 100 cm.

Mass and Exercise 4 (Unit 6): What is mass/weight? What is a kilogram (kg)? Examples: the mass of apples is 1 kg and the mass of meat is 2 kg (shown on scales). What is a gram (gm)? Examples: a pencil is 20 gm; a small toy car is 30 gm. Remember: kg for heavy things, gm for light things. Activity: (i) Measure your weight in kilograms (ii) Measure the mass of your school bag in kilograms. Exercise 4 Q1: Write the suitable unit to measure the weight (gm or kg): (a) an orange (b) a goat (c) a cucumber (d) a watermelon. Q2: Measure/estimate the weight: (a) Weight of a pencil in gm is _____ (b) Weight of a sharpener in gm is _____ (c) Weight of a seven years old boy in kg is about _____ (d) Weight of a bag of tomatoes in kg is about _____.

Answer: (the book's example), (about 20-25 kg), (c) gm (d) kg; Q2: (a) ≈20 gm (b) ≈10 gm (c) ≈22 kg (d) ≈5 kg (estimates).

Comparison of mass, Examples and Exercise 5 (Unit 6): Example 1: toy truck (50 gm) and toy car (30 gm). Example 2: bottle of water (1 kg) and school bag (2 kg). Example 3: box of pencils (20 gm) and empty lunch box (20 gm). Example 4: tomatoes (3 kg) and potatoes (2 kg). Example 5: an apple (80 g) and an orange (120 g). Example 6: sugar bag (5 kg) and flour bag (5 kg). Exercise 5: 1. A bag of apples weighs 3 kilograms, while a bag of oranges weighs 1 kilogram — which has a greater mass? (Book: 3 is greater than 1) So, _____ has greater mass. 2. A paperclip weighs 5 grams, while a pen weighs 20 grams — which has a greater mass? _____. 3. A small toy car weighs 50 grams, while a toy truck weighs 75 grams — which has a lesser mass? _____. 4. Ali's weight is 20 kg while his friend Amjad's weight is 24 kg — who is lighter? _____. 5. Compare the weight of the book (500 grams) with that of the bag of sugar (500

grams): the weight of both is _____.

Answer: mass., mass. Step 4:, lighter., equal. Final answer: Examples: truck heavier; bottle lighter; equal; tomatoes heavier; apple lighter; equal. Exercise 5: 1. Bag of apples 2. Pen 3. Toy car 4. Ali 5. Equal.

Question: Mass word problems, Examples and Exercise 6 (Unit 6): Example 1: A student packs 100 g of carrots and 80 g of apples in her lunch box — what is the total mass of the lunch box? Example 2: Asma buys 40 kg of flour and 20 kg of rice bags — what is the total mass of both bags? Example 3: Ali bought 250 g of strawberries; he ate 50 g — how many grams are left? Example 4: A bag of rice weighs 20 kg; if 5 kg of rice is taken out, what is the new mass of the bag? Exercise 6: 1. A bag of flour weighs 50 kilograms; if 20 kilograms of sugar are added, what is the total weight of the bag? 2. A packet of coffee weighs 250 grams and a packet of sugar weighs 75 g — what is the total mass of the coffee and sugar together? 3. A sack of potatoes weighs 15 kilograms; if 7 kilograms of potatoes are taken out, what is the remaining mass of the sack? 4. A box of chocolates weighs 300 grams; if 250 grams of chocolates are eaten, what is the remaining mass of the chocolates?

Answer: Step 1: Example 1: Add: 100 + 80: ones 0; tens 0 + 8 = 8; hundreds 1 → 180 g. Example 2: 40 + 20 = 60 kg. Step 2: Example 3: Subtract: 250 − 50 = 200 g left. Example 4: 20 − 5 = 15 kg. Step 3: Exercise 6: 1. 50 + 20 = 70 kilograms. 2. 250 + 75: ones 0 + 5 = 5; tens 5 + 7 = 12 → write 2, carry 1; hundreds 2 + 1 = 3 → 325 grams. Step 4: 3. 15 − 7 = 8 kilograms. 4. 300 − 250: ones 0; tens: 0 cannot take 5, borrow from hundreds → 10 − 5 = 5; hundreds: 2 − 2 = 0 → 50 grams. Final answer: Examples: 180 g; 60 kg; 200 g; 15 kg. Exercise 6: 1. 70 kg 2. 325 g 3. 8 kg 4. 50 g.

Question: Capacity, liter and milliliter: What is capacity? Example: The jug

can hold 8 glasses of water — what is the capacity of the jug? What is a liter (L)? The measuring jug shown is filled to the 1000 ml mark — what is its capacity? What is a milliliter (ml)? A small bottle can hold 1000 tiny cups, each cup is 1 ml — what is the capacity of the bottle? Example: The glass of cold drink is filled to the 200 mark on its scale — how many ml? Activities: Measure the capacity of your water bottle in liters; measure the capacity of a glass of water in milliliters using a syringe.

Answer: Step 1: Capacity is the amount of liquid a container can hold. The jug holds 8 glasses of water, so the capacity of the jug is 8 glasses. Step 2: A liter (L) is a standard unit of capacity. The measuring jug is marked 1000 ml = 1 Liter and the water reaches the top, so the capacity of the jug is 1 liter. Step 3: A milliliter (ml) is a tiny unit: the bottle holds 1000 tiny cups, each 1 ml, so the bottle can hold 1000 milliliters (ml) — that is, 1 liter = 1000 ml. Step 4: The scale on the glass goes 20, 40, ... 200 and the drink reaches 200, so the capacity of the glass is 200 ml. Step 5: The Activities are practical: measure your own bottle/glass (for example, a common bottle is 1 liter and a glass may be about 200-250 ml). Final answer: Jug: 8 glasses; measuring jug: 1 liter; bottle: 1000 ml; glass: 200 ml.

Comparison of capacity, Examples and Exercise 7 (Unit 6): Example 1: jug of water (2 liter) and packet of juice (1 liter). Example 2: glass of soda (100 milliliter) and medicine dropper (20 milliliter). Example 3: fuel tank (20 liter) and another large fuel tank (50 liter). Example 4: cup of tea (50 milliliters) and large cup of tea (100 milliliters). Example 5: container of oil (500 liter) and container of water (500 liters). Example 6: perfume bottle (50 milliliters) and perfume bottle (50 milliliters). Exercise 7: 1. A packet of juice has a capacity of 200 milliliters; a milk packet has a capacity of 250 milliliters — which has a greater capacity? _____. 2. A fuel tank A has a capacity of 60 liters; another fuel tank B has a capacity of 65 liters —

which has a lesser capacity? _____. 3. A hand wash bottle has a capacity of 250 milliliters; a shampoo bottle has a capacity of 500 milliliters — as the book's conclusion line asks, which has lesser capacity? (And which has greater?) 4. Water cooler 4 liter and fuel gallon 4 liter: the capacity of both is _____. 5. A water tank has a capacity of 700 liters; a fuel tank has a capacity of 600 liters — which has a greater capacity? _____. 6. Bottle of soap 100 milliliter and bottle of perfume 100 milliliter: the capacity of both is _____.

Answer: capacity., capacity., equal., capacity., equal. Final answer: Examples: jug greater; glass greater; tank less; cup less; equal; equal. Exercise 7: 1. Milk packet 2. Tank A 3. Hand wash bottle (lesser); shampoo (greater) 4. Equal 5. Water tank 6. Equal.

Question: Capacity word problems, Examples and Exercise 8 (Unit 6): Example 1: A juice manufacturer has 300 liters of apple juice and 50 liters of orange juice — how much juice in total? Example 2: A pharmacist has 250 mL of medicine in a bottle; she adds 100 mL more — how much medicine will the bottle contain? Example 3: A water tank can hold 500 liters of water; if 50 liters are poured out, how much water is left? Exercise 8: 1. A packet of juice has a capacity of 200 milliliters and another packet is of 100 milliliters — what is the total capacity of both packets? 2. A water tank has a capacity of 800 liters and a fuel tank has a capacity of 500 liters — what is the total capacity of both tanks? 3. A gas cylinder has a capacity of 45 liters; if 15 liters of gas are used for cooking, what is the remaining capacity? 4. A perfume bottle has a capacity of 100 milliliters; if 50 milliliters are used, what is the remaining capacity of perfume?

Answer: Step 1: Example 1: Add: 300 + 50 = 350 liters. Example 2: 250 + 100 = 350 mL. Example 3: Subtract: 500 − 50 = 450 liters left. Step 2: Exercise 8: 1. 200 + 100 = 300 milliliters. 2. 800 + 500: ones 0; tens 0;

hundreds 8 + 5 = 13 → 1300 liters. Step 3: 3. 45 − 15: ones 5 − 5 = 0; tens 4 − 1 = 3 → 30 liters. 4. 100 − 50 = 50 milliliters. Final answer: Examples: 350 L; 350 mL; 450 L. Exercise 8: 1. 300 ml 2. 1300 L 3. 30 L 4. 50 ml.

Question: Temperature and Exercise 9 (Unit 6): What is temperature and in which unit is it measured? Which instrument is used? Examples: Read the nearest temperature in the given thermometers — Examples 1, 2, 3, 4. Relate the temperature scale to the number line: what happens to temperature as we move to the right? Exercise 9: Read the thermometer: 1. The red column stops at the 25 mark 2. At the 35 mark 3. At the 45 mark 4. At the 60 mark 5. At the 80 mark (also show it on the number line) 6. At the 95 mark (also show it on the number line) — write the temperature of each.

Answer: Step 1: Temperature is a measure of how hot or cold something is; it is measured in degree Centigrade (°C) and a thermometer is used to measure it. Step 2: How to read: See at which mark the red column stops; that is the nearest temperature. Example 1: 5 degree Centigrade; Example 2: 15; Example 3: 80; Example 4: 56. Step 3: On a number line, numbers get bigger as we move to the right; in the same way, on the temperature scale, temperatures increase as we move to the right and decrease to the left (on the scale: 0°C, 5°C, 10°C, 20°C and so on). Step 4: Exercise 9: 1. 25°C 2. 35°C 3. 45°C 4. 60°C 5. 80°C — mark the point 80 on the number line 6. 95°C — mark the point 95 on the number line. Final answer: Unit °C, instrument thermometer; Examples: 5, 15, 80, 56 degrees; Exercise 9: 25°C, 35°C, 45°C, 60°C, 80°C, 95°C.

Unit 7: Time

Question: Analog clock, Examples and Exercise 1 (Unit 7): What is an analog clock? What do the short hand (hour hand) and the long hand (minute hand) tell? How many minutes is each number on the clock? Example 1: Short hand is on 7 and long hand is on 3 — tell the time. Example 2: Short hand on 5, long hand on 8. Example 3: Short hand on 6, long hand on 7. Exercise 1 (Tell the time — left column top to bottom, then right column): 1. Short hand on 12, long hand on 2 2. Short on 2, long on 5 3. Short just past 5, long on 9 4. Short on 8, long on 3 5. Short between 10 and 11, long on 6 6. Short on 1, long on 4 7. Short on 4, long on 7 8. Short just before 7, long on 10 9. Short on 9, long on 12 10. Short just before 12, long on 7.

Answer: Step 1: An analog clock shows time with hour and minute hands; its face has numbers 1 to 12. The short hand points to the hour; the long hand points to the minutes. Each number = 5 minutes (multiply the number by 5 for the long hand). Step 2: Example 1: Hour 7; minutes 3 × 5 = 15 → 7:15 (7 hours and 15 minutes). Example 2: Hour 5; minutes 8 × 5 = 40 → 5:40. Example 3: Hour 6; minutes 7 × 5 = 35 → 6:35. Step 3: Exercise 1: 1. 12:10 (2 × 5 = 10 minutes) 2. 2:25 (5 × 5 = 25) 3. 5:45 (9 × 5 = 45) 4. 8:15 (3 × 5 = 15) 5. 10:30 (6 × 5 = 30). Step 4: 6. 1:20 (4 × 5 = 20) 7. 4:35 (7 × 5 = 35) 8. 6:50 (10 × 5 = 50; the hour hand is just before 7) 9. 9:00 (long hand on 12 = exact hour) 10. 11:35 (7 × 5 = 35). Final answer: Examples: 7:15; 5:40; 6:35. Exercise 1: 12:10, 2:25, 5:45, 8:15, 10:30, 1:20, 4:35, 6:50, 9:00, 11:35.

Time interval (duration), Examples, Activity and Exercise 2 (Unit 7): What is a time interval? Example 1: A boy eats biryani at 2:00 pm and finishes it

at 2:15 pm — how long does he take? Example 2: Boys start playing football at 5:00 pm and end the game at 7:00 pm — how long is the match? Example 3: Ali starts watching a movie at 8:00 pm and it lasts till 11:00 pm — how long is the movie? Activity: (i) How long do you watch a cartoon? (ii) How long do you sleep? (iii) How long do you study at school? (iv) How long do you take to visit an amusement park? Exercise 2: 1. Khizar starts watching a cartoon at 6:00 pm and finishes at 6:30 pm — how long does it last? He takes _____ minutes to watch the cartoon. 2. Asra goes on a short trip with her parents from 10:00 am to 1:00 pm — how long has she been on the trip? She has been on the short trip for _____ hours. 3. An artist draws a picture from 3:00 pm to 5:30 pm — how long does he take to draw the picture? 4. Hina leaves Karachi at 9:00 am and arrives in Hyderabad at 11:00 pm — how long does she travel?

Answer: = 30 minutes., pm: 10 → 11 → 12 → 1 = 3

Question: Use of calendar: days of the week, months of the year and Activities (Unit 7): Write the names of the days of the week in order and tell which number day each one is. Activity 1: (i) What day is today? (ii) What day is tomorrow? (iii) What day was yesterday? Write the names of the months in a year in order. Activity 2: (i) What is the current month? (ii) What will be the next month? (iii) What was the previous month?

Answer: Step 1: The 7 days of the week: Monday — 1st day; Tuesday — 2nd day; Wednesday — 3rd day; Thursday — 4th day; Friday — 5th day; Saturday — 6th day; Sunday — 7th day. Step 2: Activity 1 (method + example): Look at today's day on the calendar; tomorrow = the next day; yesterday = the day before. Example: If today is Friday, tomorrow is Saturday and yesterday was Thursday. Step 3: The 12 months of the year: January, February, March, April, May, June, July, August, September, October, November, December. Step 4: Activity 2 (method + example):

Find the current month on the calendar; next = the one after it in the list; previous = the one before it. Example: If the current month is October, the next month is November and the previous month was September. Final answer: Days: Monday 1, Tuesday 2, Wednesday 3, Thursday 4, Friday 5, Saturday 6, Sunday 7; Months: January to December in the order above.

Question: Model Solar Calendar examples and Exercise 3 (Unit 7): Example 1: Rida's birthday is on 13th October — which day of the week will it be? Example 2: The school's annual award ceremony is scheduled for the 2nd of April — which day of the week will it be? Example 3: Sarfraz's family plans a picnic on the first Sunday of January — what date will their picnic be? Example 4: Grandmother is coming to visit on the 3rd Thursday of July — what date will it be? Exercise 3 (using the book's model calendar): 1. Ahmad's birthday is on 28th June — what day of the week will it be? 2. The school holiday is on 11th November — what day of the week is it? 3. Kamran's family plans a picnic on the second Saturday of June — what date will their next picnic be? 4. The sport festival of the school is conducted on the third Monday of February — what date will it be? 5. On what day of the week is the Independence Day of our country (14th August) celebrated?

Answer: Step 1: Method: Find the month on the calendar, find the date, and read the day written above it. Example 1: In October, 13 is in the Monday column → Monday. Example 2: In April, 2 is in the Wednesday column → Wednesday. Step 2: Example 3: In January the Sundays are 5, 12, 19, 26 — the first Sunday is the 5th → picnic on 5th January. Example 4: In July the Thursdays are 3, 10, 17, 24, 31 — the 3rd Thursday is the 17th → grandmother comes on 17th July. Step 3: Exercise 3: 1. In June, 28 is in the Saturday column (June 1 = Sunday; rows: 23 Monday … 28 Saturday, 29 Sunday) → Ahmad's birthday will be on Saturday. 2. In

November, 11 is in the Tuesday column (November 1 = Saturday; 10 Monday, 11 Tuesday) → the holiday is on Tuesday. Step 4: 3. In June the Saturdays are 7, 14, 21, 28 — the second Saturday is the 14th → picnic on 14th June. 4. In February the Mondays are 3, 10, 17, 24 — the third Monday is the 17th → festival on 17th February. 5. Independence Day is 14th August; in August, 14 is in the Thursday column (August 1 = Friday; 11 Monday, 14 Thursday) → Thursday. Final answer: Examples: Monday; Wednesday; 5th January; 17th July. Exercise 3: 1. Saturday 2. Tuesday 3. 14th June 4. 17th February 5. Thursday.

Unit 8: Geometry

Question: Two-dimensional shapes: Rectangle and Square (Unit 8): What is geometry? What are 2-D shapes? Write the definition of a rectangle: How many corners (vertices) and sides (edges) does it have? What is special about its opposite sides? What are length and breadth? Which objects are shown as examples? Example: Draw a rectangle whose length is 10 centimeters and breadth is 4 centimeters. Definition of a square: How many corners and sides? What is special about its sides? Example objects? Example: Draw a square where the length of each side is 10 centimeters.

Answer: Step 1: Geometry is the branch of mathematics that deals with shapes, sizes and positions of objects. 2-D shapes are flat shapes that have only length and width — like a bangle (circle), a speedometer (semi-circle) and pizza quarters (quarter-circle). Step 2: A rectangle is a two-dimensional shape with four corners (vertices) and four sides (edges); all sides are straight and opposite sides are equal in length. The longer side is its length and the shorter side is its breadth or width. Example objects: Mobile, Door, Book, Wooden block — all rectangular in shape. Step 3: To draw a rectangle, draw four straight sides where opposite sides are equal. Solution of the example (shape in words): First draw a straight 10 cm side (the top), then draw 4 cm sides down from both corners, and finally join the bottom 10 cm side — the rectangle is ready with length 10 cm and breadth 4 cm. Step 4: A square is a two-dimensional shape with four corners called vertices; all sides are equal in length and all sides are straight. Example objects: Clock (square face), LCD, Ludo board, Register — all square in shape. Step 5: To draw a square, draw four straight sides where all sides are equal. Solution of the example: Draw each side 10 cm

— all four sides are 10 cm and equal, so the shape is a square (difference from a rectangle: in a square all four sides are equal; in a rectangle only the opposite sides are equal). Final answer: Rectangle: 4 vertices, 4 sides, opposite sides equal; Square: 4 vertices, 4 sides, all sides equal.

Question: Circle, Triangle, Semi-Circle and Quarter-Circle (Unit 8): Write the definition of a circle and the steps to draw it; which are the example objects? Definition of a triangle: how many sides and vertices? Steps to draw a triangle. Definition of a semi-circle and steps to draw it; example objects? Definition of a quarter-circle and steps to draw it; example objects?

Answer: Step 1: A circle is a round-shaped figure that has no corners or edges. Example objects: Pizza, Clock, Ring, Wooden circular block. To draw the circle: 1. Make a small dot on your paper where you want the circle to be 2. Place the tip of your pencil on the dot 3. Move your pencil around the dot in a round and round motion, like you are making a big hug for the dot — the circle is ready. Step 2: A triangle is a 2-dimensional shape with three edges (sides) having three (corners) vertices (example figures: an ice-cream cone, a triangular biscuit, a chocolate triangle, a yellow triangular shape). To draw the triangle: 1. Draw three dots on your paper, spaced evenly apart — these will be the corners 2. Connect the first dot to the second dot with a straight line 3. Connect the second dot to the third dot with a straight line 4. Finally, connect the third dot to the first dot with a straight line — the triangle is ready. Step 3: A semicircle is a half circle formed by cutting the circle into two halves. Example objects: speedometer, protractor, a watermelon half-slice, a tunnel entrance, an igloo. To draw: 1. Start by drawing a circle 2. Draw a line through the circle, dividing it into two equal halves — each half is a semi-circle. Step 4: A quarter circle is a shape that is made by dividing a circle into four equal

parts (each part = 1/4). Example objects: a quarter slice of pizza, a quarter part of a clock face, a quarter slice of watermelon. To draw: 1. Start by drawing a big circle 2. Draw a line through the circle, dividing it into two halves 3. Draw another line, dividing one of the halves into two quarters — the quarter circle is ready (in the book's picture, one quarter of the green circle is shown in red). Final answer: Circle: 0 corners, 0 edges; Triangle: 3 sides, 3 vertices; Semi-circle: half of a circle (1/2); Quarter-circle: one-fourth of a circle (1/4).

Question: Exercise 1 (Unit 8): 1. Colour the shapes according to the key provided: Quarter circle — Purple; Triangle — Blue; Semi-circle — Green; Square — Yellow; Circle — Red. How many times does each shape appear in the big-circle picture? 2. Colour the house picture according to the key: Circle — Red; Rectangle — Blue; Triangle — Green; Square — Yellow. Which colour will each part of the house get? 3. Identify and label each shape and give a related real-life object: (a) Circle (name given in the book; object: a ring) (b) Rectangle (c) Square (d) Triangle. 4. Draw the following 2-D shapes (shown in dotted lines): square, rectangle, triangle, circle, semi-circle, quarter circle — describe each shape in words.

Answer: Step 1: Q1 — Match the key to the shape: Quarter circle → Purple; Triangle → Blue; Semi-circle → Green; Square → Yellow; Circle → Red. Counting in the big picture: Circles are 4 (top, left, right, bottom) → colour all 4 red; Squares 3 → yellow; Triangles 3 → blue; Semi-circles 2 (one standing half, one lying half) → green; Quarter circle 1 → purple. Step 2: Q2 — Parts of the house: The sun is a circle → red; the two round windows of the house are circles → red; the roof is a triangle → green; the main body of the house, the door and the two small windows are rectangles → blue; the small round door knob is a circle → red. (There is no square in this picture, so the yellow colour is not used.) Step 3: Q3: (a) Circle —

object: a ring (drawn in the book) (b) Rectangle — object: a book or a door (c) Square — object: a Ludo board or a handkerchief (d) Triangle — object: a samosa or a pizza slice. Step 4: Q4 — The shapes in words: Square: 4 equal straight sides; Rectangle: 4 sides, opposite sides equal; Triangle: 3 straight sides, 3 corners; Circle: one round closed curve, no corners; Semi-circle: half round + one straight side; Quarter circle: a quarter round part + two straight sides. Trace the dotted lines to complete each shape. Final answer: 1. Circles 4 red, squares 3 yellow, triangles 3 blue, semi-circles 2 green, quarter circle 1 purple; 2. Sun/windows red, roof green, house/door blue; 3. Circle–ring, Rectangle–book, Square–Ludo, Triangle–samosa; 4. Draw the shapes with the properties described above.

Question: 3-D shapes (Unit 8): What is a 3-D shape? What are the five common 3-D shapes? Write the definition and the book's examples of each: Cube, Cuboid, Cone, Cylinder, Sphere — and also tell the everyday object shown with each in the unit's opening picture.

Answer: Step 1: A 3-D shape is an object that has length, width and height; we can see 3-D shapes from different angles and orientations. Some objects have flat surfaces, while others are curved or rounded. Step 2: Cube: A cube is a shape with six square faces of equal size. Examples: a dice, a gift (box), a box and a dresser; in the opening picture, a Rubik's cube. Cuboid: A cuboid is a shape with six rectangular faces. Examples: a book, a brick and a match box; in the picture, a milk carton. Step 3: Cone: A cone is a shape with a circular base and a curved surface that tapers to a point. Examples: an ice cream cone, a party hat and a funnel. Cylinder: A cylinder is a shape with two circular bases connected by a curved surface. Examples: a water bottle, a gas cylinder and a tin of soft drink; in the picture, a green can. Step 4: Sphere: A sphere is a shape that is round and curved, like a ball. Examples: a basketball, a globe and an orange; in the

picture, a colourful ball. Final answer: Cube: 6 equal square faces; Cuboid: 6 rectangular faces; Cone: circular base tapering to a point; Cylinder: 2 circular bases with a curved surface; Sphere: fully round, like a ball. Question: Describing position, direction and movement (Unit 8): Write the position words with the book's examples: 1. Inside and Outside 2. Above and Below 3. Over and Under 4. Far and Near. Direction: What is clockwise? Its example? What is anti-clockwise? Its examples? Turns: quarter turn, half turn, three-quarter turn and one whole turn clockwise — where does the arrow reach? Write the book's movement examples. Answer: Step 1: Position: Objects can be found in different positions. 1. Inside/Outside: The book is inside the bag and the register is outside the bag. 2. Above/Below: The picture is above the sofa and below the clock (clock at the top, then the picture, then the sofa). Step 2: 3. Over/Under: A boy jumps over the hurdle and another crawls under the hurdle; the plane is flying over the city. 4. Far/Near: The cow is near the house and the goat is far away from the house. Step 3: Direction: Clockwise (turning right) — a clock's hands moving from 12 to 3; example: the wheel (of the car) is turning clockwise. Anti-clockwise (turning left) — a clock's hands moving from 12 to 9; example: the earth rotates around the Sun in anticlockwise direction. Step 4: Turns (starting point at the top, the arrow starts pointing up): A quarter turn clockwise: the arrow reaches the right side (the 3 position); A half turn clockwise: the arrow reaches the bottom (the 6 position); A three-quarter turn clockwise: the arrow reaches the left side (the 9 position); A one whole turn clockwise: the arrow comes back to the top (back to 12). Step 5: Movement examples: The ball rolled around the corner of the wall and into the street; the bus went up the hill to the school; the dog ran up the stairs to greet its owner; the cyclist pedaled along the bike path to the park. Final answer: Position: book inside,

register outside; picture above the sofa, below the clock; cow near, goat far. Direction: 12→3 clockwise; 12→9 anti-clockwise. Turns: quarter→right, half→bottom, three-quarter→left, whole→back to top.

Unit 9: Data and Probability

Question: Pictographs (Unit 9): What is a pictograph? Example 1: Read the pictograph of favourite sports of the students — each student icon represents 1 favourite sport: Tennis (1 icon), Football (2 icons), Cricket (3 icons) — how many students like each sport? Example 2: The pictograph represents animals — each pet icon represents 2 animals: Fish (2 icons), Cat (4 icons), Dog (5 icons) — how many animals of each? Activity: The pictograph represents vehicles for transportation — each icon represents 4 vehicles: Bus (3 icons), Bicycle (4 icons), Car (6 icons) — how many vehicles of each?

Answer: Step 1: A pictograph is a special kind of graph that uses pictures to show information. Method: Count the icons and multiply by the value of each icon. Step 2: Example 1 (each icon = 1 student): Tennis: 1 × 1 = 1 student; Football: 2 × 1 = 2 students; Cricket: 3 × 1 = 3 students. Step 3: Example 2 (each icon = 2 animals): Fish: 2 × 2 = 4; Cat: 4 × 2 = 8; Dog: 5 × 2 = 10 (the book also says 5 × 2 = 10 dogs). Step 4: Activity (each icon = 4 vehicles): Bus: 3 × 4 = 12 vehicles; Bicycle: 4 × 4 = 16; Car: 6 × 4 = 24 (the book also says 6 × 4 = 24 cars). Final answer: Example 1: Tennis 1, Football 2, Cricket 3 students; Example 2: Fish 4, Cats 8, Dogs 10; Activity: Buses 12, Bicycles 16, Cars 24.

Question: Block Graphs (Unit 9): What is a block graph? Example 1: Sleeping hours of Nasir for three days — each block represents 1 hour: Sunday 9 blocks (red), Monday 7 blocks (blue), Tuesday 8 blocks (green) — how many hours of sleep on each day? Example 2: Favourite colours — each block represents 2 students: Red 5 blocks, Blue 3 blocks, Green 4 blocks — how many students like each colour? Example 3: Favourite fruits

— each block represents 3 students: Orange 8 blocks, Apple 6 blocks, Banana 4 blocks — how many students like each fruit?

Answer: Step 1: A block graph is a graph that uses blocks or bars to represent data; each block or bar represents a certain number of items. Method: Count the blocks and multiply by the value of each block. Step 2: Example 1 (each block = 1 hour): Sunday: 9 × 1 = 9 hours of sleep; Monday: 7 × 1 = 7 hours; Tuesday: 8 × 1 = 8 hours. Step 3: Example 2 (each block = 2 students): Red: 5 × 2 = 10 students; Blue: 3 × 2 = 6 students; Green: 4 × 2 = 8 students. Step 4: Example 3 (each block = 3 students): Orange: 8 × 3 = 24 students; Apple: 6 × 3 = 18 students; Banana: 4 × 3 = 12 students. Final answer: Example 1: Sunday 9, Monday 7, Tuesday 8 hours; Example 2: Red 10, Blue 6, Green 8 students; Example 3: Orange 24, Apple 18, Banana 12 students.

Question: Possibility of occurring of any event (Unit 9): What are the special words used to describe how likely something is to happen? Write the meaning and both book examples of each: Impossible, Less Likely, More Likely, Unlikely and Certain.

Answer: Step 1: We can use special words to describe how likely something is to happen. These words are: impossible, less likely, more likely, unlikely and certain. Step 2: Impossible: Something that is impossible will never happen. Examples: It is impossible for a horse to fly; It is impossible for a cat to bark. Less Likely: Something that is less likely is not very likely to happen. Examples: It is less likely for a boy to run who has a fracture in his foot; It is less likely for a rainbow to appear in the sky on a cloudy day. Step 3: More Likely: Something that is more likely has more chances to happen. Examples: It is more likely that Ali's grandmother would feel ill in cold weather; It is more likely for a ball to bounce when dropped. Unlikely: Something that is unlikely has no chance

to happen (as the book defines it). Examples: It is unlikely for a car to fly; It is unlikely for a flower to grow in a desert. Step 4: Certain: Something that is certain will always happen. Examples: It is certain for a ball to fall when dropped; It is certain that a night comes after the day. Final answer: Impossible = never happens; Less likely = not very likely; More likely = more chances; Unlikely = no chance; Certain = will always happen.

Question: Exercise 1 (Unit 9): 1. The pictograph of traffic passing through a signal — one picture = 2 vehicles: Cars 5 pictures, Vans 4, Motor Bikes 3, Busses 2, Bicycles 4. (a) Which vehicles are more in numbers? (b) Which vehicles are least in numbers? (c) How many busses passed through the signal? (d) How many motor bikes passed through the signal? (e) How many vans and bicycles passed altogether? 2. The pictograph shows the number of animals in the zoo — each picture represents 2 animals: Lion 2 pictures, Deer 3, Monkey 4, Fox 3, Rabbit 5, Elephant 1. (a) How many lions are there? (b) How many animals are there altogether? (c) How many more foxes are there than elephants? (d) How many different types of animals are there? (e) Which animals are most in number? (f) Which animals are least in number? 3. What is the possibility of getting a sunny day tomorrow? 4. What is the possibility of a cat barking? 5. What is the possibility of getting a certain color of candy from a bag? 6. What is the possibility of a dog meowing? 7. What is the possibility of a ball bouncing when dropped?

Answer: Step 1: Q1 — Each picture = 2 vehicles: Cars 5 × 2 = 10; Vans 4 × 2 = 8; Motor bikes 3 × 2 = 6; Busses 2 × 2 = 4; Bicycles 4 × 2 = 8. (a) The biggest number is 10 → Cars are more in number. (b) The smallest is 4 → Busses are least. (c) Busses = 4. (d) Motor bikes = 6. (e) Vans + bicycles = 8 + 8 = 16. Step 2: Q2 — Each picture = 2 animals: Lion 2 × 2 = 4; Deer 3 × 2 = 6; Monkey 4 × 2 = 8; Fox 3 × 2 = 6; Rabbit 5 × 2 = 10;

Elephant 1 × 2 = 2. (a) Lions = 4. (b) Altogether: 4 + 6 = 10; 10 + 8 = 18; 18 + 6 = 24; 24 + 10 = 34; 34 + 2 = 36 → 36 animals. (c) Foxes − elephants = 6 − 2 = 4 more foxes. (d) Types: lion, deer, monkey, fox, rabbit, elephant = 6 types. (e) Most in number: Rabbits (10). (f) Least in number: Elephants (2). Step 3: Q3: A sunny day can happen but it is not certain — it is possible (depends on the weather; not certain). Q4: A cat cannot bark → Impossible. Q5: The bag has candies of many colours, so getting one certain colour is possible but not sure → Less likely. Step 4: Q6: A dog does not meow (meowing is a cat's sound; a dog barks) → Impossible. Q7: As in the book's example, a ball bounces when dropped → More likely. Final answer: 1. (a) Cars (b) Busses (c) 4 (d) 6 (e) 16; 2. (a) 4 (b) 36 (c) 4 (d) 6 (e) Rabbits (f) Elephants; 3. Possible, not certain; 4. Impossible; 5. Less likely; 6. Impossible; 7. More likely.

Unit 10: Mathematics Workbook

Question: Workbook introduction and Exercises 1-4 + Assessment 1 (Addition of two 1-digit numbers) (Unit 10): What is the workbook for and how is it used? What are the things to bring? Ex 1 (Example: 1 + 4): 2 + 4, 3 + 4, 4 + 4, 7 + 0. Ex 2 (vertical addition; Example: 2 + 5): 3 + 5, 1 + 8, 2 + 6, 4 + 0. Ex 3 (make a ten; Example: 9 + 2 = 9 + 1 + 1): 9 + 3, 8 + 3, 7 + 5, 6 + 5. Ex 4 (Example: 9 + 7 = 9 + 1 + 6): 9 + 8, 9 + 9, 2 + 9, 3 + 8. Assessment 1: 1 + 2, 6 + 4, 5 + 0, 5 + 5, 3 + 3, 9 + 3, 6 + 2, 9 + 9, 2 + 7, 3 + 9.

Answer: Step 1: The workbook is designed for students to solve short exercises on a daily basis for enhancing their basic mathematical skills; it is a 15-minutes activity during morning assembly time, one page per day; after 4 days of exercise there is 1 assessment. Things to bring: Pencil, Eraser, Color Pencils (in the pen case), Textbook, Notebook, Slate, Water bottle — all in the School Bag. Step 2: Ex 1: Example 1 + 4 = 5 (count the dots: 1 dot + 4 dots = 5). 2 + 4 = 6; 3 + 4 = 7; 4 + 4 = 8; 7 + 0 = 7 (adding 0 keeps the number the same). Ex 2: Example 2 + 5 = 7. 3 + 5 = 8; 1 + 8 = 9; 2 + 6 = 8; 4 + 0 = 4. Step 3: Ex 3 (make-a-ten method): Example 9 + 2: 9 needs 1 to make ten → 9 + 1 + 1 = 11. 9 + 3 = 9 + 1 + 2 = 12; 8 + 3 = 8 + 2 + 1 = 11; 7 + 5 = 7 + 3 + 2 = 12; 6 + 5 = 6 + 4 + 1 = 11. Ex 4: Example 9 + 7 = 9 + 1 + 6 = 16. 9 + 8 = 17; 9 + 9 = 18; 2 + 9 = 11; 3 + 8 = 11. Step 4: Assessment 1: 1 + 2 = 3; 6 + 4 = 10; 5 + 0 = 5; 5 + 5 = 10; 3 + 3 = 6; 9 + 3 = 12; 6 + 2 = 8; 9 + 9 = 18; 2 + 7 = 9; 3 + 9 = 12. (All these answers match the book's own answer key, printed page 208.) Final answer: Ex 1: 6, 7, 8, 7; Ex 2: 8, 9, 8, 4; Ex 3: 12, 11, 12, 11; Ex 4: 17, 18, 11, 11; Assessment 1: 3, 10, 5, 10, 6, 12, 8, 18, 9, 12.

Question: Workbook Exercises 5-12 and Assessments 2-3 (Addition of 2-digit numbers, with carrying) (Unit 10): Ex 5 (Example: 10 + 1): 10 + 2, 10 + 3, 10 + 4, 10 + 5. Ex 6 (Example: 15 + 3): 14 + 3, 13 + 4, 14 + 4, 13 + 5. Ex 7 (Example: 67 + 2): 32 + 5, 43 + 2, 76 + 1, 51 + 8. Ex 8 (Example: 23 + 15): 24 + 13, 32 + 24, 72 + 26, 55 + 44. Assessment 2: 10 + 7, 16 + 3, 60 + 30, 61 + 3, 45 + 24, 70 + 27, 85 + 13, 13 + 74. Ex 9 (with carrying; Example: 28 + 5, where 8 + 5 = 13): 48 + 6, 25 + 7, 48 + 3, 58 + 3. Ex 10 (Example: 28 + 5): 48 + 9, 38 + 9, 45 + 5, 84 + 9. Ex 11 (Example: 28 + 35, where 8 + 5 = 13 and 1 + 2 + 3 = 6): 39 + 26, 45 + 47, 45 + 35, 39 + 53. Ex 12 (Example: 28 + 35): 48 + 17, 26 + 49, 59 + 36, 62 + 29. Assessment 3: 28 + 5, 48 + 8, 46 + 36, 69 + 21, 25 + 37, 67 + 27.

Answer: Step 1: Method: Add the ones first, then the tens. Ex 5: Example 10 + 1 = 11. 10 + 2 = 12; 10 + 3 = 13; 10 + 4 = 14; 10 + 5 = 15. Ex 6: Example 15 + 3 = 18. 14 + 3 = 17; 13 + 4 = 17; 14 + 4 = 18; 13 + 5 = 18. Ex 7: Example 67 + 2 = 69. 32 + 5 = 37; 43 + 2 = 45; 76 + 1 = 77; 51 + 8 = 59. Step 2: Ex 8: Example 23 + 15: ones 3 + 5 = 8; tens 2 + 1 = 3 → 38. 24 + 13 = 37; 32 + 24 = 56; 72 + 26 = 98; 55 + 44 = 99. Assessment 2: 10 + 7 = 17; 16 + 3 = 19; 60 + 30 = 90; 61 + 3 = 64; 45 + 24 = 69; 70 + 27 = 97; 85 + 13 = 98; 13 + 74 = 87. Step 3: Carrying method: If the ones make 10 or more, make a ten and carry it to the tens. Ex 9: Example 28 + 5: 8 + 5 = 13 → write 3, carry 1; tens 1 + 2 = 3 → 33. 48 + 6: 8 + 6 = 14 → 4, carry 1; 1 + 4 = 5 → 54. 25 + 7: 5 + 7 = 12 → 2, carry 1; 1 + 2 = 3 → 32. 48 + 3 = 51; 58 + 3 = 61. Ex 10: 48 + 9: 8 + 9 = 17 → 57; 38 + 9 = 47; 45 + 5: 5 + 5 = 10 → 50; 84 + 9 = 93. Step 4: Ex 11: Example 28 + 35: 8 + 5 = 13 → 3, carry 1; tens 1 + 2 + 3 = 6 → 63. 39 + 26: 9 + 6 = 15; 1 + 3 + 2 = 6 → 65. 45 + 47: 5 + 7 = 12; 1 + 4 + 4 = 9 → 92. 45 + 35 = 80; 39 + 53 = 92. Ex 12: 48 + 17 = 65; 26 + 49 =

75; 59 + 36 = 95; 62 + 29 = 91. Assessment 3: 28 + 5 = 33; 48 + 8 = 56; 46 + 36: 6 + 6 = 12; 1 + 4 + 3 = 8 → 82; 69 + 21: 9 + 1 = 10; 1 + 6 + 2 = 9 → 90; 25 + 37 = 62; 67 + 27 = 94. (All answers match the book's answer key, printed page 208.) Final answer: Ex 5: 12, 13, 14, 15; Ex 6: 17, 17, 18, 18; Ex 7: 37, 45, 77, 59; Ex 8: 37, 56, 98, 99; Assessment 2: 17, 19, 90, 64, 69, 97, 98, 87; Ex 9: 54, 32, 51, 61; Ex 10: 57, 47, 50, 93; Ex 11: 65, 92, 80, 92; Ex 12: 65, 75, 95, 91; Assessment 3: 33, 56, 82, 90, 62, 94.

Question: Workbook Exercises 13-24 and Assessments 4-6 (Subtraction, with borrowing) (Unit 10): Ex 13 (Example: 8 − 3): 7 − 4, 8 − 6, 9 − 8, 7 − 0. Ex 14 (Example: 5 − 2): 5 − 3, 8 − 1, 6 − 2, 4 − 4. Ex 15 (Example: 10 − 2): 10 − 1, 10 − 3, 10 − 4, 10 − 5. Ex 16 (Example: 12 − 3 = 9, splitting 12 into 10 + 2): 11 − 2, 11 − 3, 12 − 4, 13 − 4. Assessment 4: 5 − 1, 6 − 5, 2 − 0, 9 − 9, 8 − 1, 10 − 5, 11 − 6, 15 − 8. Ex 17 (Example: 18 − 5): 18 − 2, 14 − 1, 12 − 2, 15 − 4. Ex 18 (Example: 40 − 30): 50 − 10, 60 − 50. Ex 19 (Example: 67 − 5): 79 − 6, 58 − 2. Ex 20 (Example: 38 − 15): 28 − 15, 34 − 21. Assessment 5: 19 − 6, 68 − 5, 60 − 20, 45 − 32, 75 − 22. Ex 21 (with borrowing; Example: 34 − 8, where 14 − 8 = 6): 42 − 9, 63 − 7. Ex 22 (Example: 34 − 8): 22 − 6, 71 − 3, 45 − 7, 57 − 9. Ex 23 (Example: 34 − 18): 54 − 17, 45 − 16, 36 − 19, 42 − 13. Ex 24 (Example: 34 − 18): 43 − 25, 71 − 46, 65 − 56, 53 − 49. Assessment 6: 62 − 15, 81 − 16, 56 − 28, 44 − 39, 61 − 18.

Answer: Step 1: Ex 13: Example 8 − 3 = 5 (cross out 3 of the 8 stars, 5 are left). 7 − 4 = 3; 8 − 6 = 2; 9 − 8 = 1; 7 − 0 = 7. Ex 14: Example 5 − 2 = 3. 5 − 3 = 2; 8 − 1 = 7; 6 − 2 = 4; 4 − 4 = 0. Ex 15: Example 10 − 2 = 8. 10 − 1 = 9; 10 − 3 = 7; 10 − 4 = 6; 10 − 5 = 5. Step 2: Ex 16 (splitting method): Example 12 − 3: 12 = 10 + 2; the book's method takes from the 10 part: 12 − 3 = 9. 11 − 2 = 9; 11 − 3 = 8; 12 − 4 = 8; 13 − 4 = 9.

Assessment 4: 5 − 1 = 4; 6 − 5 = 1; 2 − 0 = 2; 9 − 9 = 0; 8 − 1 = 7; 10 − 5 = 5; 11 − 6 = 5; 15 − 8 = 7. Step 3: Ex 17: Example 18 − 5: ones 8 − 5 = 3; tens 1 → 13. 18 − 2 = 16; 14 − 1 = 13; 12 − 2 = 10; 15 − 4 = 11. Ex 18: Example 40 − 30 = 10 (4 tens − 3 tens = 1 ten). 50 − 10 = 40; 60 − 50 = 10. Ex 19: Example 67 − 5 = 62. 79 − 6 = 73; 58 − 2 = 56. Ex 20: Example 38 − 15: ones 8 − 5 = 3; tens 3 − 1 = 2 → 23. 28 − 15 = 13; 34 − 21 = 13. Assessment 5: 19 − 6 = 13; 68 − 5 = 63; 60 − 20 = 40; 45 − 32 = 13; 75 − 22 = 53. Step 4: Borrowing method: If the ones are too small, borrow 1 ten (10 ones) from the tens. Ex 21: Example 34 − 8: 4 cannot take 8 → borrow: 14 − 8 = 6; tens 3 loses 1 → 2 → 26. 42 − 9: 12 − 9 = 3; tens 3 → 33. 63 − 7: 13 − 7 = 6; tens 5 → 56. Ex 22: 22 − 6 = 16; 71 − 3 = 68; 45 − 7 = 38; 57 − 9 = 48. Step 5: Ex 23: Example 34 − 18: 14 − 8 = 6; tens 2 − 1 = 1 → 16. 54 − 17: 14 − 7 = 7; 4 − 1 = 3 → 37. 45 − 16 = 29; 36 − 19 = 17; 42 − 13 = 29. Ex 24: 43 − 25 = 18; 71 − 46 = 25; 65 − 56 = 9; 53 − 49 = 4. Assessment 6: 62 − 15 = 47; 81 − 16 = 65; 56 − 28 = 28; 44 − 39 = 5; 61 − 18 = 43. (All answers match the book's answer key, printed page 208.) Final answer: Ex 13: 3, 2, 1, 7; Ex 14: 2, 7, 4, 0; Ex 15: 9, 7, 6, 5; Ex 16: 9, 8, 8, 9; Assessment 4: 4, 1, 2, 0, 7, 5, 5, 7; Ex 17: 16, 13, 10, 11; Ex 18: 40, 10; Ex 19: 73, 56; Ex 20: 13, 13; Assessment 5: 13, 63, 40, 13, 53; Ex 21: 33, 56; Ex 22: 16, 68, 38, 48; Ex 23: 37, 29, 17, 29; Ex 24: 18, 25, 9, 4; Assessment 6: 47, 65, 28, 5, 43.

Workbook Exercises 25-32 and Assessments 7-8 (Multiplication within 100) (Unit 10): Ex 25 (Example: 3 plates with 2 bananas each: 2 + 2 + 2 = 2 × 3): Exercise — 4 plates with 2 bananas each — what is the total number of bananas? Ex 26 (Example: 2 × 3): 2 × 1 to 2 × 10. Ex 27 (Example: 5 × 4): 5 × 1 to 5 × 10. Ex 28 (Example: 3 × 4): 3 × 1 to 3 × 10. Assessment 7: 2 × 4, 2 × 7, 5 × 5, 3 × 10, 2 × 6, 5 × 6, 5 × 4, 2 × 8, 5 × 9, 3 × 7. Ex 29 (Example: 4 × 4): 4 × 1 to 4 × 10. Ex 30 (Example: 10 × 4): 10 × 1 to

10 × 10. Ex 31 (vertical multiplication; Example: 4 × 3): 4 × 4, 5 × 3. Ex 32 (Example: 10 = 2 × 5 = 5 × 2): 15 = _____ × _____ = _____ × _____; 12 = _____ × _____ = _____ × _____. Assessment 8: 4 × 7, 10 × 9, 4 × 6, 4 × 5 (vertical), 6 = _____ × _____ = _____ × _____.

Answer: 3;, 3, 4, 4, 3 (as in the book's answer key)., 2, 3, 3, 2. (All answers match the book's answer key, printed page 208.) Final answer: Ex 25: 8 bananas; Ex 26: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20; Ex 27: 5, 10, 15, 20, 25, 30, 35, 40, 45, 50; Ex 28: 3, 6, 9, 12, 15, 18, 21, 24, 27, 30; Assessment 7: 8, 14, 25, 30, 12, 30, 20, 16, 45, 21; Ex 29: 4, 8, 12, 16, 20, 24, 28, 32, 36, 40; Ex 30: 10, 20, 30, 40, 50, 60, 70, 80, 90, 100; Ex 31: 16, 15; Ex 32: 15 = 3 × 5 = 5 × 3; 12 = 3 × 4 = 4 × 3; Assessment 8: 28, 90, 24, 20; 6 = 2 × 3 = 3 × 2.

Workbook Exercises 33-36, Assessment 9 and the final Assessments (Division and mixed) (Unit 10): Ex 33 (Example: 4 girls share 12 balls equally: 12 ÷ 4): Exercise — 4 girls share 8 balls equally: _____ ÷ _____ = _____. Ex 34 (Example: 12 ÷ 4, because 4 × 3 = 12): 8 ÷ 4 (with 4 × _____ = 8). Ex 35 (Example: 12 ÷ 4): 20 ÷ 4, 10 ÷ 2, 9 ÷ 3, 16 ÷ 4. Ex 36 (long division; Example: 15 ÷ 5): 12 ÷ 3 (in bracket form), 16 ÷ 4 (in bracket form). Assessment 9: 1. 4 × 3 = 3 × _____; 2. 20 ÷ 5 (with 5 × _____ = 20); 3. 25 ÷ 5; 4. 12 ÷ 4; 5. 16 ÷ 4 (bracket). Assessment Addition: 6 + 2, 9 + 8, 62 + 4, 78 + 7, 48 + 21, 57 + 26. Assessment Addition 2: 37 + 8, 45 + 7, 48 + 12, 15 + 45, 59 + 36, 62 + 29. Assessment Subtraction: 4 − 2, 14 − 8, 87 − 6, 54 − 9, 86 − 23, 71 − 17. Assessment Subtraction 2: 80 − 20, 66 − 8, 52 − 9, 81 − 45, 94 − 85, 51 − 44. Assessment Multiplication: 4 × 4, 5 × 10, 3 × 9, 18 = _____ × _____ = _____ × _____, 5 × 8 (vertical). Assessment Division: 20 ÷ 4, 4 × _____ = 20, 10 ÷ 2, 25 ÷ 5, 12 ÷ 4 (bracket), 16

÷ 4 (bracket). Level Assessment 1 (Example: 6 − 2 = 4, 6 ÷ 2 = 3, 6 × 2 = 12, 6 + 2 = 8): 6 × 3, 6 − 3, 6 + 3, 6 ÷ 3, 8 × 2, 8 − 2, 8 + 2, 8 ÷ 2, 9 − 3, 9 + 3, 9 ÷ 3, 9 × 3. Level Assessment 2: 89 + 7, 47 + 48, 83 − 5, 76 − 48, 7 × 7, 8 × 9, 32 ÷ 4 (bracket), 63 ÷ 7 (bracket).

Answer: each, girl, gets, 2, 4 (both 12)., 4, 3, 6, 6, 3 (as in the key; 2 × 9 = 9 × 2 also works);, 5

Unit 11: Numbers (Urdu Edition Exercises)

(Urdu edition — Ordinal Numbers) Activity 1: Give the finishing position of each car in the car race. Activity 2: Fill the blanks in the frogs' row: 1st, _____, 3rd, _____, _____, 6th, _____, 8th, _____, 10th. Activity 3: Read the order of the children at the bus and the school, from 1st to 20th. Activity 4: Write the ordinal numbers 1st to 20th in words.

Answer: given ones — after 1st, comes 2nd, after 3rd come, 4th and 5th, after 6th, comes 7th, after 8th comes, 9th. Final answer: 2nd, 4th, 5th, 7th, 9th.

Question: (Urdu edition) Exercise 1: Look at the animal race picture: Horse 1st, Deer 2nd, Cheetah 3rd, Dog 4th, Lion 5th, Donkey 6th, Cat 7th, Camel 8th, Rabbit 9th, Zebra 10th, Elephant 11th, Cow 12th. (1) Tick the correct position of each animal: Cat (options 6th/7th/8th), Dog (3rd/4th/5th), Lion (5th/6th/7th), Donkey (6th/7th/8th). (2) Write the ordinal number under each pictured animal.

Answer: Step 1: Read the position label above each animal in the picture. (1) Cat = 7th (✓), Dog = 4th (✓), Lion = 5th (✓), Donkey = 6th (✓). (2) Horse 1st, Cat 7th, Dog 4th, Elephant 11th; Deer 2nd, Cheetah 3rd, Camel 8th, Zebra 10th; Rabbit 9th, Lion 5th, Donkey 6th, Cow 12th. Final answer: all positions are as listed above.

Question: (Urdu edition) Exercise 1 (3): Write the ordinal numbers for the vehicles in the traffic picture (given: yellow car 1st, bus 5th, pink car 9th, fire truck 11th, red sports car 14th, big truck 17th, loader 20th). (4) In the table, write each vehicle's order in numbers and words: red car (example: 2nd), bus, fire truck, big truck, loader, orange car, yellow car, green van.

Answer: Step 1: Count the vehicles one by one starting from the traffic

light. (3) 1st yellow car, 2nd red car, 3rd blue pickup, 4th green van, 5th bus, 6th jeep, 7th black car, 8th red car, 9th pink car, 10th blue car, 11th fire truck, 12th orange car, 13th grey car, 14th red sports car, 15th purple tanker, 16th red truck, 17th big truck, 18th bicycle, 19th dump truck, 20th loader. (4) Red car 2nd, Bus 5th, Fire truck 11th, Big truck 17th, Loader 20th, Orange car 12th, Yellow car 1st, Green van 4th.

Question: (Urdu edition) Number names: the Urdu edition teaches number names from 1 to 100 (the English edition goes to 50). Write the Urdu names for 51 to 100. Exercise 2 (1): Write these numbers in (Urdu) words: 35 (example), 75, 100, 48, 29, 93, 67, 89, 47, 79.

Answer: The Urdu names for 51–100 are given in full in the Urdu answer above. Exercise 2 (1): 35 = Paintis (example), 75 = Pachhattar, 100 = Sau, 48 = Arhtalis, 29 = Untis, 93 = Tiranway, 67 = Starsath, 89 = Nawasi, 47 = Saintalis, 79 = Unasi (Urdu words written in Urdu script in the Urdu answer).

Question: (Urdu edition) Exercise 2 (2): Match the Urdu number words to the correct numbers: words — Chalis, Satawan, Chhiyanway, Unsath, Pachis, Chonsath, Ikyasi, Battis, Bahattar, Nawasi; numbers — 64, 81, 40, 32, 57, 59, 96, 89, 25, 72.

Answer: Step 1: Read each word and recognise its number. Step 2: Make the pairs — Chalis = 40, Satawan = 57, Chhiyanway = 96 (shown with an arrow in the book), Unsath = 59, Pachis = 25, Chonsath = 64, Ikyasi = 81, Battis = 32, Bahattar = 72, Nawasi = 89. Final answer: 40, 57, 96, 59, 25, 64, 81, 32, 72, 89.

Question: (Urdu edition) Place Value: Activity: Write the place value of each digit: (1) 36 (example: 6 ones, 3 tens), (2) 54, (3) 96, (4) 100. Examples: read 126 and 619 from the blocks.

Answer: (2) 54: Step 1: the ones digit 4 = 4 ones; Step 2: the tens digit 5

= 5 tens. (3) 96: 6 ones, 9 tens. (4) 100: 0 ones, 0 tens, 1 hundred. Example 126: 1 hundred flat, 2 ten rods, 6 ones — one hundred twenty-six. Example 619: 6 hundreds, 1 ten, 9 ones — six hundred nineteen.

Question: (Urdu edition) Exercise 3 (1): Count the blocks and write them in the hundreds, tens and ones boxes (three rows). (2) Write the digits of these numbers by place value: 208 (example), 180, 370, 500, 605, 347, 800, 990.

Answer: (1) Step 1: Count the big square flats as hundreds, the rods as tens and the small squares as ones. Row 1: 2 hundreds, 3 tens, 7 ones = 237. Row 2: 4 hundreds, 2 tens, 8 ones = 428. Row 3: 6 hundreds, 5 tens, 0 ones = 650. (2) 180 = 1 hundred, 8 tens, 0 ones; 370 = 3, 7, 0; 500 = 5, 0, 0; 605 = 6, 0, 5; 347 = 3, 4, 7; 800 = 8, 0, 0; 990 = 9, 9, 0.

Question: (Urdu edition) Exercise 3 (3): Count the abacus beads and write the number: (1) example 423, (2), (3), (4), (5). Exercise 4: Write the place value of the coloured (middle) digit in each number: (1) 342 (example: 4 = 40), (2) 946, (3) 964, (4) 570, (5) 189, (6) 501, (7) 333, (8) 500, (9) 835, (10) 698.

Answer: Exercise 3 (3): Step 1: Count the beads on each rod — hundreds, tens, ones. (2) 2 hundreds, 5 tens, 2 ones = 252. (3) 4 hundreds, 0 tens, 4 ones = 404. (4) 3 hundreds, 6 tens, 5 ones = 365. (5) 1 hundred, 2 tens, 8 ones = 128. Exercise 4: the middle digit is always in the tens place: (2) 4 = 40, (3) 6 = 60, (4) 7 = 70, (5) 8 = 80, (6) 0 = 0, (7) 3 = 30, (8) 0 = 0, (9) 3 = 30, (10) 9 = 90.

Question: (Urdu edition) Comparing: Activity: Colour the smaller number red in each pair: (1) 21 and 61; 28 and 24 (example), (2) 149 and 184; 169 and 165. Exercise 5: (1) Colour the box of the smaller number: 53/74, 36/44, 50/60, 24/34, 165/213, 405/210, 314/624, 510/810. (2) Colour the

box of the bigger number: 36/14, 25/46, 78/96, 64/54, 213/423, 167/314, 210/123, 718/218. (3) Write the bigger number in the blue box and the smaller in the red box: 26/70, 13/65, 36/81, 42/12, 29/79, 63/50, 84/24, 93/56, 412/360, 210/910, 244/356, 112/402.

Answer: Step 1: Compare hundreds first, then tens, then ones. Activity: (1) 21 is smaller; 24 is smaller (example). (2) 149 is smaller; 165 is smaller. Exercise 5 (1) smaller numbers: 53, 36, 50, 24, 165, 210, 314, 510. (2) bigger numbers: 36, 46, 96, 64, 423, 314, 210, 718. (3) bigger/smaller: 70/26, 65/13, 81/36, 42/12, 79/29, 63/50, 84/24, 93/56, 412/360, 910/210, 356/244, 402/112.

(Urdu edition) Activity 3: Count in order and fill the blanks: 201, 202, _____, ..., 210; 351, 352, ..., 360; 411, ..., 420; 561, ..., 570; 721, ..., 730; 881, ..., 890. Exercise 6: Complete the 10 numbers in each row: rows starting 300, 350, 430, 451, 500, 532, 661, 683, 748, 777, 816, 894, 905, 990.

Answer: order).

Question: (Urdu edition) Ascending and Descending Order: Activity 1: Tick the groups in ascending order: (8, 23, 100) example; (100, 23, 8); (23, 100, 8); (73, 42, 21, 9); (9, 21, 42, 73); (42, 21, 9, 73); (132, 112, 182, 142); (182, 142, 132, 112); (112, 132, 142, 182). Activity 2: Tick the groups in descending order: (13, 73, 93); (73, 93, 30); (93, 73, 13) example; (34, 79, 85, 58); (85, 79, 58, 34); (34, 58, 79, 85); (165, 143, 132, 156); (132, 143, 156, 165); (165, 156, 143, 132). Exercise 7: (a) Put (✓) on ascending groups: (1) 7, 19, 41, 50 (2) 45, 58, 74, 92 (3) 54, 47, 32, 25 (4) 100, 104, 111, 120 (5) 174, 163, 154, 148 (6) 224, 250, 216, 232. (b) Put (✗) on descending groups: (1) 65, 57, 88, 49 (2) 35, 53, 76, 89 (3) 94, 72, 61, 50 (4) 120, 111, 101, 94 (5) 321, 382, 365, 397 (6) 481, 474, 467, 459.

Answer: Step 1: Ascending = smallest to biggest; descending = biggest to smallest. Activity 1 (✓): (8, 23, 100), (9, 21, 42, 73), (112, 132, 142, 182) — the others are not ascending. Activity 2 (✓): (93, 73, 13), (85, 79, 58, 34), (165, 156, 143, 132). Exercise 7 (a) (✓): (1), (2), (4) — (3) and (5) are descending, (6) is mixed. (b) descending (✗): (3), (4), (6) — (1) is mixed, (2) is ascending, (5) is mixed.

Question: (Urdu edition) Exercise 8: Count back 10 steps from the given number and write (10 numbers in each row): (1) from 80 (example row 80–71), (2) from 90, (3) from 85, (4) from 110, (5) from 220. Exercise 9: Write the numbers in ascending and descending order: ascending — (36, 41, 19, 56), (180, 163, 131, 190), (390, 330, 380, 350); descending — (35, 42, 74, 89), (208, 244, 218, 256), (483, 428, 455, 419). Also the activity: balloons (131, 129, 145, 118) ascending and (167, 149, 171, 138) descending.

Answer: Exercise 8: Step 1: Take away 1 each time and write 10 numbers. (2) 90, 89, 88, 87, 86, 85, 84, 83, 82, 81. (3) 85, 84, 83, 82, 81, 80, 79, 78, 77, 76. (4) 110, 109, 108, 107, 106, 105, 104, 103, 102, 101. (5) 220, 219, 218, 217, 216, 215, 214, 213, 212, 211. Exercise 9 ascending: 19, 36, 41, 56; 131, 163, 180, 190; 330, 350, 380, 390. Descending: 89, 74, 42, 35; 256, 244, 218, 208; 483, 455, 428, 419. Activity: ascending 118, 129, 131, 145; descending 171, 167, 149, 138.

(Urdu edition) Exercise 10: Count in tens and fill the blanks: (1) 10, _____, 30, 40, _____, _____ (2) 70, _____, _____, 100, _____, 120 (3) 100, _____, 120, _____, _____, 150 (4) 250, _____, _____, _____, 290, _____ (5) 400, _____, _____, 430, _____, _____ (6) 560, 570, _____, _____, _____, 610. Exercise 11: Count in hundreds: (1) 100, _____, _____, _____, 500 (2) from 200 (3) from 300 (4) from 400 (5) from

500 — 5 numbers in each strip.

Answer: 20,, 50,, 60., 80,, 90,, 110, 120., 110,, 130,, 140, 150., 260,, 270,, 280,, 300., 410,, 420,, 440,, 450., 580,, 590,, 600,, 200,, 300,, 400, 500.

Question: (Urdu edition) Activity: Write the biggest and the smallest number: Group 1 example (35, 41, 17, 25, 14, 52 → smallest 14, biggest 52); Group 2: 112, 132, 141, 162, 150, 126. Exercise 12: (1) 215, 250, 227, 255, 245, 232 (2) 350, 320, 310, 380, 400, 370 (3) 312, 270, 350, 279, 225, 360 (4) 511, 422, 410, 505, 550, 401. Also the 999 activity: count the blocks and write the number (example 525): rows 2, 3, 4.

Answer: Step 1: Compare the hundreds of all the numbers, then the tens. Group 2: smallest 112, biggest 162. Exercise 12: (1) smallest 215, biggest 255. (2) smallest 310, biggest 400. (3) smallest 225, biggest 360. (4) smallest 401, biggest 550. 999 activity: Row 2: 7 hundreds, 5 tens, 3 ones = 753. Row 3: 9 hundreds, 9 tens, 5 ones = 995. Row 4: 9 hundreds, 9 tens, 9 ones = 999; add 1 to 999 and we get 1000 — the first four-digit number.

Question: (Urdu edition — Fractions) Activity 1: Put (✓) on shapes showing equal parts and (✗) on unequal ones: glasses (example ✓), umbrella (example ✗), bead string (parts of 3, 4, 3), circle (4 equal parts), pink bar (5 equal parts), shirt (2 equal parts). Activity 2: How many equal parts does each shape show: triangle (example 3), orange circle (2 unequal parts), blue rectangle (6 parts), green bar (5 parts).

Answer: Activity 1: Step 1: Check whether all the parts are the same size. Bead string: the parts are 3, 4, 3 — not equal, (✗). Circle: 4 equal parts — (✓). Pink bar: 5 equal parts — (✓). Shirt: 2 equal parts — (✓). Activity 2: Triangle = 3 equal parts; the orange circle's 2 parts are not equal, so equal parts = 0; rectangle = 6 equal parts; bar = 5 equal parts.

Question: (Urdu edition — Fractions) Half, one-third and one-quarter:

Shade one equal part of each shape and name the fraction (for the circle and the square: half = 1/2, one-third = 1/3, one-quarter = 1/4). Read the given examples: square with 1/4 shaded = one-quarter; bar with 1/3 = one-third; circle with 1/2 = one-half. Activities 1–3: Colour one half / one third / one quarter of each shape (triangles, rectangles, bars, circles, hexagon, big triangle).

Answer: Step 1: See the shape divided into equal parts; Step 2: colour only one of those parts. In the half shapes the coloured part = 1/2; in the third shapes = 1/3; in the quarter shapes = 1/4. Examples: square in 4 equal parts, 1 coloured = 1/4 (one-quarter); bar in 3 parts, 1 coloured = 1/3 (one-third); circle in 2 parts, 1 coloured = 1/2 (one-half).

Question: (Urdu edition — Fractions) Writing 1/2, 1/3 and 1/4 as symbols: Colour 1/2 (one half), 1/3 and 1/4 of each shape (triangles, circles, rectangles, bars, hexagon-like shapes). Then colour according to the given fraction: circle 1/2, square 1/4, diamond 1/2, diamond (4 parts) 1/4, bar (3 parts) 1/3, square (diagonal) 1/2, triangle 1/2, bar (4 parts) 1/4, circle (3 parts) 1/3, triangle (3 parts) 1/3, hexagon 1/2, circle (4 parts) 1/4.

Answer: Step 1: The bottom number of the fraction (denominator) tells how many equal parts there are in all; Step 2: the top number is 1, so colour only one part. In every shape the coloured part equals the given fraction: 1 out of 2 for 1/2, 1 out of 3 for 1/3, 1 out of 4 for 1/4.

Question: (Urdu edition — Fractions) Read the unit-fraction table from 1/2 to 1/12. Exercise 13: (a) Tick the correct unit fraction for the coloured part: (1) circle in 8 parts, 1 coloured (options 1/2, 1/7, 1/8) (2) square grid in 9 parts, 1 coloured (1/7, 1/9, 1/12) (3) big triangle in 4 small triangles, 1 coloured (1/3, 1/7, 1/4) (4) hexagon in 6 triangles, 1 coloured (1/8, 1/6, 1/5). (b) Colour the given unit fractions: (1) 1/3 bar of 3 parts (2) 1/4 grid of 4 (3) 1/8 grid of 8 (4) 1/2 bar of 2 (5) 1/6 grid of 6 (6) 1/10 grid of 10.

Answer: (a) Step 1: Count all the equal parts; Step 2: the coloured parts are 1, so the fraction = 1/(total parts). (1) 1/8 (✓ — given in the book). (2) 1/9. (3) 1/4. (4) 1/6. (b) Colour only one part in each shape: (1) 1 out of 3 = 1/3; (2) 1 out of 4 = 1/4; (3) 1 out of 8 = 1/8; (4) 1 out of 2 = 1/2; (5) 1 out of 6 = 1/6; (6) 1 out of 10 = 1/10.

Question: (Urdu edition — Fractions) Read the non-unit fraction table: 2/3 (two-thirds — 2 out of 3 coloured), 3/4 (three-quarters — 3 out of 4), 3/8 (three-eighths — 3 out of 8), 5/6 (five-sixths — 5 out of 6). Activity: Tick the fraction shown by the coloured part: (1) bar of 3 parts, 2 green (2/3, 3/4, 1/5) (2) square grid of 4 parts, 3 green (4/9, 3/4, 7/9) (3) circle of 5 parts, 4 green (3/9, 5/7, 4/5) (4) circle of 6 parts, 5 green (1/10, 5/6, 1/12). Exercise 14: Match the shapes to their fractions: (1) bar of 5, 4 yellow (2) grid of 6, 5 green (example) (3) grid of 4, 3 yellow (4) grid of 8, 7 blue (5) bar of 7, 6 pink (6) triangular shape of 3, 2 blue (7) grid of 10, 9 purple. Fractions: 9/10, 3/4, 2/3, 4/5, 5/6, 7/8, 6/7.

Answer: Step 1: Write the coloured parts on top (numerator) and all the parts below (denominator). Activity: (1) 2/3 (✓ in the book). (2) 3/4. (3) 4/5. (4) 5/6. Exercise 14: (1) 4/5; (2) 5/6 (example); (3) 3/4; (4) 7/8; (5) 6/7; (6) 2/3; (7) 9/10.

Unit 12: Algebraic Operations (Urdu Edition Exercises)

(Urdu edition — Addition) Activity 1: Count and tell the answer: apples _____ + _____ = _____ (in both baskets). Activity 2: Count the pictures and write: _____ chicks + _____ = _____; _____ balloons + _____ = _____; _____ balls + _____ = _____; _____ stars + _____ = _____. Activity 3: Colour the flower (colour key: 10 brown, 11 blue, 12 green, 13 black, 14 yellow, 15 red, 16 pink, 17 sky blue): centre 7 + 3; petals 9 + 8, 8 + 7, 6 + 4, 8 + 5, 9 + 6, 7 + 4, 5 + 9, 8 + 8; leaves 5 + 7, 6 + 6.

Answer: 4 apples; Step 2: count them together: 5, 4, 9 apples., 5, 11;, 3, 10;, 5, 14;, 8, 15.

Question: (Urdu edition) Exercise 15: (1) Solve in columns: 7 + 3 (example = 10), 8 + 2, 9 + 5, 8 + 3, 7 + 5, 6 + 6, 6 + 5, 8 + 7, 9 + 3. (2) Write: 7 + 7 (example = 14), 4 + 9, 5 + 5, 2 + 9, 7 + 8, 9 + 4.

Answer: Step 1: Add the two ones together. (1) 8 + 2 = 10, 9 + 5 = 14, 8 + 3 = 11, 7 + 5 = 12, 6 + 6 = 12, 6 + 5 = 11, 8 + 7 = 15, 9 + 3 = 12. (2) 4 + 9 = 13, 5 + 5 = 10, 2 + 9 = 11, 7 + 8 = 15, 9 + 4 = 13.

Question: (Urdu edition) Exercise 16: (a) Solve with carrying: (1) 15 pencils + 6 (example = 21), (2) 28 coloured pencils + 9, (3) 34 blocks + 9, (4) 19 sticks + 10. Column sums (5)–(16): 82 + 9 (worked = 91), 36 + 5, 47 + 7, 66 + 4, 49 + 1, 76 + 8, 56 + 9, 48 + 6, 88 + 8, 16 + 5, 34 + 6, 68 + 4. (b) Match to the correct answers: 19 + 3 (example → 22), 11 + 9, 83 + 9, 76 + 7, 47 + 5; answers: 20, 22, 52, 83, 92.

Answer: Step 1: Add the ones first; Step 2: if the ones make 10 or more, carry 1 ten into the tens. (a) (2) 28 + 9: ones 8 + 9 = 17 → write 7, carry 1; tens 2 + 1 = 3 → 37. (3) 34 + 9 = 43. (4) 19 + 10 = 29. (5)–(16): 36 + 5

= 41, 47 + 7 = 54, 66 + 4 = 70, 49 + 1 = 50, 76 + 8 = 84, 56 + 9 = 65, 48 + 6 = 54, 88 + 8 = 96, 16 + 5 = 21, 34 + 6 = 40, 68 + 4 = 72. (b) 11 + 9 = 20, 83 + 9 = 92, 76 + 7 = 83, 47 + 5 = 52.

Question: (Urdu edition) Exercise 17: (a) Solve: 18 + 12, 25 + 19, 35 + 46, 68 + 24, 39 + 58, 47 + 28, 44 + 47, 76 + 14, 37 + 58, 49 + 35, 57 + 25, 66 + 27. (b) 19 + 26, 17 + 23, 63 + 18, 57 + 34, 39 + 21, 15 + 56. Exercise 18 (word problems; example: Rashid has 18 eggs + 9 more = 27): (1) Rashida got 28 marks in English and 18 in Mathematics — total marks? (2) Hira has 49 flower buds and Hina has 37 — how many altogether? (3) Zohaib had 65 pencils and bought 18 more — total now? (4) Naeem scored 64 runs in the first match and 27 in the second — total runs? (5) Anwar bought 22 eggs in one week and 29 in the next — total? (6) 36 people came to the bazaar in one hour and 27 more in the next hour — total? (7) Sania saved 32 rupees one day and 19 rupees the next — total saving?

Answer: Exercise 17: Step 1: Add the ones (carry if needed); Step 2: add the tens. (a) 18 + 12 = 30, 25 + 19 = 44, 35 + 46 = 81, 68 + 24 = 92, 39 + 58 = 97, 47 + 28 = 75, 44 + 47 = 91, 76 + 14 = 90, 37 + 58 = 95, 49 + 35 = 84, 57 + 25 = 82, 66 + 27 = 93. (b) 19 + 26 = 45, 17 + 23 = 40, 63 + 18 = 81, 57 + 34 = 91, 39 + 21 = 60, 15 + 56 = 71. Exercise 18: (1) 28 + 18 = 46 marks. (2) 49 + 37 = 86 buds. (3) 65 + 18 = 83 pencils. (4) 64 + 27 = 91 runs. (5) 22 + 29 = 51 eggs. (6) 36 + 27 = 63 people. (7) 32 + 19 = 51 rupees.

Question: (Urdu edition) Exercise 19: (a) 3-digit + 1-digit (no carrying): 195 + 3, 174 + 4, 363 + 6, 437 + 2, 531 + 7, 710 + 9, 632 + 7, 216 + 1, 631 + 8, 900 + 7, 812 + 6, 730 + 5. (b) 100 + 2, 102 + 3, 305 + 4, 100 + 4, 601 + 7, 702 + 6. (c) Choose the correct answer: 263 + 2, 151 + 7, 373 + 6, 538 + 1, 462 + 5, 734 + 3, 670 + 4, 514 + 5, 625 + 3.

Answer: Step 1: Only the ones change; the hundreds and tens stay the same. (a) 195 + 3 = 198, 174 + 4 = 178, 363 + 6 = 369, 437 + 2 = 439, 531 + 7 = 538, 710 + 9 = 719, 632 + 7 = 639, 216 + 1 = 217, 631 + 8 = 639, 900 + 7 = 907, 812 + 6 = 818, 730 + 5 = 735. (b) 100 + 2 = 102, 102 + 3 = 105, 305 + 4 = 309, 100 + 4 = 104, 601 + 7 = 608, 702 + 6 = 708. (c) 263 + 2 = 265, 151 + 7 = 158, 373 + 6 = 379, 538 + 1 = 539, 462 + 5 = 467, 734 + 3 = 737, 670 + 4 = 674, 514 + 5 = 519, 625 + 3 = 628.

Question: (Urdu edition) Exercise 20: (a) 3-digit + 2-digit (example: 322 + 34 = 356): 125 + 14, 375 + 23, 285 + 13, 211 + 37, 289 + 10, 331 + 68, 700 + 99, 432 + 44, 650 + 40, 403 + 82, 305 + 82, 518 + 71. (b) Put (✓) for true and (✗) for false: (1) 703 + 20 = 723 (2) 119 + 50 = 159 (3) 237 + 22 = 359 (4) 333 + 40 = 370 (5) 463 + 12 = 475 (6) 661 + 38 = 699 (7) 705 + 40 = 705 (8) 777 + 20 = 790 (9) 850 + 35 = 885 (10) 809 + 90 = 890.

Answer: (a) Step 1: ones, Step 2: tens, Step 3: hundreds. 125 + 14 = 139, 375 + 23 = 398, 285 + 13 = 298, 211 + 37 = 248, 289 + 10 = 299, 331 + 68 = 399, 700 + 99 = 799, 432 + 44 = 476, 650 + 40 = 690, 403 + 82 = 485, 305 + 82 = 387, 518 + 71 = 589. (b) (1) ✓ 723 is correct. (2) ✗ correct answer 169. (3) ✗ correct 259. (4) ✗ correct 373. (5) ✓ 475. (6) ✓ 699. (7) ✗ correct 745. (8) ✗ correct 797. (9) ✓ 885. (10) ✗ correct 899.

Question: (Urdu edition) Exercise 21: 3-digit + 3-digit (example: 346 + 143 = 489): 108 + 100, 306 + 200, 230 + 205, 183 + 114, 215 + 151, 333 + 222, 550 + 330, 302 + 507, 725 + 134, 673 + 225, 577 + 103, 837 + 162. Exercise 22 (word problems; example: Adil 242 balloons + friend's 123 = 365): (1) Ahmed has 215 story books, Fatima has 271 — total? (2) A shopkeeper has 153 candies in one basket and 244 in another — total? (3) Asma has 331 cards and her sister has 625 — total? (4) Shahid made 523

kites, Babar made 113 more — total? (5) One village has 450 orange trees and another 140 — total? (6) Kanwal earned 130 rupees in the morning and 115 in the evening — total?

Answer: Exercise 21: Step 1: ones; Step 2: tens; Step 3: add the hundreds. 108 + 100 = 208, 306 + 200 = 506, 230 + 205 = 435, 183 + 114 = 297, 215 + 151 = 366, 333 + 222 = 555, 550 + 330 = 880, 302 + 507 = 809, 725 + 134 = 859, 673 + 225 = 898, 577 + 103 = 680, 837 + 162 = 999. Exercise 22: (1) 215 + 271 = 486 books. (2) 153 + 244 = 397 candies. (3) 331 + 625 = 956 cards. (4) 523 + 113 = 636 kites. (5) 450 + 140 = 590 trees. (6) 130 + 115 = 245 rupees.

Question: (Urdu edition) Exercise 23: (a) 3-digit + 1-digit with carrying (example: 194 + 7 = 201): 397 + 8, 299 + 7, 496 + 9, 396 + 7, 698 + 5, 797 + 7, 899 + 3, 494 + 8, 395 + 6, 593 + 7, 999 + 9, 591 + 9. (b) 292 + 8, 397 + 6, 896 + 4, 599 + 9, 398 + 7, 494 + 7, 598 + 9, 498 + 9, 497 + 5, 595 + 5.

Answer: Step 1: Add the ones — if they make 10 or more, carry into the tens/hundreds; Step 2: add the tens, then the hundreds. Example: 194 + 7: ones 4 + 7 = 11 → write 1, carry 1; tens 9 + 1 = 10 → write 0, carry 1; hundreds 1 + 1 = 2 → 201. (a) 397 + 8 = 405, 299 + 7 = 306, 496 + 9 = 505, 396 + 7 = 403, 698 + 5 = 703, 797 + 7 = 804, 899 + 3 = 902, 494 + 8 = 502, 395 + 6 = 401, 593 + 7 = 600, 999 + 9 = 1008, 591 + 9 = 600. (b) 292 + 8 = 300, 397 + 6 = 403, 896 + 4 = 900, 599 + 9 = 608, 398 + 7 = 405, 494 + 7 = 501, 598 + 9 = 607, 498 + 9 = 507, 497 + 5 = 502, 595 + 5 = 600.

Question: (Urdu edition) Exercise 24: (a) 3-digit + 2-digit with carrying (example: 196 + 45 = 241; (1) 195 + 39 worked = 234): (2) 287 + 37, (3) 654 + 68, (4) 358 + 76, (5) 499 + 55, (6) 493 + 18, (7) 999 + 41, (8) 845 + 87, (9) 785 + 28, (10) 667 + 36, (11) 882 + 38, (12) 792 + 99, (13) 495

+ 15, (14) 371 + 29, (15) 598 + 27, (16) 283 + 58. (b) Match: 273 + 27 (example → 300), 695 + 98, 633 + 67, 423 + 79, 195 + 29, 125 + 75, 381 + 29, 347 + 97, 397 + 17, 145 + 65; answers: 300, 502, 700, 410, 200, 224, 210, 414, 793, 444.

Answer: Step 1: Add the ones and carry; Step 2: add the tens and carry; Step 3: add the hundreds. (2) 287 + 37 = 324, (3) 654 + 68 = 722, (4) 358 + 76 = 434, (5) 499 + 55 = 554, (6) 493 + 18 = 511, (7) 999 + 41 = 1040, (8) 845 + 87 = 932, (9) 785 + 28 = 813, (10) 667 + 36 = 703, (11) 882 + 38 = 920, (12) 792 + 99 = 891, (13) 495 + 15 = 510, (14) 371 + 29 = 400, (15) 598 + 27 = 625, (16) 283 + 58 = 341. (b) 695 + 98 = 793, 633 + 67 = 700, 423 + 79 = 502, 195 + 29 = 224, 125 + 75 = 200, 381 + 29 = 410, 347 + 97 = 444, 397 + 17 = 414, 145 + 65 = 210.

Question: (Urdu edition) Exercise 25: 3-digit + 3-digit with carrying (examples: 234 + 679 = 913 and 234 + 386 = 620 — note: the book's heading for the first example prints 243 but its working uses 234): 565 + 369, 287 + 327, 395 + 378, 729 + 175, 777 + 199, 555 + 395, 633 + 267, 468 + 338, 319 + 691, 392 + 589, 663 + 258, 831 + 179. Exercise 26 (word problems; examples: Salma 318 + 219 = 537 pages; Ali's father 685 + 766 = 1451 rupees): (1) Sadia had 159 buttons and bought 244 more — total? (2) 376 litres of milk from one shop and 465 litres from another — total? (3) Farzana took 155 rupees from her mother and 147 from her father — total? (4) Bismah had 268 keychains and bought 153 — total? (5) A bat costs 395 rupees and a ball 127 rupees — price together? (6) The school had 289 benches and 372 more came — total? (7) Ali had saved 586 rupees and added 579 more — total? (8) Adnan sold 354 tickets and 259 the next day — total?

Answer: Exercise 25: Step 1: ones with carrying; Step 2: tens with carrying; Step 3: add the hundreds. 565 + 369 = 934, 287 + 327 = 614,

395 + 378 = 773, 729 + 175 = 904, 777 + 199 = 976, 555 + 395 = 950, 633 + 267 = 900, 468 + 338 = 806, 319 + 691 = 1010, 392 + 589 = 981, 663 + 258 = 921, 831 + 179 = 1010. Exercise 26: (1) 159 + 244 = 403 buttons. (2) 376 + 465 = 841 litres. (3) 155 + 147 = 302 rupees. (4) 268 + 153 = 421 keychains. (5) 395 + 127 = 522 rupees. (6) 289 + 372 = 661 benches. (7) 586 + 579 = 1165 rupees. (8) 354 + 259 = 613 tickets.

(Urdu edition — Commutative Property of Addition) Activity 1: Count the mangoes: 4 mangoes + 5 mangoes = _____ and 5 mangoes + 4 mangoes = _____. Examples: 27 + 12 = 39 and 12 + 27 = 39; 28 + 22 = 50 and 22 + 28 = 50. Exercise 27 (a): Count the pictures and write: (1) bananas 3 + 4 and 4 + 3 (2) apples 7 + 2 and 2 + 7 (3) stars 5 + 3 and 3 + 5. (b) (1) 4 + 3, 3 + 4 (2) 4 + 6, 6 + 4 (3) 8 + 0, 0 + 8 (4) 3 + 7, 7 + 3 (5) 32 + 24, 24 + 32 (6) 9 + 5, 5 + 9 (7) 64 + 24, 24 + 64 (8) 6 + 8, 8 + 6. (c) Verify that both answers are equal: 8 + 2 and 2 + 8; 9 + 1 and 1 + 9; 28 + 30; 24 + 36; 42 + 38; 40 + 50; 45 + 35; 31 + 49 — make the pairs for the last four yourself.

Answer: 9, 9.

Question: (Urdu edition — Subtraction) Example: Solve 23 − 7 step by step. Activity: Solve 35 − 8 and 40 − 6 on the abacus. Exercise 28: 37 − 9, 43 − 5, 61 − 8, 74 − 6, 85 − 7, 30 − 4.

Answer: Example 23 − 7: Step 1: 7 cannot be taken from ones 3, so borrow 1 ten: 23 = 1 ten + 13 ones. Step 2: ones: 13 − 7 = 6. Step 3: tens: 1 − 0 = 1. Final answer: 23 − 7 = 16. Activity: 35 − 8: 8 cannot be taken from 5, borrow: 15 − 8 = 7; tens left 2 = 27. 40 − 6: 10 − 6 = 4; tens 3 = 34. Exercise 28: 37 − 9 = 28, 43 − 5 = 38, 61 − 8 = 53, 74 − 6 = 68, 85 − 7 = 78, 30 − 4 = 26.

Question: (Urdu edition) Exercise 29: (a) 32 − 15, 70 − 25, 90 − 31, 76 − 47, 80 − 52, 42 − 24, 67 − 48, 55 − 28, 80 − 32, 75 − 57, 89 − 78, 70 −

51. (b) 82 − 31, 72 − 18, 53 − 29, 44 − 15, 35 − 27, 90 − 82, 95 − 77, 66 − 49, 37 − 28, 85 − 13. Word examples: Ayaz had 20 oranges and ate 7 — how many left? Rabia is 33 years old and Khalida is 19 — how many years younger?

Answer: Step 1: Subtract the ones (borrow if needed); Step 2: subtract the tens. (a) 32 − 15 = 17, 70 − 25 = 45, 90 − 31 = 59, 76 − 47 = 29, 80 − 52 = 28, 42 − 24 = 18, 67 − 48 = 19, 55 − 28 = 27, 80 − 32 = 48, 75 − 57 = 18, 89 − 78 = 11, 70 − 51 = 19. (b) 82 − 31 = 51, 72 − 18 = 54, 53 − 29 = 24, 44 − 15 = 29, 35 − 27 = 8, 90 − 82 = 8, 95 − 77 = 18, 66 − 49 = 17, 37 − 28 = 9, 85 − 13 = 72. Examples: 20 − 7 = 13 oranges left; 33 − 19 = 14 years younger.

Question: (Urdu edition) Exercise 30 (word problems): (1) Danish had 32 oranges and gave 3 to a friend — how many left? (2) Zareen had 23 hens and sold 5 — how many left? (3) Faraz has 12 kites — if 7 are red, how many are blue? (4) Nadeem scored 32 in cricket and needed 15 more to win — the other team's score? (5) Areeba collected 45 shells and gave 18 to her sister — how many left? (6) Kanwal picked 25 flowers from the garden and brought 16 fewer from the shop — how many flowers left? (7) Kareem had 35 seeds and sowed 8 — how many left? (8) Paras has 82 candies and his brother has 35 fewer — how many does the brother have? (9) Bismah had 24 bananas and gave 12 to a friend — how many left? (10) Saleem had 33 parrots and 20 flew away — how many left?

Answer: Step 1: In each question subtract the smaller number from the bigger one. (1) 32 − 3 = 29 oranges. (2) 23 − 5 = 18 hens. (3) 12 − 7 = 5 blue kites. (4) 32 − 15 = 17 (by the book's method, Nadeem has 17 more points). (5) 45 − 18 = 27 shells. (6) 25 − 16 = 9 flowers. (7) 35 − 8 = 27 seeds. (8) 82 − 35 = 47 candies. (9) 24 − 12 = 12 bananas. (10) 33 − 20 = 13 parrots.

Question: (Urdu edition) Exercise 31: (a) 3-digit − 1-digit (example: 269 − 7 = 262): 381 − 1, 455 − 4, 369 − 6, 578 − 7, 746 − 5, 637 − 3, 868 − 5, 779 − 3. (b) True (✓) / False (✗): 119 − 2 = 117; 507 − 5 = 502; 313 − 2 = 315; 429 − 7 = 322; 738 − 4 = 424. Exercise 32: (a) 3-digit − 2-digit (example: 589 − 38 = 551): 675 − 64, 487 − 76, 399 − 88, 431 − 20, 888 − 88, 567 − 60, 978 − 43, 336 − 23. (b) Match the pairs with equal answers: 125 − 20, 235 − 14, 385 − 71, 566 − 36, 666 − 66 and 561 − 31, 137 − 32, 677 − 77, 395 − 81, 247 − 26.

Answer: Exercise 31 (a): Step 1: Subtract only the ones. 381 − 1 = 380, 455 − 4 = 451, 369 − 6 = 363, 578 − 7 = 571, 746 − 5 = 741, 637 − 3 = 634, 868 − 5 = 863, 779 − 3 = 776. (b) 119 − 2 = 117 ✓; 507 − 5 = 502 ✓; 313 − 2 = 315 ✗ (correct 311); 429 − 7 = 322 ✗ (correct 422); 738 − 4 = 424 ✗ (correct 734). Exercise 32 (a): 675 − 64 = 611, 487 − 76 = 411, 399 − 88 = 311, 431 − 20 = 411, 888 − 88 = 800, 567 − 60 = 507, 978 − 43 = 935, 336 − 23 = 313. (b) 125 − 20 = 105 and 137 − 32 = 105; 235 − 14 = 221 and 247 − 26 = 221; 385 − 71 = 314 and 395 − 81 = 314; 566 − 36 = 530 and 561 − 31 = 530; 666 − 66 = 600 and 677 − 77 = 600.

Question: (Urdu edition) Exercise 33: (a) 3-digit − 3-digit (example: 387 − 265 = 122): 276 − 163, 369 − 232, 666 − 643, 725 − 304, 999 − 444, 883 − 262, 556 − 312, 428 − 115. (b) 257 − 141, 385 − 383, 467 − 157, 500 − 400, 611 − 600, 733 − 730. Exercise 34 (word problems; activity example: Saima's shoes 238 rupees, Raza's 225 rupees — difference 13 rupees): (1) Nazir scored 137 runs, Saleem 26 — difference? (2) A basket had 256 candies, 144 were sold — how many left? (3) Akram invited 299 friends, 242 came — how many did not come? (4) Rizwan had 986 books, 763 were sold — how many left? (5) 153 books are in the cupboard and there is space for 265 — how many more can be kept? (6) Kiran had 765

rupees and spent 545 — how much left? (7) Farah has 273 roses and 62 tulips — how many more roses?

Answer: Exercise 33: Step 1: ones; Step 2: tens; Step 3: subtract the hundreds. (a) 276 − 163 = 113, 369 − 232 = 137, 666 − 643 = 23, 725 − 304 = 421, 999 − 444 = 555, 883 − 262 = 621, 556 − 312 = 244, 428 − 115 = 313. (b) 257 − 141 = 116, 385 − 383 = 2, 467 − 157 = 310, 500 − 400 = 100, 611 − 600 = 11, 733 − 730 = 3. Exercise 34: (1) 137 − 26 = 111 runs. (2) 256 − 144 = 112 candies. (3) 299 − 242 = 57 friends did not come. (4) 986 − 763 = 223 books. (5) 265 − 153 = 112 more books. (6) 765 − 545 = 220 rupees. (7) 273 − 62 = 211 more roses.

Question: (Urdu edition) Exercise 35: (a) Subtract borrowing across zero (examples: 841 − 9 = 832 and 705 − 8 = 697): 325 − 9, 433 − 7, 571 − 4, 730 − 8, 902 − 6, 805 − 7, 604 − 7, 402 − 5, 723 − 6. (b) 552 − 3, 902 − 8, 303 − 5, 507 − 9, 685 − 8, 703 − 4, 800 − 2, 906 − 7, 396 − 9, 801 − 5, 203 − 4, 900 − 9. Exercise 36: (a) 3-digit − 2-digit with borrowing (examples: 739 − 64 = 675 and 845 − 78 = 767; activity: 900 − 84 = 816): 235 − 46, 567 − 88, 388 − 99, 666 − 77, 653 − 92, 760 − 81, 703 − 95, 334 − 98, 485 − 77, 700 − 36, 600 − 49, 550 − 62. (b) 330 − 45, 560 − 81, 446 − 97, 600 − 96, 364 − 75, 541 − 63.

Answer: Exercise 35: Step 1: If the ones are too small, borrow from the tens; if the tens are 0, borrow from the hundreds. (a) 325 − 9 = 316, 433 − 7 = 426, 571 − 4 = 567, 730 − 8 = 722, 902 − 6 = 896, 805 − 7 = 798, 604 − 7 = 597, 402 − 5 = 397, 723 − 6 = 717. (b) 552 − 3 = 549, 902 − 8 = 894, 303 − 5 = 298, 507 − 9 = 498, 685 − 8 = 677, 703 − 4 = 699, 800 − 2 = 798, 906 − 7 = 899, 396 − 9 = 387, 801 − 5 = 796, 203 − 4 = 199, 900 − 9 = 891. Exercise 36 (a): 235 − 46 = 189, 567 − 88 = 479, 388 − 99 = 289, 666 − 77 = 589, 653 − 92 = 561, 760 − 81 = 679, 703 − 95 = 608, 334 − 98 = 236, 485 − 77 = 408, 700 − 36 = 664, 600 − 49

= 551, 550 − 62 = 488. (b) 330 − 45 = 285, 560 − 81 = 479, 446 − 97 = 349, 600 − 96 = 504, 364 − 75 = 289, 541 − 63 = 478.

Question: (Urdu edition) Exercise 37: (a) 3-digit − 3-digit with borrowing (example: 582 − 294 = 288): 426 − 287, 345 − 186, 462 − 378, 712 − 425, 860 − 386, 822 − 278, 542 − 256, 613 − 336, 953 − 625. (b) 874 − 287, 847 − 758, 621 − 388, 513 − 239, 626 − 378, 931 − 832. Exercise 38 (word problems; example: 872 books in the library, 198 taken out — 674 left; note: the book's text prints 118 but its working uses 198): (1) Ali had 125 hens and sold 9 — how many left? (2) A mall has 135 shops and only 7 are open — how many are closed? (3) The secondary school has 650 students, the primary 153 — how many more in secondary? (4) 364 students came, 57 are absent — how many present? (5) There were 137 mangoes in the garden and people ate 47 — how many left? (6) The library has 932 books, 155 went out — how many left? (7) Faraz needs 181 rupees for a bicycle and has 172 — how many more does he need?

Answer: Exercise 37: Step 1: ones with borrowing; Step 2: tens with borrowing; Step 3: subtract the hundreds. (a) 426 − 287 = 139, 345 − 186 = 159, 462 − 378 = 84, 712 − 425 = 287, 860 − 386 = 474, 822 − 278 = 544, 542 − 256 = 286, 613 − 336 = 277, 953 − 625 = 328. (b) 874 − 287 = 587, 847 − 758 = 89, 621 − 388 = 233, 513 − 239 = 274, 626 − 378 = 248, 931 − 832 = 99. Exercise 38: (1) 125 − 9 = 116 hens. (2) 135 − 7 = 128 shops closed. (3) 650 − 153 = 497 more. (4) 364 − 57 = 307 present. (5) 137 − 47 = 90 mangoes. (6) 932 − 155 = 777 books. (7) 181 − 172 = 9 more rupees needed.

(Urdu edition — Missing Numbers) Examples: _____ books + 4 books = 12 books; _____ − 3 = 17; _____ + 50 rupees = 200 rupees; Sada's money − 25 rupees = 375 rupees. Exercise 39: (1) Iqbal: 7 added to which number makes 20? (2) 14 taken from Nazir's number leaves 30 —

the starting number? (3) 250 added to my number makes 700 — my number? (4) 320 taken from Haleema's number leaves 700 — the starting number? (5) Aslam had 300 rupees; he spent 80 — how much is left? (6) Rani had 500 rupees; she got 90 more — how much now? (7) Hasan's marks are 16 less than his sister's 80 marks — Hasan's marks? (8) The sister's marks are 24 more than Hasan's 97 marks — the sister's marks? (9) Umair's marks are 16 more than 92 — Umair's marks?

Answer: 12 − 4 = 8 books; 17, 20; 200 − 50 = 150 rupees; Sada: 375

(Urdu edition — Multiplication) Introduction: 4 bicycles × 2 wheels = 2 + 2 + 2 + 2 = 8 wheels. Activity 1: 5 plates × 3 balls in each plate = 3 + 3 + 3 + 3 + 3 = _____; 5 × 3 = _____. Activity 2: 2 stacks with 7 pencils in each stack — 2 × 7 = _____. Activity 3: 5 fans, 4 blades in each fan — 5 × 4 = _____. Exercise 40: Write as repeated addition (example: 4 × 2 = 2 + 2 + 2 + 2 = 8): 3 × 2, 5 × 10, 5 × 5, 4 × 5, 6 × 2, 5 × 2, 7 × 2.

Answer: 15;, 15., 14., 20.

(Urdu edition — Tables of 2 and 3) Table of 2 (2, 4, 6, ..., 20 given in the book's chart). Activity 1: Fill the empty circles by jumping in 2s (rows 2–10, 12–20, 22–30). Activity 2: 3 pairs of roses — 3 × 2 = _____ roses. Activity 3: 4 pairs of buttons — 4 × 2 = _____. Table of 3 (3, 6, 9, ..., 30). Activity 1: Write the multiples of 3 on the stars (row 1: 3, 6, _____, _____, _____; row 2: _____, ...; row 3: _____, ... up to 45). Activity 2: 4 bunches × 3 balloons = _____ balloons. Activity 3: 6 bunches × 3 bananas = _____ bananas.

Answer: 6, 8 buttons., 9,, 12,, 15; Row, 18, 21, 24, 27, 30; Row, 33, 36, 39, 42, 45., 12, 18

(Urdu edition — Tables of 4, 5 and 10) Table of 4, Activity 1: Write the count in 4s on the balloons (row 1: 4 to 56; row 2: 60 to 112). Activity 2: 6 cars × 4 wheels = _____ wheels. Activity 3: 5 nests × 4 eggs =

_____ eggs. Table of 5, Activity 1: Each flower has 5 petals: write the petals for 1 to 8 flowers. Activity 2: Count in 5s in the grid (5–50, 55–100, 105–150). Activity 3: Count in 5s from 5 to 60. Table of 10, Activity 1: Count in 10s in the grid (10–80, 90–160, 170–240, 250–320, 330–400). Activity 2: Count in 10s from 10 to 120. Write the tables of 4, 5 and 10 (up to 10 × 10).

Answer: 24, 20

(Urdu edition) Exercise 41: (1) Complete the multiplication sentences for the pictures (rows of wheels): ×5: 1 _____, 2 _____, ..., 10 _____; ×10: 1 _____, ..., 10 _____. Complete the tables: ×2, ×3, ×4 (from 1 to 10). (2) Solve: 7 × 5, 4 × 2, 9 × 3, 9 × 3, 8 × 2, 8 × 3, 10 × 5, 6 × 5, 5 × 5, 8 × 3, 4 × 10, 6 × 4, 5 × 10, 8 × 4, 9 × 2, 8 × 7, 7 × 10, 10 × 10.

Answer: 5, 10, 15,, 20, 25, 30,, 35, 40, 45, 50., 10, 20, 30, 40, 50, 60, 70, 80, 90, 100. ×2: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20., ×3: 3, 6, 9, 12, 15, 18, 21, 24, 27, 30. ×4: 4, 8, 12, 16, 20, 24, 28, 32, 36, 40.

Question: (Urdu edition) Exercise 42: (1) Multiply in columns (examples: 3 × 2 = 6 and 5 × 10 = 50 in vertical form): 9 × 2, 5 × 2, 5 × 5, 6 × 3, 8 × 4, 3 × 3, 2 × 4, 3 × 5, 5 × 10, 5 × 3, 10 × 10, 9 × 10, 4 × 10, and on the board 9 × 10. (2) Balloon questions: 3 × 5, 4 × 2, 5 × 5, 4 × 4, 2 × 3, 2 × 6, 5 × 3, 2 × 4, 4 × 5, 4 × 9, 5 × 6, 4 × 8. (3) Body questions: 1. How many eyes do you have? 2. Eyes of three children? 3. Hands of six children? 4. Fingers in one hand? 5. Fingers in five hands? (4) Solve: 3 × 2, 7 × 5, 8 × 2, 3 × 4, 5 × 2, 3 × 3, 9 × 3, 8 × 4, 6 × 4, 7 × 5, 5 × 5, 6 × 10, 7 × 10, 8 × 5, 8 × 3. (5) Count in 3s from 30 to 60. (6) Count in 10s from 50 to 150.

Answer: (1) Step 1: Use the table: 9 × 2 = 18, 5 × 2 = 10, 5 × 5 = 25, 6 × 3 = 18, 8 × 4 = 32, 3 × 3 = 9, 2 × 4 = 8, 3 × 5 = 15, 5 × 10 = 50, 5 × 3

= 15, 10 × 10 = 100, 9 × 10 = 90, 4 × 10 = 40, board: 9 × 10 = 90. (2) 3 × 5 = 15, 4 × 2 = 8, 5 × 5 = 25, 4 × 4 = 16, 2 × 3 = 6, 2 × 6 = 12, 5 × 3 = 15, 2 × 4 = 8, 4 × 5 = 20, 4 × 9 = 36, 5 × 6 = 30, 4 × 8 = 32. (3) 1. Eyes = 2. 2. 3 × 2 = 6 eyes. 3. 6 × 2 = 12 hands. 4. Fingers = 5. 5. 5 × 5 = 25 fingers. (4) 3 × 2 = 6, 7 × 5 = 35, 8 × 2 = 16, 3 × 4 = 12, 5 × 2 = 10, 3 × 3 = 9, 9 × 3 = 27, 8 × 4 = 32, 6 × 4 = 24, 7 × 5 = 35, 5 × 5 = 25, 6 × 10 = 60, 7 × 10 = 70, 8 × 5 = 40, 8 × 3 = 24. (5) 30, 33, 36, 39, 42, 45, 48, 51, 54, 57, 60. (6) 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150. (Urdu edition — Commutative Property of Multiplication) Examples: 3 × 4 = 12 and 4 × 3 = 12; 2 × 4 = 8 and 4 × 2 = 8; 6 × 3 = 18 and 3 × 6 = 18. Exercise 43: Complete the equal pairs: (1) 5 × 4 = _____ × _____ (2) 7 × 3 (3) 6 × 5 (4) 4 × 2 (5) 3 × 7 (6) 2 × 8 (7) 9 × 3 (8) 4 × 5 (9) 6 × 10. Exercise 44 (word problems; example: 1 balloon costs 5 rupees; 3 balloons = 3 × 5 = 15 rupees): (1) 5 ribbons at 6 rupees each — total? (2) 8 flats with 4 rooms in each flat — total rooms? (3) 10 fans with 3 blades in each fan — total blades? (4) 9 bicycles — how many wheels? (5) 40 cows — how many legs? (6) 6 children — 10 fingers each, total fingers? (7) 5 shirts with 7 buttons on each shirt — total buttons? (8) 7 children — how many arms? (9) 5 children with 4 pencils each — total?

Answer: 4, 5 = 20.

(Urdu edition — Division) Introduction Example 1: Share 8 candies equally in 4 bags: repeated subtraction 8 − 2 = 6, 6 − 2 = 4, 4 − 2 = 2, 2 − 2 = 0 (4 times), so 8 ÷ 2 = 4 and the inverse 4 × 2 = 8. Example 2: For 12 stars: 12 ÷ 3 = _____ and 12 ÷ 4 = _____. Example 3: 6 tomatoes: in 3 boxes _____ in each; in 2 boxes _____ in each. Exercise 45: (1) 12 balls in 3 plates (example: 4 in each plate), (2) 15 balloons in 5 groups — how many in each group?

Answer: 4 (4 stars in each group);, 3 (3 in each group)., =, tomatoes

Question: (Urdu edition) Exercise 46: (a) Divide (example: 15 ÷ 3 = 5): 16 ÷ 2, 10 ÷ 2, 80 ÷ 10, 16 ÷ 4, 24 ÷ 3, 14 ÷ 2, 45 ÷ 5, 18 ÷ 3, 27 ÷ 3, 50 ÷ 5, 24 ÷ 3, 12 ÷ 3, 18 ÷ 2, 30 ÷ 3, 10 ÷ 10. (b) By long division (example: 12 ÷ 3 = 4): 15 ÷ 3, 2 ÷ 2, 24 ÷ 3, 10 ÷ 2, 14 ÷ 2, 27 ÷ 3, 20 ÷ 4, 40 ÷ 5, 36 ÷ 4, 50 ÷ 5, 70 ÷ 10, 40 ÷ 5, 24 ÷ 4, 50 ÷ 10.

Answer: Step 1: Find the answer backwards from the table — e.g. for 16 ÷ 2 think 2 × _____ = 16, the answer is 8. (a) 16 ÷ 2 = 8, 10 ÷ 2 = 5, 80 ÷ 10 = 8, 16 ÷ 4 = 4, 24 ÷ 3 = 8, 14 ÷ 2 = 7, 45 ÷ 5 = 9, 18 ÷ 3 = 6, 27 ÷ 3 = 9, 50 ÷ 5 = 10, 24 ÷ 3 = 8, 12 ÷ 3 = 4, 18 ÷ 2 = 9, 30 ÷ 3 = 10, 10 ÷ 10 = 1. (b) 15 ÷ 3 = 5, 2 ÷ 2 = 1, 24 ÷ 3 = 8, 10 ÷ 2 = 5, 14 ÷ 2 = 7, 27 ÷ 3 = 9, 20 ÷ 4 = 5, 40 ÷ 5 = 8, 36 ÷ 4 = 9, 50 ÷ 5 = 10, 70 ÷ 10 = 7, 40 ÷ 5 = 8, 24 ÷ 4 = 6, 50 ÷ 10 = 5.

Question: (Urdu edition) Exercise 47 (word problems; example: Arif has 32 pictures, 4 on each page — how many pages? 32 ÷ 4 = 8 pages): (1) 50 candies shared equally among 5 children — how many for each child? (2) 18 books shared equally between 2 brothers — how many each? (3) 30 chairs in 3 rows — how many in each row? (4) 15 apples among 3 brothers — how many each? (5) Rafia has 24 stickers, 4 on each page — how many pages? (6) 28 candies, 4 in each box — how many boxes? (7) Pens for 80 rupees at 10 rupees per pen — how many pens? (8) Rafay travelled 20 km equally in 2 days — how many km each day?

Answer: Step 1: In each question divide the total by the number of equal parts. (1) 50 ÷ 5 = 10 candies for each child. (2) 18 ÷ 2 = 9 books each. (3) 30 ÷ 3 = 10 chairs in each row. (4) 15 ÷ 3 = 5 apples each. (5) 24 ÷ 4 = 6 pages. (6) 28 ÷ 4 = 7 boxes. (7) 80 ÷ 10 = 8 pens. (8) 20 ÷ 2 = 10 km each day.

Question: (Urdu edition — Money) Recognising Pakistani notes: 500, 1000 and 5000 rupee notes (read the front and back). Activity 1: Which groups

total 100 rupees? Tick them: (1) 30 + 20 + 50 (2) 26 + 54 + 40 (3) 80 + 10 + 10 (4) 95 + 10 + 5 (5) 50 + 10 + 30. Activity 2: Write the total amount: (1) 4 × 50 rupees (example = 200), (2) 2 × 500, (3) 6 × 20, (4) 7 × 1000, (5) 4 × 100.

Answer: Activity 1: Step 1: Add each group. (1) 30 + 20 + 50 = 100 ✓. (2) 26 + 54 + 40 = 120 ✗. (3) 80 + 10 + 10 = 100 ✓. (4) 95 + 10 + 5 = 110 ✗. (5) 50 + 10 + 30 = 90 ✗. Activity 2: (2) 2 × 500 = 1000 rupees. (3) 6 × 20 = 120 rupees. (4) 7 × 1000 = 7000 rupees. (5) 4 × 100 = 400 rupees.

(Urdu edition — Shop Prices) Prices: toy car 325 rupees, story book 350, football 210, shoes 499, jeep 500, doll 300, helicopter 395, lamp 550, school bag 465. Example 1: lamp + helicopter = _____? Example 2: bag − football = _____? (Note: the book's last line prints 225; the correct difference is 255.) Exercise 48: (a) Total price: (1) shoes + football (example = 709), (2) book + doll, (3) football + jeep, (4) helicopter + car, (5) lamp + car, (6) jeep + doll. (b) Difference of prices: (1) jeep − book, (2) car − doll, (3) shoes − football, (4) helicopter − car, (5) jeep − football, (6) lamp − doll.

Answer: 945 rupees., 255 rupees.

(Urdu edition) Exercise 48 (3)–(8): (3) Umair bought a toy for 25 rupees and had 100 rupees — how much left? (4) A child spent 100 rupees on a book and 500 on a notebook — total spent? (5) Ahmar had to pay 680 rupees; he paid 500 — how much is still to pay? (6) Sana spent 145 rupees on a toy and had 200 — how much left? (7) Raza earned 250, 135 and 350 rupees in three days — total? (8) A cake for 365 rupees and chocolates for 150 rupees — total spent? Multiplication example: 1 pencil costs 10 rupees; 8 pencils = _____? Exercise 49: (1) 6 biscuits at 5 rupees per biscuit (2) 4 ribbons at 8 rupees per ribbon (3) 9 notebooks at 10 rupees per notebook (4) 5 bags of fish at 7 rupees per bag (5) 3 scales at 6

rupees per scale (6) 2 bananas at 9 rupees per banana.

Answer: 80 rupees.

(Urdu edition) Division example: 2 pencils cost 16 rupees — 1 pencil = 16 ÷ 2 = _____? Exercise 50: (1) 3 ribbons cost 18 rupees — 1 ribbon? (2) 4 pens cost 40 rupees — 1 pen? (3) 5 balls cost 35 rupees — 1 ball? (4) 10 brushes cost 100 rupees — 1 brush? (5) 2 flowers cost 16 rupees — 1 flower?

Answer: 8 rupees per pencil.

Unit 13: Measurement (Urdu Edition Exercises)

Question: (Urdu edition — Measurement) Non-standard units: measuring with handspan, footstep, arm and foot. Paper-clip activity: chalk = 2 clips (example) — how many clips for the sharpener, the pencil and the paint brush? 1 metre = 100 centimetres. Activity 1: Choose the correct unit: (1) length of a mobile phone (2) length of a chalk (3) cloth for clothes (4) height of a cupboard.

Answer: Clip activity: Step 1: Lay clips along the object's length and count (relative estimates from the picture): sharpener ≈ 1 clip; pencil ≈ 3 clips; paint brush ≈ 4 clips. Activity 1: Step 1: Use centimetres for small things and metres for big things. (1) Mobile → centimetres. (2) Chalk → centimetres. (3) Cloth → metres. (4) Cupboard height → metres.

(Urdu edition) Activity 2: Write the lengths: desk = 1 metre (example), blackboard = _____ metres; pencil on the tilted scale ≈ _____ cm; book ≈ _____ cm. Exercise 51 (1): Measure with the scale and write: (1) nail/screw (2) matchstick (3) spoon (4) comb (5) pencil (6) toothpaste (7) toothbrush (8) key. (b) Lengths of the lines: (1) example 4 cm (2) (3) (4) (5). (c) Correct unit: (1) length of a desk (2) mathematics book (3) pencil (4) eraser (5) class bench. Activity: Which of the two bicycles is longer? Exercise 52: Tick the things longer than 1 metre: (1) school bus (2) book (3) height of the classroom (4) glass (5) your height (6) teacher's height (7) pencil.

Answer: 2, 14, 25 cm (relative picture readings).

Question: (Urdu edition — Mass) Activity 1: Put (✓) on the heavy object and (✗) on the light one: (1) kettle/cup (2) tennis ball/cricket bat (3) elephant/rabbit (4) jug/glass. 1 kilogram = 1000 grams. Activity (suitable

unit): (1) mango (example: tick on grams), (2) orange, (3) baby goat, (4) cucumber, (5) watermelon. Exercise 53: Write the suitable unit: (1) boy (example: kilograms), (2) goat, (3) sack of sugar, (4) pencil, (5) ring, (6) chalk, (7) hen, (8) paper clip.

Answer: Activity 1: Step 1: Compare which object is heavier: (1) kettle ✓ heavy, cup ✗ light; (2) tennis ball ✗, bat ✓; (3) elephant ✓, rabbit ✗; (4) jug ✓, glass ✗. Activity: (2) orange → grams; (3) baby goat → kilograms; (4) cucumber → grams; (5) watermelon → kilograms. Exercise 53: (2) goat → kilograms, (3) sack of sugar → kilograms, (4) pencil → grams, (5) ring → grams, (6) chalk → grams, (7) hen → kilograms, (8) paper clip → grams.

Question: (Urdu edition — Capacity) Comparisons: a bottle holds more water than a glass; a jug more than a cup. Activity: Tick the vessel that holds more: jug/glass (example: jug ✓); 1-litre bottle/mug; 1-litre milk carton/ink pot; glass/small cup; jug/250 ml milk; big bowl/small bowl. 1 litre = 1000 millilitres. Exercise 54: (1) If 1 litre fills 4 cups, how much water is needed for 8 cups? (2) Sara: a 1-litre box fills 3 glasses — how many litres for 6 glasses? (3) Ahmed: a 1-litre jug fills 2 glasses — how many glasses from 2 jugs? (4) In which will the bucket's 1 litre of water fit: (1) two 250 ml cups (2) two 500 ml bottles?

Answer: Activity: Choose the bigger vessel in each pair: bottle ✓ over mug; milk carton ✓; glass ✓ over the small cup; jug ✓; big bowl ✓. Exercise 54: (1) Step 1: 8 cups = 2 groups of 4 cups; Step 2: 1 litre for each group — 2 litres in all. (2) 6 glasses = 2 groups of 3 glasses — 2 litres. (3) Each jug = 2 glasses, so 2 jugs = 4 glasses. (4) (1) 250 + 250 = 500 ml — less than 1 litre ✗; (2) 500 + 500 = 1000 ml = 1 litre ✓.

Unit 14: Time (Urdu Edition Exercises)

Question: (Urdu edition — Time) One day = 24 hours, divided into two 12-hour parts. The long hand shows minutes, the short hand shows hours; example clock 5:00. Activity: Fill the numbers 1–12 in the flower clock — what time does it show? Facts: 60 minutes = 1 hour; 24 hours = 1 day; 7 days = 1 week; 4 weeks ≈ 1 month. Reading in 5 minutes: 8:15 = eight fifteen; 8:50 = eight fifty. Example clocks: 5:00, 5:05, 6:00, 6:05 (write in words).

Answer: Activity: Flower clock: the short hand is on 3 and the long hand is on 12 — the time is 3 O'clock. Example clocks: 5:00 = five o'clock; 5:05 = five oh five (five past five); 6:00 = six o'clock; 6:05 = six oh five. Remember: every number on the clock shows 5 minutes (1 = 5 minutes, 2 = 10 minutes, ..., 12 = 0/60 minutes).

(Urdu edition) Exercise 55: Write the times of the clocks in words: worked: 4:05 (four oh five), 11:25 (eleven twenty-five); star clock _____; square clock _____; flower clock _____; the three clocks below: purple clock _____, scroll clock _____, sun clock _____.

Answer: short, hand, and, minutes from the long hand (number × 5 = minutes). Star clock: long hand on 2 = 10 minutes, short hand at 3 — 3:10 (three ten). Square clock: long hand on 3 = 15 minutes, short hand just, past 6 — 6:15 (six fifteen). Flower clock: long hand on 7 = 35 minutes, short hand between 8 and 9 — 8:35 (eight thirty-five). Purple clock: long hand on 10 = 50 minutes, short hand just before 3, — 2:50 (two fifty). Scroll clock: long hand on 9 = 45 minutes, short hand near 6 — 5:45 (five forty-five). Sun clock: long hand on 5 = 25 minutes, short hand just past 1 — 1:25 (one twenty-five).

(Urdu edition — Quarter past, half past, quarter to) Half past: 3:30; one hour = _____ (60) minutes. Quarter past: 5:15; half an hour = _____ (30) minutes. Quarter to: 7:45; a quarter of an hour = _____ (15) minutes. Exercise 56: worked: half past three = 3:30; quarter past three = 3:15; quarter to four = 3:45. Fill in: half past ten = _____; half past seven = _____; quarter past ten = _____; quarter past eight = _____; quarter to eleven = _____; quarter to nine = _____.

Answer: 60 minutes;, 30 minutes;, 15, minutes.)

(Urdu edition — a.m./p.m.) From midnight 12 to noon = a.m.; from noon to midnight = p.m. Examples: 12:00 night (sleeping), 6:00 a.m. (waking up), 12:00 noon (playing), 6:00 p.m. (TV). Activity: Write a.m. or p.m.: Sunny's breakfast 7:00 _____; Bilal and Maria go to school 7:45 _____; Nasir's lunch 1:30 _____; Azam and Fozia play 5:15 _____; Salman goes to bed 10:00 _____.

Answer: = a.m.;, = a.m.;, = p.m.;, = p.m.;, = p.m.

Question: (Urdu edition) Exercise 57: Draw the hands on the clocks (describe them in words): worked 5:35; 7:10, 11:25, 3:05, 8:15, 6:30, 9:40, 10:20, 2:55.

Answer: Step 1: Put the long minute hand on the correct number (minutes ÷ 5 = number); Step 2: put the short hour hand between the two numbers accordingly. 5:35 (worked: five thirty-five; short hand between 5 and 6, long hand on 7). 7:10: short hand just past 7, long hand on 2. 11:25: short hand between 11 and 12, long hand on 5. 3:05: short hand just past 3, long hand on 1. 8:15: short hand just past 8, long hand on 3. 6:30: short hand halfway between 6 and 7, long hand on 6. 9:40: short hand between 9 and 10 (near 10), long hand on 8. 10:20: short hand just past 10, long hand on 4. 2:55: short hand just before 3, long hand on 11.

(Urdu edition — Calendar) Activity: Write the solar months in order (1

January is given): 2 _____, 3 _____, ..., 12 _____. Exercise 58 (from the book's printed solar calendar): (1) On which date and day does your birthday fall? (2) On which date is Pakistan's Independence Day? (3) On which date is the first Friday of May? (4) On which day is 23 March? (5) December starts on which day?

Question: (Urdu edition — Lunar Calendar) Write the lunar months in order. Activity questions: (1) Which is the first Islamic month? (2) Which is the month of fasting? (3) Which is the last month of the lunar year? (4) In which month does Eid-ul-Adha come? Exercise 59 (from the book's printed lunar calendar): (1) On which day is 27 Rajab? (2) Which day is one day before 10 Muharram? (3) Which day is one day after 5 Zilquad? (4) On which date is Hajj? (5) Which is the first lunar month? (6) Which month comes after Shaban?

Answer: Lunar months: 1 Muharram, 2 Safar, 3 Rabi-ul-Awwal, 4 Rabi-us-Sani, 5 Jamadi-ul-Awwal, 6 Jamadi-us-Sani, 7 Rajab, 8 Shaban, 9 Ramadan, 10 Shawwal, 11 Zilquad, 12 Zil Hajj. Activity: (1) Muharram. (2) Ramadan. (3) Zil Hajj. (4) Zil Hajj. Exercise 59: (1) Step 1: 1 Rajab = Wednesday (in the book's calendar); Step 2: 27 Rajab = Monday. (2) 1 Muharram = Sunday, so 9 Muharram = Monday — one day before 10 Muharram is Monday. (3) 1 Zilquad = Wednesday, 5 Zilquad = Sunday, so 6 Zilquad = Monday. (4) Hajj = 9 Zil Hajj. (5) Muharram. (6) Ramadan.

Unit 15: Geometry (Urdu Edition Exercises)

(Urdu edition — 2-D Shapes) Names of the shapes: square, rectangle, circle, semi-circle, quarter-circle, triangle. Activity 1: Colour with the given colours (examples at the top: red square, green triangle, yellow rectangle, blue circle): (1) square (2) rectangle (3) triangle (4) circle — write the colour for each. Activity 2: Write the name of the shape matching each object: book (square-like) _____; car tyre _____; samosa _____; register _____.

Answer: = square;, = circle;, = triangle;, = rectangle.

(Urdu edition — Folding) Fold a paper circle in half and it becomes a _____; fold it again and it becomes a _____. Exercise 60 (1): Join the dots to complete each shape and write its name: dotted circle (example: circle), _____ (rectangle-like), _____ (right triangle), _____ (square-like), _____ (semi-circle-like), _____ (quarter-circle-like). (2) Write the name of the left shape and tick the same shape on the right: Row 1: triangle (example, ✓ on the triangle) — choices: square, triangle, quarter circle; Row 2: _____ — choices: semi-circle, rectangle, circle; Row 3: _____ — choices: square, triangle, rectangle; Row 4: _____ — choices: wedge piece, semi-circle, circle; Row 5: _____ — choices: rectangle, square, semi-circle; Row 6: _____ — choices: semi-circle, wedge piece, triangle.

Answer: → semi-circle; folded, → quarter-circle (one-quarter part)., rectangle,, triangle,, square,, semi-circle,, quarter-circle., first shape, (the square)., left = semi-circle; (✓) on the second shape (the semi-circle)., left = rectangle; (✓) on the first shape (the rectangle)., left = quarter-circle (wedge piece); (✓) on the second shape.

(Urdu edition) Activity 1: Adil has 3 big shapes (square, circle, triangle) and 3 small shapes; he puts one small shape inside one big shape (picture: a small square inside a big square) — how many shapes can he make this way? _____ _____ _____. Activity 2: Lallu saw a cartoon: its body was like a rectangle, its legs were like triangles and there were small circles all over its body — describe the cartoon's shape in words.

Answer: rectangle;, the, two legs are

Question: (Urdu edition — Sides and Vertices) Triangle: 3 sides and 3 vertices. Rectangle: 4 sides and 4 vertices. Square: 4 sides and 4 vertices. Exercise 61: Write the name of each shape and its sides and vertices: (1) ABC right triangle (example: this is a triangle) (2) ABCD rectangle (3) ABCD square.

Answer: Step 1: Count the sides — 3 sides means a triangle; 4 sides means a rectangle or a square (all sides of a square are equal). (1) This is a triangle; it has 3 sides and 3 vertices. (2) This is a rectangle; it has 4 sides and 4 vertices. (3) This is a square; it has 4 sides and 4 vertices.

Question: (Urdu edition — Straight and Curved Lines) A straight road and a curved road; a tightly tied rope shows a straight line, a loosely tied rope shows a curved line; read the examples of the straight line AB and the curved line AB. Exercise 62: Put (✓) on straight lines and (✗) on curved ones: (1) a snake's movement (example: ✗) (2) a kite's loose string (3) the edges of a door (4) the edges of a blackboard (5) railway tracks (6) a coil (round and round wire). Activity: Tick the type of each line: (1) a straight double-arrow line (given: ✓ on straight) (2) a curved line (given: ✓ on curved) (3) a loop line (4) a slanted straight line (5) a looped curved line (6) a wavy line. Exercise 63: Write the number of straight and curved lines in each shape: (1) a rectangle with one side curved inwards (2) a shape with two ends curved outwards (3) a three-quarter circle (270-degree

shape).

Answer: Exercise 62: Step 1: See whether the line goes straight without bending or bends along the way. (2) A kite's loose string = curved ✗. (3) Edges of a door = straight ✓. (4) Edges of a blackboard = straight ✓. (5) Railway tracks = straight ✓. (6) A coil = curved ✗. Activity: (3) ✓ on curved line. (4) ✓ on straight line. (5) ✓ on curved. (6) ✓ on curved. Exercise 63: (1) Straight lines = 3 (top, bottom, right side), curved = 1 (the left curved side). (2) Straight = 4 (top, bottom and two slanted sides), curved = 2 (the two bulging ends). (3) Straight = 2 (the two radii), curved = 1 (the big arc).

Question: (Urdu edition — Drawing Line Segments) Read the steps for drawing a line segment with a scale: Step 1: place the scale; Step 2: draw the line with a pencil; Step 3: write the length (in the book's example Nasir is said to draw 4 cm but the drawn segment is labelled 5 cm — note this inconsistency in the book). Exercise 64: (1) Draw segments of the given measures (describe the method): 4 cm, 6 cm, 2 cm, 3 cm, 9 cm, 7 cm. (2) Hina drew a segment of 8 cm and Shania drew one of 9 cm — whose segment is shorter?

Answer: Method: Step 1: Place the 0 of the scale at the starting point; Step 2: mark the second point at the required measure; Step 3: join both points along the edge of the scale, draw the line and write the measure above it. In this way segments of 4 cm, 6 cm, 2 cm, 3 cm, 9 cm and 7 cm are drawn. (2) Step 1: Compare the measures: 8 is smaller than 9. Final answer: Hina's 8 cm segment is shorter.